# V. I. Arnold: Ordinary Differential Equations
# Chapters 1–3 — Worked Solutions

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**Abstract**

Solutions to all explicitly numbered problems in Chapters 1–3, "Basic Concepts," "Basic Theorems," and "Linear Systems," of the third edition, translated by Roger Cooke (Springer, 1992), together with answers to selected embedded proof questions. Statements are summarized; the textbook remains the reference for their full wording. Each identifier has the form section.subsection/problem, so repeated problem numbers are unambiguous. Page references are the printed pages, not the PDF viewer's page numbers (the library scan has an offset of four).

**Scope and conventions.** This is a complete three-chapter draft for mathematical review. Solutions are real unless indicated otherwise. A phase curve is the image of a maximal solution; an integral curve includes the time coordinate. Formulas obtained by division are always supplemented by the solutions lost in that division. A flow is global only when its solutions exist for every real time. Stars in the textbook are retained in the relevant headings.

**Reading cautions.** The projective-extension wording on p. 24 requires clarification; see the corresponding solution. Angular velocity for the oscillator is signed $-1$ in the usual orientation. Degenerate zero-radius "circles" and "helices" are singletons and straight lines, respectively. All diagrams below are original schematic drawings.

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# Chapter 1 — Basic Concepts

## 1 Phase spaces (§1)

### 1.2/1: Two roads and two circular wagons

*Source: printed page 14–15.*

Parametrize the roads continuously by $a, b : [0,1] \to \mathbb{R}^2$, from $A$ to $B$. A joint configuration is $(s,u) \in [0,1]^2$. The tethered cars trace a continuous path from $(0,0)$ to $(1,1)$ on which $|a(s)-b(u)| < 2l$. Wagons moving in opposite directions trace a path from $(0,1)$ to $(1,0)$. Paths in a square joining these alternating boundary vertices must intersect. At an intersection the distance between wagon centers is less than $2l$, so the disks overlap. Thus passage without touching is impossible.

Here the elementary crossing lemma follows, for simple polygonal paths, from the Jordan curve theorem by adjoining a boundary arc. General continuous paths follow by uniform polygonal approximation: if their compact images were disjoint, their positive distance would allow disjoint approximations, contradicting the polygonal case. Perturb the approximations slightly into the square except at their endpoints to handle contacts with its boundary.

**Review note.** A free rigid body has three translational and three rotational configuration coordinates, and six velocity coordinates. Hence $n$ unconstrained rigid bodies have a $12n$-dimensional phase space. Locally the configuration space of one body is $\mathbb{R}^3 \times SO(3)$ and its tangent bundle has dimension twelve.

### 1.6/1: Isothermal atmospheric density

*Source: printed page 22–23.*

Hydrostatic balance and the isothermal ideal-gas law give

$$p'(h) = -g\rho(h), \qquad p(h) = c\rho(h), \qquad \rho(h) = \rho_0 e^{-h/H}, \quad H = \frac{c}{g} = \frac{p_0}{\rho_0 g}.$$

Consequently $h_{1/2} = H \ln 2$. Taking the sea-level pressure implicit in the book's numerical answer, $p_0 \simeq 10^5\,\mathrm{Pa}$, $\rho_0 = 1.25\,\mathrm{kg/m^3}$ and $g \simeq 10\,\mathrm{m/s^2}$, gives $H \simeq 8\,\mathrm{km}$ and $h_{1/2} \simeq 5.55\,\mathrm{km}$. Density alone does not determine $H$; a surface pressure or temperature is an additional physical assumption. With $101325\,\mathrm{Pa}$ and $9.81\,\mathrm{m/s^2}$ the value is about $5.73\,\mathrm{km}$.

### 1.7/1*: Extending a monomial field to the projective line

*Source: printed page 24.*

Take $n$ to be an integer for which $x^n\partial_x$ is smooth on all of $\mathbb{R}$, so $n \geq 0$. Near infinity use $y = 1/x$. Then

$$\dot{y} = -x^{-2}\dot{x} = -y^{2-n}.$$

This is smooth at $y = 0$ precisely for $n = 0, 1, 2$. Thus the smooth extensions are exactly those in the printed answer. The phrase "without singular points" cannot literally mean "without zeros": $n = 0$

has a zero at infinity, $n = 1$ has zeros at zero and infinity, and $n = 2$ has a zero at zero. Literally no member of this list is everywhere nonzero. The intended meaning is absence of a singularity obstructing smooth extension.

### 1.8/1: The logistic curve

*Source: printed page 25.*

On $0 < x < 1$, separate variables:

$$\log \frac{x}{1-x} = t + C, \qquad x(t) = \frac{1}{1 + Ae^{-t}}, \quad A > 0.$$

For $x(t_0) = x_0$, $A = e^{t_0}(1 - x_0)/x_0$. These solutions are defined for all $t$, approach zero as $t \to -\infty$, and approach one as $t \to +\infty$. The constant solutions $x = 0, 1$ must also be included. The same rational formula describes other branches whenever its denominator is nonzero.

### 1.8/2: The branch above carrying capacity

*Source: printed page 25.*

For $x_0 > 1$ put $q = (x_0 - 1)/x_0 \in (0, 1)$. Then

$$x(t) = \frac{1}{1 - qe^{-(t-t_0)}}, \qquad t > t_* := t_0 + \ln q.$$

As $t \downarrow t_*$ the denominator tends to zero from above, giving a vertical asymptote. As $t \to \infty$, $x(t) \downarrow 1$. The maximal interval of this branch is $(t_*, \infty)$.

### 1.13/1: Free particle with initial state $(a, b)$

*Source: printed page 31.*

Integrating $\dot{x}_2 = 0$ and then $\dot{x}_1 = x_2$ gives $x_2(t) = b$, $x_1(t) = a + bt$. For $b \neq 0$ the phase curve is the entire horizontal line at height $b$; for $b = 0$ it is the singleton $(a, 0)$.

### 1.14/1: Free-fall phase curves

*Source: printed page 32.*

For $g > 0$, $\dot{x}_1 = x_2$, $\dot{x}_2 = -g$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}(x_2^2 + 2gx_1) = 0, \qquad x_1 = \frac{C - x_2^2}{2g}.$$

These are parabolas in the phase plane, not the particle's trajectory in physical space. Since $x_2 = b - gt$ ranges over $\mathbb{R}$, each full parabola is one maximal phase curve.

### 1.15/1: Circular phase curves of the oscillator

*Source: printed page 32–33.*

For $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1$, differentiation gives $(x_1^2 + x_2^2)' = 0$. On a positive-radius circle the field never vanishes, and the explicit solution below traverses the entire circle. The origin is a separate equilibrium phase curve.

### 1.15/2: Angular speed

*Source: printed page 33.*

At radius $r > 0$,

$$\dot{\theta} = \frac{x_1\dot{x}_2 - x_2\dot{x}_1}{r^2} = -1.$$

Thus the speed of rotation is one and its direction is clockwise. The signed angular velocity is $-1$, not $+1$, with the usual counterclockwise convention. At the origin angle is undefined.

### 1.15/3: Oscillator initial-value problem

*Source: printed page 33.*

$$x_1(t) = a\cos t + b\sin t, \qquad x_2(t) = -a\sin t + b\cos t.$$

Differentiating verifies both equations, and evaluation at zero gives $(a, b)$.

### 1.15/4: Energy for arbitrary stiffness

*Source: printed page 33.*

For $\dot{x}_1 = x_2$, $\dot{x}_2 = -kx_1$,

$$E = \tfrac{1}{2}x_2^2 + \tfrac{k}{2}x_1^2, \qquad \dot{E} = x_2(-kx_1) + kx_1x_2 = 0.$$

For $k > 0$ the nonzero energy curves are ellipses; their frequency is $\sqrt{k}$.

### 1.15/5: Helices in extended phase space

*Source: printed page 33.*

Writing the solution as $(t, r\cos(\theta_0 - t), r\sin(\theta_0 - t))$ shows that it is a circular helix about the time axis, with radius $r$ and pitch of magnitude $2\pi$. For $r = 0$ it degenerates to the time axis. Its projection onto the phase plane is the phase circle.

### 1.16/1: Equilibria of the mathematical pendulum

*Source: printed page 34.*

The field $(x_2, -\sin x_1)$ vanishes at $(0,0)$ and $(\pi,0)$, so their constant solutions have these singletons as phase curves. On the covering plane all equilibria are $(m\pi, 0)$; on the physical cylinder $x_1 \bmod 2\pi$ there are two.

### 1.18/1: Spherical-pendulum energy spheres

*Source: printed page 35.*

Let $X = (x_1, x_2, x_3, x_4)$ and $JX = (x_2, -x_1, x_4, -x_3)$. Since $J^T = -J$, the system $\dot{X} = JX$ satisfies $(|X|^2)' = 2\langle X, JX\rangle = 0$. Nonzero phase curves lie on three-spheres $S^3_r$; the zero-energy set is one point.

### 1.18/2: Great circles

*Source: printed page 35.*

Since $J^2 = -I$,

$$X(t) = \cos t\, X_0 + \sin t\, JX_0.$$

For $X_0 \neq 0$, $X_0$ and $JX_0$ are orthogonal and have equal length. The orbit is the intersection of their two-dimensional linear span with $S^3_r$, hence a great circle. Only $J$-invariant two-planes produce phase circles.

### 1.18/3*: The sphere of phase circles

*Source: printed page 35.*

Normalize $r = 1$ and put $z_1 = x_1 + ix_2$, $z_2 = x_3 + ix_4$. The flow is $z \mapsto e^{-it}z$. Define

$$H(z_1, z_2) = \bigl(2\Re(z_1\bar{z}_2),\, 2\Im(z_1\bar{z}_2),\, |z_1|^2 - |z_2|^2\bigr).$$

Its norm is $(|z_1|^2 + |z_2|^2) = 1$, and $H$ is unchanged by common multiplication by a unit complex number. Conversely, equality of $H$ determines the rank-one Hermitian matrix $zz^*$, so two unit vectors with the same image differ by precisely such a factor. Every point of $S^2$ occurs: the corresponding matrix

$$\frac{1}{2}\begin{pmatrix} 1+Z & X+iY \\ X-iY & 1-Z \end{pmatrix}$$

is positive semidefinite of rank one and trace one and hence equals $zz^*$. Thus the fibers are exactly the phase circles. The induced continuous bijection from the compact quotient to the Hausdorff sphere is a homeomorphism. In the charts $z_1 \neq 0$ and $z_2 \neq 0$, the coordinates $z_2/z_1$ and $z_1/z_2$ also give the smooth quotient $\mathbb{C}P^1 \simeq S^2$.

### 1.18/4*: Linking of two phase circles

*Source: printed page 35.*

Use the complex orientation of $\mathbb{C}^2$ and the boundary orientation of $S^3$. A unitary transformation, isotopic to the identity, takes the first fiber to $K_1 = \{(e^{i\theta}, 0)\}$. The other is $K_2 = \{e^{i\theta}(a, b)\}$ with $b \neq 0$. On $S^3 \setminus K_1$ the map

$$(z_1, z_2) \longmapsto z_2/|z_2|$$

measures linking with $K_1$: a page of this angular map is a spanning disk of $K_1$, and the degree along a disjoint loop counts its signed intersections with that page. On $K_2$ its degree is $+1$. Thus the linking number is $+1$ when both fibers have the common complex orientation. The actual oscillator flow reverses both orientations, which leaves the linking number unchanged. Reversing only one orientation, or reversing the ambient orientation, changes the sign; the orientation-independent assertion is absolute linking number one.

## 2 Vector fields on the line (§2)

### 2.1/1: The gap in the existence–uniqueness proof

*Source: printed page 36.*

At a zero $x_0$ of $v$, exhibiting $x(t) = x_0$ proves existence but not uniqueness. Separation by $v$ is unavailable there. For the continuous field $v(x) = |x|^{2/3}$ both $x = 0$ and $x = (t/3)^3$ pass through zero. Smoothness, or a suitable substitute, must be used to exclude this behavior.

### 2.3/1: Faster-than-exponential approach

*Source: printed page 38.*

If a nonconstant solution approaches an equilibrium $a$, local Lipschitz continuity gives $|v(x)| \leq L|x - a|$ near $a$. For all sufficiently large $t$, integration of the differential inequality yields

$$|x(t) - a| \geq |x(T) - a|e^{-L(t-T)}.$$

Thus it cannot approach faster than every exponential. For two nonstationary solutions the answer can be yes on an open phase interval. On $U = (0, 1)$ the smooth field $v(x) = x \log x$ has solutions

$$x_a(t) = \exp(-ae^t), \qquad a > 0.$$

For $a \neq b$, $|x_a(t) - x_b(t)| \leq \exp(-\min(a, b)e^t)$, which decays faster than every exponential. Their common limit zero is outside $U$; the field does not extend smoothly there. This domain qualification matters: if the limiting point is an interior equilibrium of a smooth field, the preceding Lipschitz obstruction applies.

### 2.3/2: A derivative that is not continuous

*Source: printed page 38.*

Yes, for the scalar autonomous equation, differentiability of $v$ suffices. Away from zeros, continuity and separation give uniqueness. At a zero $a$, existence of the finite derivative implies $v(x) = v'(a)(x-a) + o(|x-a|)$, hence $|v(x)| \le C|x-a|$ near $a$. A nonconstant trajectory would need infinite time to leave or reach $a$, by comparison with $\int \mathrm{d}x/|x-a|$. Continuity of $v'$ is unnecessary. This argument is specific to the scalar autonomous setting.

### 2.3/3: The divergent travel-time criterion

*Source: printed page 38.*

At an equilibrium $a$, interpret divergence as infinite one-sided travel time on every side on which a departure or arrival could occur, not as a principal value with cancellations. If $v$ has no zeros between $a$ and $b$, separation would give a finite departure time equal to the divergent integral $|\int_a^b \mathrm{d}\xi/v(\xi)|$, an immediate contradiction.

The same conclusion holds with intervening zeros, using a regularized integral. For a proposed departure to $b > a$, the trajectory must cross every intermediate value toward the right, so $v \ge 0$ throughout $[a, b]$. For $\varepsilon > 0$ define $F_\varepsilon(x) = \int_a^x \mathrm{d}\xi/(v(\xi) + \varepsilon)$. Along the portion from $a$ to its first arrival at $b$,

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\varepsilon(x(t)) = \frac{v(x(t))}{v(x(t)) + \varepsilon} \le 1, \qquad \int_a^b \frac{\mathrm{d}\xi}{v(\xi) + \varepsilon} \le t_b - t_a.$$

Letting $\varepsilon \downarrow 0$ contradicts infinite travel time, even if the zeros accumulate. The other spatial direction and backward time are identical after reversing signs. Thus only the constant solution passes through $a$. At a nonzero point uniqueness already follows by inverting the strictly monotone travel-time function.

### 2.3/4: Lipschitz uniqueness

*Source: printed page 38.*

For solutions $x, y$ with the same initial value,

$$|x(t) - y(t)| \le L \int_{t_0}^{t} |x(s) - y(s)| \, \mathrm{d}s \quad (t \ge t_0).$$

Gronwall's inequality gives zero. Alternatively, on an interval of length $h < 1/L$, taking the supremum gives $M \le LhM$, hence $M = 0$, and successive intervals extend the conclusion. Reverse time for $t < t_0$. This also proves the vector-valued version with a norm.

### 2.3/5: Nonautonomous uniqueness

*Source: printed page 38.*

Let $\phi$ be one solution and put $z = x - \phi(t)$. Then

$$\dot{z} = v(t, \phi(t) + z) - v(t, \phi(t)), \qquad z(t_0) = 0.$$

On a compact neighborhood of the initial point, smoothness gives a bound $L$ for the derivative with respect to $x$, so the right-hand side has norm at most $L|z|$. The preceding integral inequality proves $z = 0$ locally. Continuing through overlapping neighborhoods proves equality on the common interval. For systems, replace the derivative by its operator norm.

### 2.3/6: Invariance of the predator–prey quadrant

*Source: printed page 38.*

For $\dot{x} = x(k - ay)$ and $\dot{y} = y(-l + bx)$,

$$x(t) = x(t_0) \exp \int_{t_0}^{t} (k - ay(s))\, \mathrm{d}s, \qquad y(t) = y(t_0) \exp \int_{t_0}^{t} (-l + bx(s))\, \mathrm{d}s.$$

Positive coordinates remain positive at every finite time in the existence interval. The axes are invariant by uniqueness; no interior phase curve can meet them. Phase curves initially on an axis of course remain there.

### 2.3/7: Uniqueness on the whole common domain

*Source: printed page 39.*

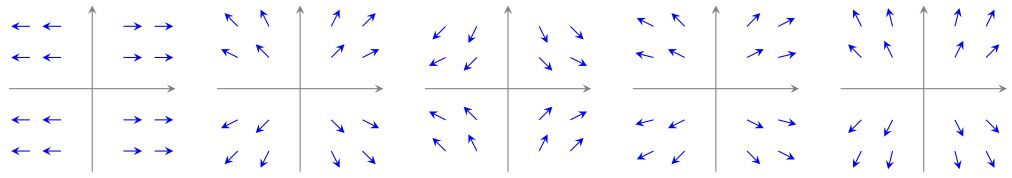
The intersection $I$ of two solution intervals containing $t_0$ is an interval. The set $E = \{t \in I : x(t) = y(t)\}$ is nonempty and relatively closed by continuity. Local uniqueness makes it relatively open. Connectedness gives $E = I$.

### 2.5/1: Five product fields

*Source: printed page 39.*

For $(\dot{x}, \dot{y}) = (x, ky)$, the solution is $(ae^t, be^{kt})$. The following table and schematic portraits specify the field and its orientation.

| $k$ | Off-axis curves | Forward motion |
|---|---|---|
| 0 | $y = b$ | Away from the equilibrium line $x = 0$ |
| 1 | Rays through zero | Radially outward |
| $-1$ | $xy = C$ | Toward the $x$-axis, away from the $y$-axis |
| $1/2$ | $x = Cy^2$ in each half-plane | Outward, tangent to $y$-axis near zero |
| 2 | $y = Cx^2$ in each half-plane | Outward, tangent to $x$-axis near zero |




### 2.5/2: A level set need not be one phase curve

*Source: printed page 40.*

The set $|y| = C|x|^k$ generally includes several orbits: signs of nonzero coordinates are preserved, and no nonconstant orbit can contain an equilibrium. For example, with $k = 2$ a whole parabola contains its two open half-branches and the equilibrium at the origin. For $k = 0$, a horizontal line consists of two open half-lines and its equilibrium on the $y$-axis. A level-set equation describes invariant sets; their maximal connected moving pieces, with equilibria separated, are the phase curves.

### 2.5/3: All phase curves of a product field

*Source: printed page 41.*

For $k \neq 0$, off-axis orbits are the individual quadrant branches

$$y = b\left(\frac{x}{a}\right)^k, \qquad x/a > 0, \quad ab \neq 0.$$

There are also four coordinate half-axes and the origin. For $k > 1$ branches approach zero tangent to the $x$-axis; for $0 < k < 1$ they are tangent to the $y$-axis; for $k = 1$ they are rays. All move outward for $k > 0$. For $k < 0$ they are unbounded at both ends, approaching the axes asymptotically, with $|x|$ increasing and $|y|$ decreasing in forward time. For $k = 0$, every $(0, b)$ is an equilibrium and the other curves are $\{(x, b) : x > 0\}$ and $\{(x, b) : x < 0\}$. In particular $y = x^2$ consists of exactly three phase curves.

### 2.5/4: A very anisotropic node and saddle

*Source: printed page 41.*

For $k = .01$, $|y| = C|x|^{.01}$, equivalently $|x| = (|y|/C)^{100}$: near zero non-axial curves are almost vertical, although forward growth in $x$ is much faster than in $y$. For $k = -.01$, $|y| = C|x|^{-.01}$: the $y$-axis is stable and the $x$-axis unstable; $x$ grows on time scale one, while $y$ decays on time scale one hundred. Small $|k|$ does not turn the node or saddle into the $k = 0$ equilibrium line.

### 2.5/5: Inverted pendulum

*Source: printed page 41.*

Set $X = x_1 + x_2$, $Y = x_1 - x_2$. Then $\dot{X} = X$, $\dot{Y} = -Y$. With initial condition $(a, b)$,

$$x_1 = a\cosh t + b\sinh t, \qquad x_2 = a\sinh t + b\cosh t.$$

The first integral is $x_1^2 - x_2^2 = XY = a^2 - b^2$. Nonzero levels have two separate hyperbola branches. The zero level splits into the origin and four rays: $x_2 = -x_1$ is stable and $x_2 = x_1$ unstable. The unique velocity leading a displacement $a$ toward the upper equilibrium is $b = -a$.

### 2.6/1: Four separable direction fields

*Source: printed page 43.*

Integration gives the following curves, always restricted to the domain of the displayed differential equation:

| | | | |
|---|---|---|---|
| $y' = y/x$ | $y = Cx$ | $x \neq 0$ | (rays/line segments on each half-plane) |
| $y' = x/y$ | $y^2 - x^2 = C$ | $y \neq 0$ | (hyperbola branches) |
| $y' = -y/x$ | $xy = C$ | $x \neq 0$ | (rectangular hyperbolas; also $y = 0$) |
| $y' = -x/y$ | $x^2 + y^2 = C$ | $y \neq 0$ | (upper/lower semicircles) |

At $C = 0$ in the second equation the lines $y = \pm x$ are split at the excluded origin. In the fourth equation $C > 0$; the origin is not in the equation's domain. Vertical tangencies of the implicit curves require switching independent variable rather than pretending $y(x)$ stays differentiable there.

### 2.6/2: Powers and trigonometric examples

*Source: printed page 43.*

For $y' = kx^\alpha y^\beta$, on a real smooth branch with $y \neq 0$ define

$$F_\gamma(s) = \begin{cases} s^{1-\gamma}/(1-\gamma), & \gamma \neq 1, \\ \log|s|, & \gamma = 1. \end{cases}$$

Then the curves are $F_\beta(y) = kF_{-\alpha}(x) + C$. This includes the logarithmic cases $\alpha = -1$ or $\beta = 1$. If $y = 0$ belongs to the domain and the right-hand side vanishes there, add that solution; if the right-hand side is not locally Lipschitz there, additional glued solutions may occur. For $k = 0$ all admissible horizontal lines are solutions. Arbitrary real powers should normally be read on $x > 0, y > 0$, unless a different real branch is specified.

For the remaining equations,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin y}{\sin x} \quad \Longrightarrow \quad \log|\tan(y/2)| = \log|\tan(x/2)| + C,$$

where separation is valid, together with $y = m\pi$; the domain excludes $x = n\pi$. Equivalently $\tan(y/2) = A\tan(x/2)$ on each admissible chart. For

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin x}{\sin y}, \qquad \cos y - \cos x = C,$$

restrict to $\sin y \neq 0$. These implicit level sets describe the requested sketches; their repeated cells follow from $2\pi$-periodicity. Each connected graph branch is taken separately.

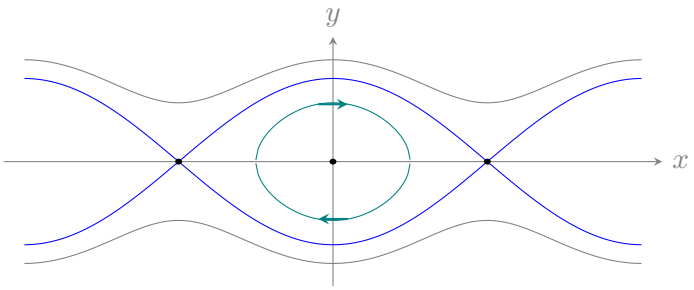
### 2.6/3: Pendulum phase portrait

*Source: printed page 43.*

For $\dot{x} = y$, $\dot{y} = -\sin x$, the energy

$$E = \frac{y^2}{2} + 1 - \cos x$$

is constant. At $E = 0$ are the centers $(2m\pi, 0)$. For $0 < E < 2$, closed ovals surround each center, traversed clockwise. At $E = 2$ the saddles $((2m+1)\pi, 0)$ and the separatrix branches $y = \pm 2|\cos(x/2)|$ are distinct orbits. For $E > 2$, $y = \pm\sqrt{2(E - 1 + \cos x)}$ are rotations: the upper branches run right, the lower left. They are not closed on the covering plane, but are closed on the cylinder $x \bmod 2\pi$.




### 2.7/1: Divergence of the Lotka–Volterra period

*Source: printed page 44.*

Assume $a, b, k, l > 0$, and initial states strictly inside the positive quadrant. Put $x_* = l/b$, $y_* = k/a$, $u = \log(x/x_*)$, $v = \log(y/y_*)$. Then

$$\dot{u} = k(1 - e^v), \quad \dot{v} = l(e^u - 1), \qquad E = l(e^u - u - 1) + k(e^v - v - 1)$$

is conserved. As $(x, y) \to (0, 0)$, $E \to \infty$. At the leftmost point of an energy oval, $v = 0$ and $u = u_- < 0$, where $l(e^{u_-} - u_- - 1) = E$, so $u_- = -E/l + O(1)$. During the arc from this point to $u = 0$ in $v < 0$, $\dot{u} \le k$. Hence

$$T(E) \ge \frac{-u_-}{k} = \frac{E}{kl} + O(1) \longrightarrow \infty.$$

States on the axes themselves are not periodic, so the limiting statement concerns interior states.

### 2.7/2: Logarithmic divergence near an axis

*Source: printed page 44.*

At $(x_*, \varepsilon)$, the energy above satisfies $E = k\log(1/\varepsilon) + O(1)$. The preceding estimate gives $T \ge cE$ for large $E$. We prove a matching upper bound, which makes "logarithmically" precise as $T = \Theta(\log(1/\varepsilon))$.

Fix $\delta > 0$. On an energy oval, $u, v$ range between $-O(E)$ and $O(\log E)$. In each quadrant, outside the strips $|u| < \delta$ and $|v| < \delta$, both $|\dot{u}|$ and $|\dot{v}|$ are bounded below by a positive constant depending

only on the parameters and $\delta$. Both coordinates are monotone on each quadrant arc. Its time there is at most a constant times its coordinate range, hence $O(E)$. Inside $|u| < \delta$, large energy forces $|v| > \delta$, so $|\dot{u}|$ is bounded below and passage across each such strip takes $O(1)$. Similarly passage through $|v| < \delta$ takes $O(1)$ using $\dot{v}$. A quadrant arc meets each strip at most once. Adding the finitely many pieces proves $T \le CE$. This establishes upper and lower logarithmic bounds without claiming an unproved exact asymptotic coefficient.

### 2.7/3: Birth and collision of limit cycles

*Source: printed page 46–47.*

Write $s = r^2$. Positive-radius cycles satisfy $s^2 - s + c = 0$, hence

$$s_\pm = \frac{1 \pm \sqrt{1 - 4c}}{2}.$$

At a positive root the derivative of $F(r) = cr - r^3 + r^5$ is $F'(r) = 2s(2s - 1)$. Thus $r_- = \sqrt{s_-}$ is attracting and $r_+ = \sqrt{s_+}$ repelling whenever both exist.

| Parameter | Behavior |
|---|---|
| $c < 0$ | Origin attracting; one repelling cycle $r_+ > 1$. |
| $c = 0$ | Origin still attracting ($\dot{r} \sim -r^3$); repelling cycle $r = 1$. |
| $0 < c < 1/4$ | Origin repelling; attracting inner and repelling outer cycles. |
| $c = 1/4$ | Double cycle $r = 1/\sqrt{2}$, attracting from inside and repelling from outside. |
| $c > 1/4$ | No cycles; every $r > 0$ increases. |

As $c \downarrow 0$ through positive values, $r_- \sim \sqrt{c}$. At $c = 1/4$ the cycles merge; their squared-radius separation is $\sqrt{1 - 4c}$. Each cycle has period $2\pi$, since $\dot{\varphi} = 1$. The origin is an equilibrium of the associated smooth Cartesian system, despite the polar angle being undefined there.

## 3 Linear equations (§3)

**Review note.** On an interval, all solutions of $y' = f(x)y$ are $Ce^{\int_{x_0}^{x} f}$, so the solution space is one-dimensional. For $T$-periodic $f$, set $\lambda = e^{\int_0^T f} > 0$. Writing $x = nT + s$, $0 \le s < T$, gives $y(x) = \lambda^n y(s)$. For a nonzero solution, $|y(s)|$ has positive minimum and finite maximum on $[0, T]$, proving decay for $\lambda < 1$, divergence in absolute value for $\lambda > 1$, and periodicity for $\lambda = 1$. A negative solution diverges to $-\infty$, not $+\infty$.

### 3.2/1: From a multiplier to stability of a cycle

*Source: printed page 51.*

In coordinates near the cycle write $Y' = a(X, Y)$, $a(X, 0) = 0$, with period $T$ in $X$. Let $P(\eta)$ be the return map. Differentiating the solution with respect to its initial transverse coordinate gives $z' = a_Y(X, 0)z$, $z(0) = 1$, so

$$P(0) = 0, \qquad P'(0) = \lambda = \exp \int_0^T a_Y(X, 0)\, \mathrm{d}X.$$

If $\lambda < 1$, choose $q$ between $\lambda$ and one. Near zero the mean-value theorem gives $|P(\eta)| \le q|\eta|$, and iterates converge to zero. Smooth dependence over the compact time interval $[0, T]$ bounds intermediate deviations by a constant times the section deviation, so the full phase curve spirals toward the cycle. If $\lambda > 1$, $|P(\eta)| \ge q|\eta|$ for some $q > 1$ while it remains nearby, so nonzero deviations eventually leave a fixed neighborhood; the inverse return map is contracting. This is repulsion, or outward spiraling. At $\lambda = 1$ linearization alone decides neither case.

### 3.2/2: A cycle with alternating radial attraction and repulsion

*Source: printed page 51.*

Since $\dot{\varphi} = 1$, use $\varphi$ as time in $\dot{r} = (r^2 - 1)(2r\cos\varphi - 1)$. At $r = 1$ the transverse linear coefficient is

$$f(\varphi) = 2(2\cos\varphi - 1) = 4\cos\varphi - 2.$$

Therefore $\lambda = \exp\int_0^{2\pi} f \, \mathrm{d}\varphi = e^{-4\pi} < 1$, and the cycle is asymptotically stable. Instantaneous radial expansion on some arcs does not outweigh the contraction over a full revolution.

**Review note.** For a nonnegative delta-shaped sequence $h_N$ supported in $[-\varepsilon_N, \varepsilon_N]$, with integral one and $\varepsilon_N \to 0$, continuity gives $|\int h_N(x)g(x)\,\mathrm{d}x - g(0)| \le \sup_{|x|\le\varepsilon_N} |g(x) - g(0)| \to 0$. This proves the embedded claim on p. 54. No ordinary function with integral one and support at a single point exists.

### 3.4/1: A rectangular impulse and its limit

*Source: printed page 56.*

Let $a_N = 1 - 1/(2N)$, $b_N = 1 + 1/(2N)$, and let $h_N = N$ on $(a_N, b_N)$ and zero elsewhere. The integrating-factor formula gives

$$y_N(x) = \begin{cases} 0, & x \le a_N, \\ N(e^{x-a_N} - 1), & a_N \le x \le b_N, \\ N(e^{x-a_N} - e^{x-b_N}), & x \ge b_N. \end{cases}$$

This is continuous and piecewise differentiable, and solves the equation almost everywhere; the discontinuous forcing prevents a classical derivative at the two switching points. Pointwise,

$$y_N(x) \longrightarrow \begin{cases} 0, & x < 1, \\ 1/2, & x = 1, \\ e^{x-1}, & x > 1. \end{cases}$$

At $x = 1$, use $N(e^{1/(2N)} - 1) \to 1/2$. Convergence is locally uniform away from one, not across the jump. The distributional limit is the causal Green function $H(x-1)e^{x-1}$; its value at the single jump point is immaterial distributionally and depends on the regularization.

### 3.5/1: Periodic forcing with damping

*Source: printed page 56–57.*

A particular solution of $y' = -y + \sin x$ is

$$y_p(x) = \tfrac{1}{2}(\sin x - \cos x), \qquad y(x) = y_p(x) + Ce^{-x}.$$

Only $C = 0$ is periodic, since a nonzero $Ce^{-x}$ cannot have period $2\pi$. Every solution converges exponentially to $y_p$ as $x \to \infty$. The multiplier is $e^{-2\pi}$.

**Review note.** For general periodic $f, g$, put $A(x) = e^{\int_0^x f}$. Then $y(T) = A(T)y(0) + A(T)\int_0^T A(s)^{-1}g(s)\,\mathrm{d}s$, an affine map as claimed on p. 56. If $A(T) \neq 1$ it has exactly one fixed point, giving the unique periodic solution. If $A(T) = 1$, it has either no fixed points or every point is fixed, according to whether the added constant is nonzero or zero.

### 3.5/2: The inhomogeneous variational equation

*Source: printed page 57.*

Differentiate $\partial_t \phi(t, \varepsilon) = v(t, \phi(t, \varepsilon); \varepsilon)$ at $\varepsilon = 0$. Smoothness permits interchanging derivatives and applying the chain rule:

$$\dot{\psi} = v_x(t, \phi_0(t); 0)\psi + v_\varepsilon(t, \phi_0(t); 0).$$

Thus $f = v_x$ and $g = v_\varepsilon$ along the reference solution. If the initial value depends on $\varepsilon$, the initial condition is $\psi(t_0) = \partial_\varepsilon \phi(t_0, 0)$; if it is fixed, this is zero. For systems $f$ becomes the Jacobian matrix.

## 4 Phase flows (§4)

### 4.1/1: Which maps are transformations?

*Source: printed page 57.*

A transformation here is a bijection of the specified set onto itself. The exponential misses the nonpositive real numbers, $x^3$ is a bijection of $\mathbb{R}$ with inverse $\sqrt[3]{x}$, and $z^2$ is not injective on $\mathbb{C}$ since $z$ and $-z$ have the same square. Only the second is a transformation. Smoothness of its inverse is not yet required.

### 4.1/2: Noncommuting transformations

*Source: printed page 58.*

On $\mathbb{R}$ take $f(x) = x + 1$, $g(x) = 2x$. Then $(fg)(x) = 2x + 1$ while $(gf)(x) = 2x + 2$, so $fg \neq gf$.

### 4.1/3: Three reflections

*Source: printed page 58.*

No. The three vertex-axis reflections of an equilateral triangle do not include the identity, although the square of each is the identity. The product of two distinct reflections is a rotation by $\pm 2\pi/3$, also absent. The six-element group generated by them does include these maps.

### 4.1/4: Triangle isometries and tetrahedron rotations

*Source: printed page 58.*

An equilateral triangle's three rotations (including the identity) and three reflections give six isometries. A regular tetrahedron has the identity, eight rotations by $\pm 2\pi/3$ about its four vertex–opposite-face axes, and three half-turns about opposite-edge-midpoint axes: twelve in total. Equivalently these realize all even permutations of its four vertices.

### 4.1/5: The additive group of real numbers

*Source: printed page 58.*

Addition is associative, zero is an identity, and each $a$ has inverse $-a$; closure is automatic. To fit the book's transformation-group definition, use $T_a(x) = x + a$. Then $T_a T_b = T_{a+b}$, $T_a^{-1} = T_{-a}$, and the representation is faithful.

### 4.1/6: Permutations of coordinate axes by cube isometries

*Source: printed page 59.*

Every permutation $\sigma \in S_3$ is realized by the orthogonal permutation matrix $e_i \mapsto e_{\sigma(i)}$, which preserves $\max(|x|, |y|, |z|)$. Thus all six permutations occur. Even if only rotations are allowed, signs can be chosen to make the signed permutation matrix have determinant $+1$ without changing its action on the unoriented axes.

### 4.1/7: Action on matrices

*Source: printed page 59.*

Under the coordinate change $y = gx$, an operator $x \mapsto Mx$ becomes $y \mapsto gMg^{-1}y$. Hence $T_g(M) = gMg^{-1}$. Moreover $T_g T_h = T_{gh}$ and $T_{g^{-1}} = T_g^{-1}$. If one instead describes a change of basis by giving the new basis in old coordinates, the inverse convention produces $g^{-1}Mg$; the distinction is purely the meaning assigned to $g$.

### 4.1/8: Rotation orbits

*Source: printed page 59.*

Rotations preserve distance from zero and act transitively on every circle of fixed positive radius. The orbits are therefore the origin and all circles centered at the origin.

### 4.1/9: Orbits partition a set

*Source: printed page 59.*

If $z = gx = hy$ belongs to both $Gx$ and $Gy$, then $y = h^{-1}gx$, so every $ky$ lies in $Gx$. This gives $Gy \subseteq Gx$; reversing the argument gives equality. Otherwise the two orbits are disjoint.

### 4.1/10: Unordered pairs of tetrahedron edges

*Source: printed page 59.*

For two distinct edges there are two invariant possibilities: they share a vertex, or they are opposite. Vertex permutations act transitively on each type, so there are two orbits (of sizes twelve and three). If "pairs" were to allow a repeated edge, the six double-edge pairs form a third orbit. The standard interpretation is a two-element subset, giving two.

### 4.1/11: Six distinct colors on a cube

*Source: printed page 59.*

There are 6! labelings using every color once. A rotation is determined by the image of one face (six choices) and then one adjacent face (four choices), so the rotation group has order twenty-four. Its action on distinct-color labelings is free: a color-preserving rotation fixes all faces and is the identity. Every orbit thus has size twenty-four, giving $6!/24 = 30$ colorings up to rotation.

### 4.3/1: Diffeomorphisms of the real line

*Source: printed page 61.*

Only $x$ and $-x$ are diffeomorphisms of $\mathbb{R}$ onto itself. The square is not injective, the cube has an inverse not differentiable at zero, and $\arctan x$ is not onto $\mathbb{R}$. The last map is a diffeomorphism onto $(-\pi/2, \pi/2)$, which is a different codomain.

### 4.3/2: Must a linear flow split into one-dimensional factors?

*Source: printed page 63.*

No, over $\mathbb{R}$. A nontrivial rotation flow has no common invariant real line, whereas a direct product would have two. A shear $g^t(x, y) = (x + ty, y)$ supplies another obstruction: its infinitesimal matrix $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is not diagonalizable. A linear splitting would diagonalize this matrix, an impossibility.

### 4.4/1: Generators of three line flows

*Source: printed page 63.*

Differentiate at $t = 0$: $x + t$ gives $v(x) = 1$, $e^t x$ gives $v(x) = x$, and $e^{-t} x$ gives $v(x) = -x$.

### 4.4/2: Zeros of the generator are fixed points

*Source: printed page 63.*

If $v(a) = 0$, the constant curve at $a$ solves $\dot{x} = v(x)$. The orbit $g^t a$ solves the same initial-value problem, so uniqueness yields $g^t a = a$. Conversely differentiating a fixed orbit gives $v(a) = 0$.

### 4.4/3: Every solution is an orbit of the given flow

*Source: printed page 64.*

For a solution with $\phi(0) = x_0$, the curve $g^t x_0$ satisfies the same equation and initial condition. Uniqueness identifies them on the entire domain of $\phi$. More generally, $\phi(t) = g^{t-t_0}\phi(t_0)$.

### 4.4/4: Four explicit flows

*Source: printed page 64.*

Solving with initial value $x$ at time zero gives

$$\begin{array}{c|c} \dot{x} = 0 & g^t(x) = x \\ \dot{x} = 1 & g^t(x) = x + t \\ \dot{x} = x - 1 & g^t(x) = 1 + (x-1)e^t \\ \dot{x} = \sin x, \quad 0 < x < \pi & g^t(x) = 2\arctan(e^t \tan(x/2)). \end{array}$$

For the last formula integrate $\mathrm{d}x/\sin x = \mathrm{d}t$; the chosen branch stays in $(0, \pi)$ for every finite $t$. Each formula has inverse $g^{-t}$ and satisfies the group law.

### 4.4/5: Three planar flows

*Source: printed page 64–65.*

For initial state $(x, y)$, successive integration gives

$$\begin{array}{c|c} (\dot{x}, \dot{y}) = (y, 0) & g^t(x, y) = (x + ty, y) \\ (\dot{x}, \dot{y}) = (y, 1) & g^t(x, y) = (x + ty + t^2/2, y + t) \\ (\dot{x}, \dot{y}) = (\sin y, 0) & g^t(x, y) = (x + t\sin y, y). \end{array}$$

Differentiating verifies the equations and setting $t = 0$ verifies the initial states. These polynomial/trigonometric formulas exist for every real $t$.

### 4.4/6: At most linear growth implies completeness

*Source: printed page 65.*

Suppose $|v(x)| \le a + b|x|$ with $a, b \ge 0$. Gronwall gives, on any finite interval measured from zero,

$$|x(t)| \le (|x(0)| + a/b)e^{b|t|} - a/b \quad (b > 0),$$

and $|x(t)| \le |x(0)| + a|t|$ if $b = 0$. The same bound holds backward by reversing time. Thus no solution escapes to infinity in finite time. A solution confined to a compact interval can be extended by local existence, so every maximal solution is global. Smooth dependence and uniqueness give smooth maps $g^t$, the group law, and inverse $g^{-t}$; hence a one-parameter diffeomorphism group.

### 4.4/7: The field $e^x \sin x$

*Source: printed page 65.*

Yes. Every $n\pi$ is an equilibrium, and uniqueness confines any other solution to its initial interval $(n\pi, (n+1)\pi)$. Its closure is compact and the field is smooth there, so no finite-time escape occurs. A zero of a smooth scalar field cannot be reached in finite time either. All solutions are complete even though the field has no global linear-growth bound.

### 4.4/8: Translation on a polynomial space

*Source: printed page 66.*

Let $P_{<n}$ be the vector space of polynomials of degree less than $n$. The map $(g^t p)(x) = p(x + t)$ is linear in $p$, preserves this space, satisfies $g^s g^t = g^{s+t}$, and has inverse $g^{-t}$. Its coefficients depend polynomially on $t$, so the flow is smooth. Its generator is $Dp = p'$; in the basis $1, x, \ldots, x^{n-1}$ this sends $x^j$ to $jx^{j-1}$. Since $D^n = 0$, the flow is the finite sum $e^{tD} = \sum_{j=0}^{n-1} t^j D^j / j!$.

## 5 Changes of variables and fields (§5)

### 5.1/1: Independence of the curve representing a tangent vector

*Source: printed page 67.*

If $\phi(0) = \psi(0) = x$ and $\phi'(0) = \psi'(0) = v$, then $\phi(t) - \psi(t) = o(|t|)$. Smoothness of $f$ gives a local Lipschitz bound, so $f(\phi(t)) - f(\psi(t)) = o(|t|)$. Their derivatives at zero therefore agree. Equivalently, the chain rule gives $(f \circ \phi)'(0) = Df_x v$ for either curve.

### 5.1/2: A unit vector under squaring

*Source: printed page 67.*

For $f(x) = x^2$, $Df_a(v) = 2av$. A positive unit vector has component one, so the image has component $2a$, attached at $a^2$. It is negative when $a < 0$ and zero when $a = 0$.

### 5.1/3: Motions along different axes with the same velocity

*Source: printed page 67.*

Yes, but only with zero common velocity, since the two coordinate-axis tangent lines intersect only in zero. The curves $\phi(t) = (t^2, 0)$ and $\psi(t) = (0, t^2)$ both leave the origin with zero derivative while moving on different axes.

### 5.1/4: Linearity of the derivative

*Source: printed page 68.*

Differentiability means $f(x + h) = f(x) + Ah + o(|h|)$ for a linear map $A = Df_x$. Applying this to a curve with $h = tv + o(t)$ gives $f_{*x}v = Av$. Hence $f_{*x}(av + bw) = af_{*x}v + bf_{*x}w$. In coordinates $(f_*v)_i = \sum_j (\partial f_i/\partial x_j)v_j$.

### 5.1/5: Derivative of a circle parametrization

*Source: printed page 68.*

For $f(x) = (\sin x, \cos x)$ and $v = 10\partial_x$ at $\alpha$,

$$f_{*\alpha}v = (10\cos\alpha, -10\sin\alpha),$$

attached at $(\sin\alpha, \cos\alpha)$. Its norm is ten and it is tangent to the circle.

### 5.1/6: Critical points and values of the cusp map

*Source: printed page 68.*

For $f(x_1, x_2) = (x_1^3 + x_1x_2, x_2)$,

$$Df = \begin{pmatrix} 3x_1^2 + x_2 & x_1 \\ 0 & 1 \end{pmatrix}.$$

Its determinant vanishes on $x_2 = -3x_1^2$. Parametrizing that parabola by $s = x_1$ gives critical values $(y_1, y_2) = (-2s^3, -3s^2)$, hence

$$\frac{y_1^2}{4} + \frac{y_2^3}{27} = 0.$$

Conversely every real point on this semicubical parabola has such a parameter, so the equation describes the entire critical-value set.

### 5.1/7: A diffeomorphism induces tangent-space isomorphisms

*Source: printed page 70.*

Apply the chain rule to $f^{-1} \circ f = \mathrm{id}$ and $f \circ f^{-1} = \mathrm{id}$. It gives $D(f^{-1})_{f(x)}Df_x = I$ and $Df_xD(f^{-1})_{f(x)} = I$. Thus $Df_x$ is an isomorphism with the indicated inverse. In terms of curves, applying $f^{-1}$ recovers the original tangent-vector class.

### 5.1/8: Invertible derivatives do not imply a global diffeomorphism

*Source: printed page 70.*

The map $f(u, v) = (e^u \cos v, e^u \sin v)$ from $\mathbb{R}^2$ onto $\mathbb{R}^2 \setminus \{0\}$ has Jacobian determinant $e^{2u} > 0$, but $f(u, v + 2\pi) = f(u, v)$, so it is not injective. The inverse-function theorem yields local inverses, not a global one.

### 5.2/1: Pushforward by dilation

*Source: printed page 71.*

For $g(x) = 2x$ and $v = \partial_x$, the field at $y = g(x)$ is $Dg_x v(x) = 2$. Thus $g_* v = 2\partial_y$.

### 5.2/2: Pushforward of the Euler field by the exponential

*Source: printed page 71.*

For $v = x\partial_x$ and the diffeomorphism $y = e^x$ from $\mathbb{R}$ onto $(0, \infty)$, the chain rule gives

$$g_* v = xe^x \partial_y = y \log y \, \partial_y.$$

The coefficient must be expressed at the image point, so substitute $x = \log y$ after multiplying by $g'(x) = e^x$. The equilibrium at $x = 0$ becomes the equilibrium at $y = 1$.

### 5.2/3: Linear invariance of the Euler field

*Source: printed page 71.*

For an invertible linear map $g(x) = Ax$ and $v(x) = x$, one has $(g_* v)(y) = A(A^{-1}y) = y$. Thus rotations, hyperbolic rotations, and all invertible linear transformations preserve the Euler field. Invertibility is required because the question concerns diffeomorphisms.

### 5.2/4: Phase curves versus time-parametrized trajectories

*Source: printed page 72.*

If $g_* v = w$ and $x' = v(x)$, then $(g \circ x)' = Dg_x v(x) = w(g(x))$, so phase curves map to phase curves with time preserved. The converse fails: the identity maps all phase curves of $x\partial_x$ onto those of $2x\partial_x$ (the two open rays and zero), but does not map the first field to the second. Their speeds differ by two.

### 5.3/1: Polar solution of the oscillator

*Source: printed page 72.*

For $r > 0$, $\dot r = (x_1\dot x_1 + x_2\dot x_2)/r = 0$ and $\dot\theta = (x_1\dot x_2 - x_2\dot x_1)/r^2 = -1$. Thus

$$x_1 = r_0\cos(\theta_0 - t), \qquad x_2 = r_0\sin(\theta_0 - t).$$

Add the equilibrium at zero, where polar coordinates are not a chart.

### 5.3/2: A stable circular limit cycle

*Source: printed page 72–73.*

For the displayed system, polar coordinates give $\dot r = r(1 - r^2)$, $\dot\theta = -1$. For $r_0 > 0$,

$$r(t) = \left[1 + (r_0^{-2} - 1)e^{-2t}\right]^{-1/2}, \qquad \theta(t) = \theta_0 - t,$$

on the maximal interval where the bracket is positive. The origin is repelling, the unit circle is a clockwise periodic orbit of period $2\pi$, and every other nonzero trajectory spirals toward it in forward time. If $r_0 < 1$ the solution is global and tends to zero backward; if $r_0 > 1$ it blows up at a finite negative time. Thus forward attraction does not imply a global two-sided flow on the plane.

### 5.4/1: A direction field without a nonvanishing vector representative

*Source: printed page 74.*

On the punctured plane write a point as $(r, \theta)$ and prescribe the unoriented line

$$L_{r,\theta} = \operatorname{span}(\cos(\theta/2), \sin(\theta/2)).$$

It is single-valued and smooth as a line field because increasing $\theta$ by $2\pi$ changes the vector's sign but not its span. A nonvanishing continuous representative on the unit circle would be $a(\theta)(\cos(\theta/2), \sin(\theta/2))$, with $a$ continuous and nonzero. Periodicity forces $a(2\pi) = -a(0)$, contradicting the intermediate-value theorem. The field therefore cannot be represented by a nowhere-zero vector field. Allowing the zero field would make the question vacuous: zero does not determine a direction.

### 5.4/2: Separating a homogeneous equation

*Source: printed page 74.*

For $\mathrm{d}x/\mathrm{d}t = (x^2 - t^2)/(x^2 + t^2)$, use $u = x/t$ on either half-plane $t \neq 0$. Then

$$u + t\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{u^2 - 1}{u^2 + 1}, \qquad \frac{u^2 + 1}{-u^3 + u^2 - u - 1}\,\mathrm{d}u = \frac{\mathrm{d}t}{t}.$$

This separates the variables. Roots of $-u^3 + u^2 - u - 1 = 0$ yield the straight solutions $x = ut$ lost by division. The original equation is undefined only at $(0, 0)$; use the complementary chart $t/x$ where needed. Alternatively, polar coordinates $t = r\cos\theta, x = r\sin\theta$ give

$$\frac{\mathrm{d}r}{r} = \frac{\cos\theta - \sin\theta\cos 2\theta}{-\sin\theta - \cos\theta\cos 2\theta}\,\mathrm{d}\theta,$$

where the denominator is nonzero; its zero directions are again handled separately.

### 5.4/3: Straightening $x' = x - 1$ globally

*Source: printed page 74.*

The map $F(t, x) = (t, a)$, $a = e^{-t}(x - 1)$, is a global diffeomorphism with inverse $(t, a) \mapsto (t, 1 + ae^t)$. Along a solution, $a' = e^{-t}(x' - x + 1) = 0$, so the integral curves become the parallel lines $a =$ constant.

### 5.4/4: A plane direction field that cannot be straightened globally

*Source: printed page 74–75.*

Consider the smooth nowhere-zero vector field $V(t, x) = (\cos x, \sin x)$ on the entire plane. The lines $x = 0$ and $x = \pi$ are distinct leaves. In the strip $0 < x < \pi$ the other leaves satisfy

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \cot x, \qquad t = \log(\sin x) + C.$$

Their two ends approach the two boundary leaves as $t \to -\infty$. More precisely, any saturated open neighborhoods of the two boundary leaves intersect: take small neighborhoods of $(0, 0)$ and $(0, \pi)$. For $C$ sufficiently large, the leaf $t = \log(\sin x) + C$ crosses both at $t = 0$, since $\sin x = e^{-C}$ has roots close to both endpoints. Thus the two leaves cannot be separated in the quotient space of leaves. A foliation by parallel lines has leaf space $\mathbb{R}$, which is Hausdorff. A diffeomorphism mapping leaves to leaves induces a homeomorphism of leaf spaces, so no such global straightening exists.

### 5.4/5: Global straightening of $x' = x^2$

*Source: printed page 75.*

Despite finite-time explosion, the answer for its direction field in the $(t, x)$-plane is yes. Rescale its vector representative to

$$W(t, x) = \frac{(1, x^2)}{1 + x^2}.$$

Both components are bounded, so $W$ is complete. The function $s = t + x$ satisfies $W(s) = 1$. Therefore every orbit meets the transversal $\Gamma = \{(-a, a) : a \in \mathbb{R}\}$ exactly once: following the complete flow for time $-s$ reaches $\Gamma$. If $\Phi_W$ is that flow, then

$$(s, a) \longmapsto \Phi_W^s(-a, a)$$

is a global smooth bijection with smooth inverse, by the flow property and transversality. It takes horizontal lines $a =$ constant to the desired integral curves.

For an explicit expression for the second coordinate of the inverse, one may use

$$a(t, x) = \frac{2x}{1 - tx + \sqrt{(1 - tx)^2 + 4x^2}}.$$

The denominator is strictly positive (also at $x = 0$), and this is smooth. For $x \neq 0$ it is obtained by equating the first integral $t + 1/x$ to its value $-a + 1/a$ on $\Gamma$; it extends with $a(t, 0) = 0$. Thus $F(t, x) = (t + x, a(t, x))$ is the promised explicit diffeomorphism. This straightens a direction field, not the original time-parametrized incomplete flow.

### 5.5/1: Conjugation preserves the flow laws

*Source: printed page 75.*

For $h^t = f g^t f^{-1}$,

$$h^s h^t = f g^s f^{-1} f g^t f^{-1} = f g^{s+t} f^{-1} = h^{s+t}, \quad h^0 = \mathrm{id}, \quad (h^t)^{-1} = h^{-t}.$$

Each map is a diffeomorphism and the dependence on $(t, y)$ is smooth. Hence this is a one-parameter diffeomorphism group.

### 5.5/2: Circular and hyperbolic rotations are not conjugate

*Source: printed page 76.*

Every nonzero orbit of a circular rotation flow is compact and periodic. Every nonzero orbit of a hyperbolic rotation $(x, y) \mapsto (e^t x, e^{-t} y)$ is nonperiodic and noncompact. A conjugacy preserves periodicity (indeed the exact periods), so these flows cannot be conjugate. Also the circular flow has $g^{2\pi} = \mathrm{id}$, whereas the hyperbolic flow does not.

### 5.5/3: Classifying five scalar vector fields

*Source: printed page 76.*

Write

$$v_1 = \sin x, \quad v_2 = 2 \sin x, \quad v_3 = \sin^2 x, \quad v_4 = \sin 2x, \quad v_5 = 2 \sin x + \sin^2 x.$$

For a diffeomorphism $h$ satisfying $h' v = w \circ h$, zeros correspond and differentiation at a zero $a$ gives $v'(a) = w'(h(a))$. The zero derivatives of $v_1$ alternate $\pm 1$, those of $v_2, v_4, v_5$ alternate $\pm 2$, and those of $v_3$ are zero. Thus the only possible equivalences between different members are within $\{v_2, v_4, v_5\}$. The map $h(x) = x/2$ takes $v_2$ to $v_4$ directly.

Here is a full existence argument for the harder equivalence of $v_2$ and $v_5$. Both have exactly the simple zeros $n\pi$, the same sign on each interval between zeros, and matching derivatives there. We use two elementary constructions.

*Local construction at a simple zero.* If $v(a) = 0$, $v'(a) = \lambda \neq 0$, write $z = x - a$. The function

$$q_v(x) = z \exp\left( \int_0^z \left[ \frac{\lambda}{v(a+s)} - \frac{1}{s} \right] \mathrm{d}s \right)$$

is smooth near $a$, has derivative one at $a$, and satisfies $q_v' v = \lambda q_v$. The integrand extends smoothly at zero: write $v(a+s) = s(\lambda + s b(s))$. Thus matching fields with the same $\lambda$ are locally conjugate by $q_w(h(x)) = c q_v(x)$ for any $c > 0$. Every increasing conjugacy on one side near the zero has exactly this form, because the ratio of the two linearizing coordinates is constant along the scalar equation.

*Extension across a zero-free interval.* On $(n\pi, (n+1)\pi)$ the time coordinates $\int \mathrm{d}x / v(x)$ and $\int \mathrm{d}y / w(y)$ are monotone onto $\mathbb{R}$; the logarithmic divergences at the endpoints ensure this. Equating them up to an additive constant therefore defines an increasing diffeomorphism of that interval. Starting with a local conjugacy near zero, this extends uniquely through $(0, \pi)$ and approaches $\pi$. Near $\pi$ it has the local form just proved, with a positive constant $c$, so it extends smoothly across $\pi$ with nonzero derivative. Continue inductively in both directions. Since the zeros have no finite

accumulation point, this yields a smooth increasing diffeomorphism $h : \mathbb{R} \to \mathbb{R}$ mapping each $n\pi$ to itself and satisfying $h'v_2 = v_5 \circ h$ everywhere. It is onto because it maps every consecutive interval onto itself.

The equivalence classes among the five displayed fields are therefore $\{v_1\}$, $\{v_3\}$, and $\{v_2, v_4, v_5\}$. No periodicity condition is imposed on $h$; the proof is for the line as requested, not a prescribed conjugacy on a fixed circle.

# 6 Symmetries (§6)

### 6.1/1: Preserving phase curves need not preserve a field

*Source: printed page 76.*

For $v = \partial_x$ on $\mathbb{R}$, the diffeomorphism $h(x) = 2x$ preserves its sole maximal phase curve, the whole line, but $h_*v = 2\partial_x \neq v$. Preserving the paths does not preserve speed. The same example on the plane, with $v = \partial_x$ and $h(x, y) = (2x, y)$, preserves each horizontal phase line.

### 6.1/2: Preserving integral curves preserves a direction field

*Source: printed page 77.*

At each point, the prescribed line is the tangent line to the integral curve through that point. The derivative of a diffeomorphism taking integral curves to integral curves takes their tangent lines to their tangent lines. It therefore preserves the direction field. Speed and orientation along a line are irrelevant here.

**Review note.** The identity is a symmetry; if $f_*v = v$ and $g_*v = v$, then $(f \circ g)_*v = f_*(g_*v) = v$; applying the inverse pushforward shows $f^{-1}$ is a symmetry. The same proof applies to direction fields, answering the embedded group question on p. 77.

### 6.1/3: All global symmetries of the Euler field

*Source: printed page 77.*

A symmetry $h : \mathbb{R}^2 \to \mathbb{R}^2$ of $v(x) = x$ fixes its unique zero and commutes with its flow, so $h(0) = 0$ and $h(e^t x) = e^t h(x)$. For $s > 0$ this gives $h(sx) = sh(x)$. Differentiability at zero now forces

$$h(x) = \lim_{s \downarrow 0} \frac{h(sx) - h(0)}{s} = Dh_0 x.$$

Thus $h$ is an invertible linear map. Conversely every such map preserves the Euler field. The full group is $GL(2, \mathbb{R})$. The inclusion of the origin and differentiability there are essential; on the punctured plane there are nonlinear homogeneous symmetries.

### 6.2/1: Reducing dimension using a one-parameter symmetry

*Source: printed page 79.*

Work locally where the symmetry generator is nonzero. Straighten its flow to translation in a coordinate $u$, with transverse coordinates $z \in \mathbb{R}^{n-1}$. An invariant line field can locally be represented by a nonvanishing vector whose components depend only on $z$:

$$\dot{u} = A(z), \qquad \dot{z} = B(z).$$

To see this, normalize one nonzero component of any representative to one; translation invariance of the line makes all ratios of components independent of $u$. Where $B \neq 0$, solve the reduced $(n-1)$-dimensional direction field spanned by $B$. Choose its parametrization so that $\dot{z} = B(z)$ and then recover $u$ by the quadrature $u = u_0 + \int A(z(t))\,\mathrm{d}t$. If $B(z_0) = 0$, the leaf there is a symmetry orbit $z = z_0$, and the remaining equation is one-dimensional. The reduction is local; a global orbit space need not be a manifold, and fixed points of the symmetry require separate treatment.

### 6.3/1: Three homogeneous equations

*Source: printed page 79.*

Each right-hand side is unchanged by $(x, y) \mapsto (\lambda x, \lambda y)$ for $\lambda > 0$:

$$\frac{\lambda y}{\lambda x} = \frac{y}{x}, \quad \frac{\lambda x}{\lambda y} = \frac{x}{y}, \quad \log(\lambda x) - \log(\lambda y) = \log x - \log y.$$

Thus all three are homogeneous on their stated domains. In the logarithmic example both coordinates are positive.

### 6.3/2: A homogeneous Riccati-type equation

*Source: printed page 80.*

Set $y = xv$ in $y' = y/x + y^2/x^2$ with $x > 0$. Then $xv' = v^2$. For $v \neq 0$,

$$-1/v = \log x + C, \qquad y = -\frac{x}{\log x + C}.$$

Each solution is restricted to a connected interval avoiding $x = e^{-C}$. The missing solution is $y = 0$ on all $x > 0$.

### 6.3/3: Sketching $y' = 2y/x + y^2/x^2$

*Source: printed page 80.*

For $x \neq 0$, put $v = y/x$. Then $xv' = v + v^2$. The constant solutions for $v$ give $y = 0$ and $y = -x$. Otherwise,

$$\log\left|\frac{v}{1+v}\right| = \log|x| + C, \qquad y = \frac{Cx^2}{1 - Cx}, \quad C \neq 0.$$

Each graph is split at $x = 0$ and at its vertical asymptote $x = 1/C$. Near zero it is tangent to the $x$-axis, with $y \sim Cx^2$. At large $|x|$, $y = -x - 1/C + O(1/x)$, giving an oblique asymptote of slope $-1$. These curves and the two exceptional lines give the complete sketch on both half-planes. All are dilations of representative curves within their respective sectors.

### 6.3/4: Smoothness of the real blow-up surface

*Source: printed page 81.*

The closure of the graph of the direction map is the incidence surface

$$\Sigma = \{((x, y), [a : b]) \in \mathbb{R}^2 \times \mathbb{RP}^1 : xb = ya\}.$$

In the chart $a \neq 0$, put $v = b/a$; then $y = vx$, and $(x, v) \mapsto ((x, xv), [1 : v])$ is a smooth parametrization with smooth inverse. In the chart $b \neq 0$, put $u = a/b$; then $x = uy$ and $(u, y)$ are smooth coordinates. The charts cover $\Sigma$ and have smooth transition $u = 1/v$, $y = xv$ where both apply. Hence $\Sigma$ is a smooth two-dimensional manifold, including the exceptional circle over the origin.

### 6.3/5: The blow-up is an open Möbius band

*Source: printed page 81.*

A point of $\Sigma$ is a line $\ell$ through zero together with a vector on $\ell$, so $\Sigma$ is the tautological real line bundle over $\mathbb{RP}^1$. Concretely,

$$(\theta, r) \longmapsto ((r\cos\theta, r\sin\theta), [\cos\theta : \sin\theta])$$

identifies $(\theta + \pi, r)$ with $(\theta, -r)$. Thus $\Sigma$ is $[0, \pi] \times \mathbb{R}$ with its ends glued by a reversal of the fiber coordinate. This is an open Möbius band (the total space of the line bundle), not the compact band with boundary. A diffeomorphism $\mathbb{R} \to (-1, 1)$ in each fiber identifies it with the interior of the usual band.

### 6.3/6: Resolving the semicubical cusp

*Source: printed page 82.*

Parametrize $x^2 = y^3$ by $(x, y) = (s^3, s^2)$. In the blow-up chart $x = uy$, its equation becomes $y^2(u^2 - y) = 0$. Removing the exceptional factor leaves the strict transform $y = u^2$, parametrized by $(u, y) = (s, s^2)$. This is already smooth, so one blow-up resolves the singularity of the curve itself.

To obtain the full normal-crossing picture suggested by the book's three-step drawing, continue. The strict transform $y = u^2$ is tangent to the exceptional curve $y = 0$. Blow up their contact point with $u = v$, $y = vw$. The strict transform becomes $w = v$. The two exceptional components are $v = 0$ and $w = 0$; these and $w = v$ meet at one point. Blow up this triple point. In the chart $w = vs$, the strict transform is $s = 1$, transverse to the new exceptional curve $v = 0$ at $(0, 1)$. The old component $w = 0$ has strict transform $s = 0$, meeting the new exceptional curve at $(0, 0)$; the other old component meets it in the complementary chart. The three intersections are distinct, leaving only ordinary pairwise transverse intersections. Distinguish resolution of the curve (one step) from this stronger embedded normal-crossing resolution (three steps).

### 6.3/7: Lifting a field vanishing at the origin

*Source: printed page 82.*

Write $V = P(x,y)\partial_x + Q(x,y)\partial_y$ with $P(0,0) = Q(0,0) = 0$. In the chart $y = xv$, away from $x = 0$,

$$\dot{x} = P(x, xv), \qquad \dot{v} = \frac{Q(x,xv) - vP(x,xv)}{x}.$$

Hadamard's lemma gives smooth functions $A, B, C, D$ with $P = xA + yB$, $Q = xC + yD$; for instance obtain them by integrating the first derivatives along the segment from zero to $(x,y)$. Hence

$$\dot{v} = C(x,xv) + vD(x,xv) - vA(x,xv) - v^2 B(x,xv),$$

which is smooth even at $x = 0$. The complementary chart $x = uy$ works in the same way. The extensions agree on overlaps because they agree off the exceptional set and are continuous. Thus the lifted field extends smoothly to all of $\Sigma$; the exceptional circle is invariant since $\dot{x} = 0$ on $x = 0$.

### 6.4/1: Weights for $y' = -x/y^3$

*Source: printed page 83.*

The derivative has weight $\beta - \alpha$, while the right-hand side has weight $\alpha - 3\beta$. Equality gives $\alpha = 2\beta$. Choose $(\alpha, \beta) = (2, 1)$; any common nonzero multiple works. The equation is considered where $y \neq 0$.

### 6.4/2: Weights of $x^2y + y^4$

*Source: printed page 83.*

Degree one requires $2\alpha + \beta = 1$ and $4\beta = 1$. Solving gives $\beta = 1/4$, $\alpha = 3/8$.

### 6.4/3: Weighted Euler identity

*Source: printed page 84.*

If $f(e^{\alpha_1 s}x_1, \ldots, e^{\alpha_n s}x_n) = e^{rs}f(x)$, differentiate in $s$ at zero to get

$$\sum_i \alpha_i x_i \frac{\partial f}{\partial x_i} = rf.$$

Conversely, along the dilation orbit put $F(s) = f(e^{\alpha_1 s}x_1, \ldots, e^{\alpha_n s}x_n)$. The identity gives $F' = rF$, hence $F(s) = e^{rs}F(0)$. The conclusion holds for all orbit segments within the domain; for a global homogeneity statement the domain must be invariant under these dilations. The unweighted Euler theorem on p. 80 is the case $\alpha_i = 1$.

### 6.4/4: Weights for Newton phase curves

*Source: printed page 84.*

For $\ddot{x} = Cx^k$, put $y = \dot{x}$. Away from $y = 0$, the phase-curve equation is $\mathrm{d}y/\mathrm{d}x = Cx^k/y$. Matching weights gives

$$\beta - \alpha = k\alpha - \beta, \qquad 2\beta = (k+1)\alpha.$$

For example $(\alpha, \beta) = (2, k+1)$ works. The undivided system includes turning points with $y = 0$; powers are interpreted only on domains where they are real and smooth. For $C = 0$ the reduced equation is zero and no such restriction on weights is necessary.

### 6.5/1: Degree of differentiation along a weighted field

*Source: printed page 84.*

If the coordinate $x_i$ has weight $\alpha_i$, then $\partial_i f$ has degree $r - \alpha_i$. A vector field of degree $s$ has component $v_i$ of degree $s + \alpha_i$, because $\partial_i$ has weight $-\alpha_i$. Every term $v_i \partial_i f$ therefore has degree $s + r$. Their sum $v(f)$ has that degree as well (the zero function is compatible with every degree).

### 6.5/2: Scaling periods for homogeneous polynomial systems

*Source: printed page 84–85.*

Let $V$ be homogeneous of degree $m$ in its components and $x(t)$ a periodic solution. For $\lambda > 0$, define $y(t) = \lambda x(\lambda^{m-1} t)$. Then

$$\dot{y} = \lambda^m V(x(\lambda^{m-1} t)) = V(y).$$

The new trajectory is the dilation of the old one and its period is $T' = \lambda^{1-m} T$. With $\lambda = e^s$, this is the book's $e^{s(1-m)}T$. If $T$ is the least positive period, the same argument applied inversely shows that $T'$ is also least.

### 6.5/3: Scaling periods for a weighted field

*Source: printed page 85.*

Let $D_s = \mathrm{diag}(e^{\alpha_i s})$ and let the field have degree $r$, meaning $v(D_s x) = e^{rs} D_s v(x)$. Then $y(t) = D_s x(e^{rs} t)$ solves the same equation:

$$\dot{y} = e^{rs} D_s v(x(e^{rs} t)) = v(y).$$

Its orbit is $D_s \gamma$ and its period is $e^{-rs} T$.

### 6.5/4: The cubic oscillator's period

*Source: printed page 85.*

For $\ddot{x} = -x^3$, weights $(\deg x, \deg y) = (1, 2)$ make the phase field have degree one. Doubling the amplitude therefore halves the period: $T(A) = T(1)/A$ for $A > 0$. Energy gives the explicit constant:

$$\frac{\dot{x}^2}{2} + \frac{x^4}{4} = \frac{A^4}{4}, \qquad T(A) = \frac{4\sqrt{2}}{A} \int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}}.$$

The integral converges at one since its singularity is proportional to $(1-u)^{-1/2}$. Zero amplitude is an equilibrium and has no least positive period.

### 6.5/5: Scaling Newtonian trajectories

*Source: printed page 85.*

Suppose $\ddot{x} = F(x)$ with $F(\lambda x) = \lambda^m F(x)$. Seek the dilated solution $y(t) = \lambda x(t/\tau)$. Then

$$\ddot{y} = \lambda \tau^{-2} F(x(t/\tau)), \qquad F(y) = \lambda^m F(x(t/\tau)).$$

Choose $\tau = \lambda^{(1-m)/2}$. Thus the similar motion traverses the dilated trajectory in time $T' = \lambda^{(1-m)/2} T$, with its initial velocity scaled by $\lambda/\tau = \lambda^{(1+m)/2}$. A corresponding initial velocity is necessary; scaling position alone does not prescribe the similar motion.

### 6.5/6: Kepler scaling

*Source: printed page 85.*

For a fixed inverse-square gravitational field, $m = -2$. Hence $T' = \lambda^{3/2} T$ and $(T'/T)^2 = \lambda^3$. Similar trajectories have linear dimensions in ratio $\lambda$, so their periods squared have the ratio of the cubes of their dimensions. This scaling argument proves the assertion for similar orbits; the stronger Kepler law across nonsimilar ellipses requires more information.

### 6.5/7: Linear and cubic restoring forces

*Source: printed page 85–86.*

For $\ddot{x} = -kx$ with $k > 0$, the force degree is one and $T = 2\pi/\sqrt{k}$ is independent of positive amplitude. For $\ddot{x} = -kx^3$, the force degree is three and

$$T(A) = \frac{4\sqrt{2}}{\sqrt{k}\, A} \int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}}.$$

Thus the cubic oscillator's period is inversely proportional to amplitude. These are statements about the specified idealized force laws, not about the finite-amplitude nonlinear sine pendulum.

### 6.5/8: Daily versus annual thermal penetration

*Source: printed page 86.*

The heat equation $u_t = au_{xx}$ is invariant under $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$ with temperature unchanged. At fixed boundary-temperature amplitude and otherwise corresponding conditions, penetration depth therefore scales like the square root of the forcing period. A daily period is $1/365$ of an annual one, giving

$$d_{\text{daily}} = \frac{1\,\text{m}}{\sqrt{365}} \simeq 0.0523\,\text{m} = 5.23\,\text{cm}.$$

This uses the linear heat-conduction model in the question, with the same mean temperature and material parameters; it is not a separate model of latent heat or actual freeze–thaw fronts.

### 6.6/1: Radial harmonic functions and their point sources

*Source: printed page 88.*

For $u(x) = U(r)$, $r = |x| > 0$ in dimension $n$,

$$\Delta u = U''(r) + \frac{n-1}{r}U'(r) = r^{1-n}\frac{\mathrm{d}}{\mathrm{d}r}(r^{n-1}U'(r)).$$

Thus $r^{n-1}U'$ is constant, and

$$n = 2: \quad U(r) = A\log r + B = C\log(1/r) + B, \qquad n = 3: \quad U(r) = A/r + B.$$

These are harmonic off zero. If smoothness at zero on all of $\mathbb{R}^n$ is required, only the constants remain.

For the embedded distributional question, let $\phi \in C_c^\infty(\mathbb{R}^n)$ and apply Green's second identity on the complement of the ball of radius $\varepsilon$. Since $U$ is harmonic there,

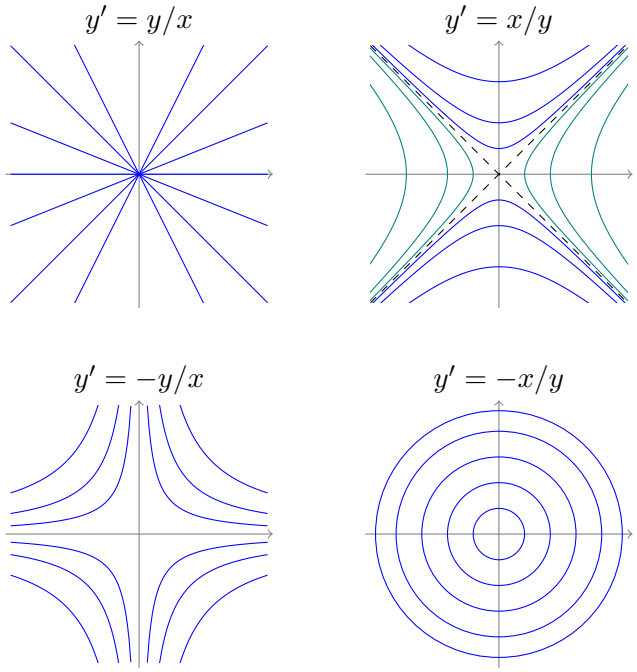
$$\int_{r>\varepsilon} U\Delta\phi\,\mathrm{d}x = \int_{r=\varepsilon}(-U\partial_r\phi + \phi\partial_r U)\,\mathrm{d}S.$$

The normal points toward the deleted ball. The first term tends to zero: it is $O(\varepsilon|\log\varepsilon|)$ in dimension two and $O(\varepsilon)$ in dimension three. For $U = \log(1/r)$, $\partial_r U = -1/r$, so the second term tends to $-2\pi\phi(0)$. For $U = 1/r$, it tends to $-4\pi\phi(0)$. Both $U$ are locally integrable, so the left side tends to their distributional action on $\Delta\phi$. Therefore

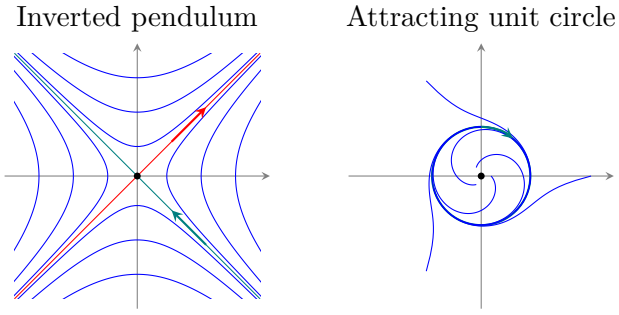
$$\boxed{\Delta\log(1/r) = -2\pi\delta_0 \quad (n=2), \qquad \Delta(1/r) = -4\pi\delta_0 \quad (n=3).}$$

## Additional phase-curve sketches

These drawings supplement the curve equations and domain restrictions in the solutions. A geometric curve may contain several distinct orbits; excluded axes and equilibrium points must still be treated as described in the text.




**Problem 2.6/1.** Rays, hyperbolas, rectangular hyperbolas, and circles. In the second and fourth panels the points on $y = 0$ are outside the displayed equation's domain; in the first and third panels $x = 0$ is excluded. The plots show the geometric completions to make their shapes visible.




**Problems 2.5/5 and 5.3/2.** The saddle has a stable diagonal (teal) and an unstable diagonal (red); the second portrait has clockwise spirals approaching the unit circle.

## Chapter 1: review guide and coverage

The main logical distinctions to check while reading are: a phase curve versus an integral curve; an invariant level set versus a single maximal orbit; preservation of directions versus preservation of velocity; local solvability versus a complete flow; and a formula obtained by division versus all solutions of the original equation.

| Book section | Subsections containing numbered problems | Number |
|---|---|---|
| §1 | 2, 6, 7, 8, 13, 14, 15, 16, 18 | 17 |
| §2 | 1, 3, 5, 6, 7 | 19 |
| §3 | 2, 4, 5 | 5 |
| §4 | 1, 3, 4 | 21 |
| §5 | 1, 2, 3, 4, 5 | 22 |
| §6 | 1, 2, 3, 4, 5, 6 | 24 |
| Total | Explicitly numbered problems | 108 |

The counts include problems for which the textbook already gives a solution. Embedded questions about rigid-body dimension, the dimension of a linear solution space, Floquet bounds, delta sequences, affine period maps, symmetry groups, and distributional Laplacians are also addressed. Subsections with no numbered exercises do not receive artificial problem numbers.

**Suggested first review.** Check 1.7/1 (the projective-line wording), 1.18/3–4 (quotient and linking), 2.3/1 (domain at the limiting point), 2.7/2 (two-sided logarithmic estimates), 5.2/2 (expressing the pushforward at its image point), 5.4/4–5 (global straightening), and 5.5/3 (smooth gluing of scalar conjugacies). These are the places where a short answer most readily hides an assumption.

**Source.** Vladimir I. Arnold, *Ordinary Differential Equations*, third edition, translated by Roger Cooke, Springer-Verlag, 1992, Chapter 1, pp. 13–88. Prepared against the scanned copy in the user's library. These notes are an independent worked companion, not an official solution manual.

# Chapter 2 — Basic Theorems

This chapter covers all 96 numbered problems in Sections 7–12 (printed pp. 89–151), including the starred problems. The identifiers continue the textbook's numbering. Smooth means $C^\infty$ when no finite differentiability class is specified. Perturbation remainders are uniform on fixed compact time intervals, unless a longer interval is explicitly justified.

## 7 Rectification theorems (§7)

### 7.1/1: Two local straightening maps

*Source: printed page 90.*

Retain $t$ as the first coordinate. For $\dot{x} = t$, put $y = x - t^2/2$; then $\dot{y} = 0$ and the inverse is $x = y + t^2/2$. For $\dot{x} = x^2$, put

$$y = \frac{x}{1 + tx}, \qquad x = \frac{y}{1 - ty}.$$

Near $(0, 0)$ the denominators are nonzero, $y_x = (1 + tx)^{-2}$, and $y_t + x^2 y_x = 0$. These are local diffeomorphisms carrying integral curves to horizontal lines.

### 7.1/2: Global straightening can fail

*Source: printed page 90.*

No. In fact the smooth everywhere nonvertical example in Problem 7.1/4 below already supplies a counterexample on the plane. Its space of maximal leaves is not Hausdorff, whereas the leaves of the horizontal foliation of the whole plane are parametrized by a real coordinate and have Hausdorff leaf space. A global diffeomorphism onto the plane cannot change this property. Here and in 7.1/4, global straightening means straightening on the whole space onto the whole space; a coordinate chart onto a proper open set is a different requirement.

### 7.1/3: A plane field that cannot be straightened

*Source: printed page 90.*

On $\mathbb{R}^3$ consider the planes $\ker(dz + y\,dx)$. Their projection to the $(x, y)$-plane is an isomorphism at every point. A surface tangent to these planes would therefore locally be a graph $z = f(x, y)$, with

$$f_x = -y, \qquad f_y = 0.$$

The second equation makes $f$ independent of $y$, contradicting the first. Thus there are no local integral surfaces. A field of parallel planes has integral surfaces, and their inverse images under a straightening diffeomorphism would give the impossible surfaces above. Equivalently, $(dz + y\,dx) \wedge d(dz + y\,dx)$ is nowhere zero.

### 7.1/4*: An everywhere nonvertical global counterexample

*Source: printed page 91.*

Consider $\dot{x} = tx^2$ on the entire $(t, x)$-plane. Its nonzero solutions satisfy

$$\frac{1}{x} = C - \frac{t^2}{2}.$$

For $C < 0$ this is one complete negative solution. At $C = 0$ there are two different maximal leaves $L_-$ and $L_+$, given by $x = -2/t^2$ for $t < 0$ and $t > 0$, respectively. They are not one leaf: neither extends through $t = 0$.

No disjoint saturated open sets can contain $L_-$ and $L_+$. Indeed, the points on the $C < 0$ leaves at $t = -1$ and $t = 1$ tend to $(-1, -2)$ and $(1, -2)$ as $C \uparrow 0$. Any such two open sets therefore both meet the same leaf for $C$ sufficiently close to zero; saturation makes them intersect. Thus the leaf space is not Hausdorff, ruling out a diffeomorphism to the horizontal foliation of the whole plane. Taking extra constant dependent variables gives the same obstruction in higher dimension. This example also shows why local existence alone does not provide global straightening.

### 7.1/5: Straightening without changing time

*Source: printed page 91.*

Fix $(t_0, a_0)$ and let $\Phi(t; t_0, a)$ be the local solution with initial value $a$. The map

$$(t, a) \longmapsto (t, \Phi(t; t_0, a))$$

has invertible derivative at $(t_0, a_0)$ because $D_a\Phi(t_0; t_0, a_0) = I$. Its local inverse has the form $(t, x) \mapsto (t, y(t, x))$. Along a solution $y$ is precisely its initial value at $t_0$, hence $\dot{y} = 0$. The inverse function theorem therefore gives the required time-preserving local straightening.

### 7.1/6: A global time-preserving example

*Source: printed page 91.*

For $\dot{x} = x + t$, set $y = e^{-t}(x + t + 1)$. Direct differentiation gives $\dot{y} = 0$. The inverse $x = e^t y - t - 1$ exists for every $(t, y) \in \mathbb{R}^2$, so this is a global time-preserving diffeomorphism.

### 7.1/7: Why the Riccati equation cannot be globally straightened with time fixed

*Source: printed page 91.*

For $\dot{x} = x^2$, the nonzero solution $x(t) = a/(1 - at)$ has a finite endpoint of its maximal time interval. A time-preserving diffeomorphism of the entire plane taking this equation to $\dot{y} = 0$ would have an inverse defined along the entire horizontal line $y = y_0$, producing a solution for every $t \in \mathbb{R}$. This contradicts the finite-time blow-up. The local transformation in 7.1/1 is not defined globally.

### 7.1/8: Newton's series for the exponential

*Source: printed page 91.*

Write $x = \sum_{j\ge 0} a_j t^j$ in $\dot{x} = x$, $x(0) = 1$. Comparing coefficients gives $a_0 = 1$ and $(j+1)a_{j+1} = a_j$, hence $a_j = 1/j!$. The ratio test proves convergence for every real or complex $t$, and uniform convergence on compact sets justifies termwise differentiation. The resulting solution is $x = e^t$.

### 7.3/1: Variation of a scalar initial condition

*Source: printed page 94.*

For $\dot{x} = x^2 + x\sin t$, $x(0) = a$, the solution at $a = 0$ is zero. Its derivative $z = \partial_a x|_{a=0}$ solves $\dot{z} = (\sin t)z$, $z(0) = 1$. Thus

$$\boxed{z(t) = e^{1-\cos t}}, \qquad x(t;a) = ae^{1-\cos t} + O(a^2).$$

The error statement is on each fixed compact interval, for sufficiently small $a$.

### 7.3/2: A perturbed Lotka–Volterra phase curve

*Source: printed page 94.*

For $\dot{x} = x(1 - ya(x,y))$, $\dot{y} = y(x-1)$, divide by $\dot{x}$ near $(1,0)$:

$$\frac{dy}{dx} = \frac{y(x-1)}{x(1-ya(x,y))}, \qquad y(1) = \varepsilon.$$

Set $y = \varepsilon z + O(\varepsilon^2)$. Then $z' = (1 - 1/x)z$, $z(1) = 1$, so

$$\boxed{y(x) = \varepsilon \frac{e^{x-1}}{x} + O(\varepsilon^2)}.$$

The estimate holds on compact intervals of $x > 0$ where this perturbative branch is defined. It describes the portion near the $x$-axis, not an entire closed orbit.

### 7.3/3: Initial-angle derivative of the pendulum

*Source: printed page 94.*

For $\ddot{\theta} = -\sin\theta$, $\theta(0) = a$, $\dot{\theta}(0) = 0$, differentiation at $a = 0$ gives $\ddot{z} = -z$, $z(0) = 1$, $\dot{z}(0) = 0$. Consequently $z = \cos t$ and $\partial_a \dot{\theta}|_0 = -\sin t$. Uniqueness and the odd symmetry of the equation imply $\theta(t;-a) = -\theta(t;a)$; smooth dependence therefore improves the fixed-time expansion to $\theta = a\cos t + O(a^3)$. This does not bound the error uniformly for arbitrarily large time.

### 7.3/4: The first variation of a soft oscillator

*Source: printed page 95.*

For $\ddot{x} = -x^3$, $x(0) = 0$, $\dot{x}(0) = a$, the first variation satisfies $\ddot{z} = 0$, $z(0) = 0$, $\dot{z}(0) = 1$. Thus $x = at + O(a^3)$ on fixed time intervals. Substituting an odd expansion gives more precisely

$$x(t; a) = at - \frac{a^3 t^5}{20} + O(a^5)$$

on such intervals: the cubic coefficient has second derivative $-t^3$ and zero initial data.

### 7.3/5: The time scale on which the relative error is small

*Source: printed page 95.*

Let $X'' = -X^3$, $X(0) = 0$, $X'(0) = 1$. For $a \neq 0$, uniqueness and scaling give exactly

$$x(t; a) = \operatorname{sgn}(a)\sqrt{|a|}\, X(\sqrt{|a|}t).$$

Near zero, $X(s) = s - s^5/20 + O(s^9)$. Hence, interpreting the quotient at $t = 0$ by continuity,

$$\frac{x(t; a) - at}{at} = -\frac{|a|^2 t^4}{20} + O(|a|^4 t^8).$$

This tends to zero whenever $\sqrt{|a|}|t| \to 0$, proving the requested regime $|t| \ll |a|^{-1/2}$, including intervals whose lengths grow as $a \to 0$. Energy gives amplitude $2^{1/4}\sqrt{|a|}$, consistent with this scale.

### 7.4/1: The period map and the order of composition

*Source: printed page 97.*

Write $G(t, s)$ for the evolution map and $A = G(T, 0)$. Periodicity and uniqueness give $G(t+T, s+T) = G(t, s)$; the evolution law gives $G(t, s)G(s, r) = G(t, r)$. Since all maps are globally defined and invertible, these imply $G(kT, 0) = A^k$ for every integer $k$. Moreover

$$G(kT + s, 0) = G(kT + s, kT)G(kT, 0) = G(s, 0)A^k.$$

Thus the second displayed ordering in the book is the valid one. There is no general reason for $A$ to commute with $G(s, 0)$.

### 7.5/1: A parameter derivative of the logistic solution

*Source: printed page 98.*

For $\dot{x} = x(a - x)$, $x(0) = 1$, the base solution at $a = 1$ is $x = 1$. Differentiating with respect to $a$ yields $\dot{z} = 1 - z$, $z(0) = 0$. Therefore $\partial_a x(t; 1) = 1 - e^{-t}$.

### 7.5/2: A small constant torque

*Source: printed page 99.*

For $\ddot{\theta} = a - \sin\theta$ with zero initial angle and velocity, the derivative at $a = 0$ obeys $\ddot{z} = 1 - z$, $z(0) = \dot{z}(0) = 0$. It is $z = 1 - \cos t$. Hence $\theta = a(1 - \cos t) + O(a^2)$ on each fixed finite interval (odd symmetry in fact gives $O(a^3)$).

### 7.5/3: Varying an oscillator's frequency

*Source: printed page 99.*

The solution is $\theta(t;\omega) = \cos(\omega t)$, so $\partial_\omega \theta|_{\omega=1} = -t\sin t$. Thus $\theta(t; 1+\varepsilon) = \cos t - \varepsilon t \sin t + O(\varepsilon^2 t^2)$ by Taylor's theorem. The growing first-order term is a phase error, not an unbounded exact oscillation. A small phase error requires $|\varepsilon t| \ll 1$.

### 7.6/1: A compact set need not be exited by an arbitrary direction field

*Source: printed page 101.*

Take the nonzero field $(y, -x)$ on the punctured plane and the compact annulus $1 \le x^2 + y^2 \le 2$. A circle of radius strictly between 1 and $\sqrt{2}$ stays in the interior forever and never meets the boundary. The extension-to-boundary assertion in extended phase space uses the strictly increasing time coordinate; it does not hold for an arbitrary direction field.

### 7.6/2: Linear growth prevents finite-time escape

*Source: printed page 101.*

Fix a finite time window $|t - t_0| \le T$. Smoothness on its product with $|x| \le r$ gives a bound $M_T$ there. Together with the assumed estimate outside the ball, this gives $|v(t,x)| \le M_T + k|x|$ everywhere in this window. The integral equation and Gronwall's inequality give, for both forward and backward time,

$$|x(t)| \le (|x(t_0)| + M_T|t - t_0|)e^{k|t-t_0|}.$$

Thus the graph stays in a compact subset of the full domain on every finite time interval. The extension theorem excludes a finite maximal endpoint, proving completeness in both directions. No bound uniform in all time on the interior ball is needed.

### 7.6/3: Complete solutions define a flow

*Source: printed page 103.*

Put $g^t(x) = \Phi(t; 0, x)$. Completeness defines this for all $(t, x)$. Uniqueness gives $g^0 = \mathrm{id}$ and $g^{s+t} = g^s \circ g^t$. In particular $g^{-t}$ is the inverse of $g^t$. Local smooth dependence, applied along any finite orbit segment and composed over finitely many subintervals, makes $(t, x) \mapsto g^t(x)$ smooth. These maps are therefore a global phase flow.

### 7.7/1: Straightening the oscillator's vector field

*Source: printed page 104.*

Near $(x_1, x_2) = (1, 0)$ use $x_1 = r\cos\theta$, $x_2 = -r\sin\theta$, with the local branch of $\theta$ near zero. For $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1$, direct differentiation gives $\dot{r} = 0$, $\dot{\theta} = 1$. Thus the vector field becomes $\partial_\theta$, with its speed as well as its direction straightened.

### 7.7/2: Three explicit vector-field rectifications

*Source: printed page 104.*

In each case the new coordinates $(s, z)$ satisfy $\dot{s} = 1$, $\dot{z} = 0$.

1. For $x_1\partial_{x_1} + 2x_2\partial_{x_2}$ on $x_1 > 0$, choose $s = \log x_1$, $z = x_2/x_1^2$. The inverse is $(x_1, x_2) = (e^s, ze^{2s})$.
2. For $\partial_{x_1} + \sin x_1\partial_{x_2}$, choose $s = x_1$, $z = x_2 + \cos x_1$. The inverse is $(s, z - \cos s)$.
3. For $x_1\partial_{x_1} + (1 - x_1^2)\partial_{x_2}$ on $|x_1| < 1$, choose $s = x_2 + x_1^2/2$, $z = x_1e^{-s}$. Then $\dot{s} = 1$ and $\dot{z} = 0$, including at $x_1 = 0$. The inverse is $x_1 = ze^s$, $x_2 = s - z^2e^{2s}/2$, with image domain $|z|e^s < 1$.

## 8 Higher-order equations (§8)

### 8.1/1: The second finite difference

*Source: printed page 106.*

For $\Delta_h\varphi(t) = \varphi(t + h) - \varphi(t)$, apply the same operator once more:

$$\Delta_h^2\varphi(t) = [\varphi(t + 2h) - \varphi(t + h)] - [\varphi(t + h) - \varphi(t)] = \varphi(t + 2h) - 2\varphi(t + h) + \varphi(t).$$

### 8.1/2: Vanishing differences and polynomial sequences

*Source: printed page 106.*

Fix $t$ and write $a_k = \varphi(t + kh)$. If $a_k$ is a polynomial in $k$ of degree less than $n$, each forward difference reduces its degree, so $\Delta^n a = 0$. Conversely, define

$$P(k) = \sum_{j=0}^{n-1} \binom{k}{j} (\Delta^j a)_0, \qquad \binom{k}{j} = \frac{k(k-1)\cdots(k-j+1)}{j!}.$$

Since $\Delta\binom{k}{j} = \binom{k}{j-1}$, this polynomial has the prescribed first $n$ difference data. Thus $a - P$ has its first $n$ values zero and satisfies $\Delta^n(a - P) = 0$. The recurrence $\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}(a - P)_{k+j} = 0$ determines values uniquely both forwards and backwards. Hence $a_k = P(k)$ for every integer $k$.

**Review note.** The assertion is on each lattice $t + h\mathbb{Z}$. It does not say $\varphi$ is a polynomial as a function on $\mathbb{R}$: an $h$-periodic function already has $\Delta_h\varphi = 0$.

### 8.1/3: Vanishing ordinary derivatives

*Source: printed page 107.*

Repeated integration of $x^{(n)} = 0$ yields

$$x(t) = \sum_{j=0}^{n-1} \frac{x^{(j)}(t_0)}{j!}(t - t_0)^j$$

on the connected interval of definition. Conversely the $n$th derivative of each such polynomial is zero. The constants are arbitrary, giving an $n$-dimensional solution space.

### 8.1/4: Radial solutions of the planar Helmholtz equation

*Source: printed page 107.*

For $u(x, y) = f(r)$, $r = \sqrt{x^2 + y^2} > 0$, direct differentiation gives $\Delta u = f'' + f'/r$. Hence $\Delta u + u = 0$ becomes

$$f''(r) + \frac{1}{r} f'(r) + f(r) = 0.$$

It is a smooth linear second-order equation on $(0, \infty)$. The values $f(r_0), f'(r_0)$ at any $r_0 > 0$ determine a unique solution on that interval. Thus the radial solution space has dimension two. Requiring smooth extension through $r = 0$, which the problem excludes, would impose an additional restriction.

### 8.2/1: An equation admitting both the sine and the identity

*Source: printed page 107.*

At $t = 0$, $t$ and $\sin t$ have the same derivatives of orders zero, one, and two, but they are different functions. Uniqueness for a smooth equation solved for its highest derivative rules out orders $n \leq 3$ on a domain containing these initial data. Every $n \geq 4$ is possible: both solve $x^{(n)} + x^{(n-2)} = 0$. The information therefore gives $n \geq 4$, not one uniquely determined order.

### 8.2/2: The impossible configuration in Figure 77

*Source: printed page 108.*

For $\ddot{x} + p(t)\dot{x} + q(t)x = 0$, every scalar multiple $c\varphi_1$ is also a solution. The configuration in the figure forces a multiple $c\varphi_1$ to be tangent to $\varphi_2$ at an interior point while the two curves do not coincide. One way to see the tangency is to take an interior extremum of $\varphi_2/\varphi_1$ in the depicted positive region. At this point both values and first derivatives of $c\varphi_1$ and $\varphi_2$ agree. Second-order uniqueness makes the solutions identical, a contradiction.

### 8.2/3: Why a singular equation loses uniqueness

*Source: printed page 108.*

Both $x = 0$ and $x = t^2$ satisfy $t^2\ddot{x} = 2x$ and $x(0) = \dot{x}(0) = 0$. For $t \neq 0$ the normal form is $\ddot{x} = 2x/t^2$, whose right-hand side is not defined near the initial point $t = 0$. At $t = 0$ the original equation does not determine $\ddot{x}$. The hypotheses of the normal-form uniqueness theorem are therefore absent.

### 8.2/4: A quadratic sequence from difference data

*Source: printed page 108.*

With step $h = 1$, $\Delta^3\varphi = 0$ gives a quadratic sequence. Newton's formula from 8.1/2 gives

$$\varphi(k) = 0 + 0\binom{k}{1} + 2\binom{k}{2} = k(k-1), \qquad k \in \mathbb{Z}.$$

Indeed $\Delta\varphi(k) = 2k$ and $\Delta^2\varphi(k) = 2$, so all initial conditions hold.

### 8.3/1: First-order correction to a fall with weak resistance

*Source: printed page 108.*

For $\ddot{x} = -g + \varepsilon F(x, \dot{x})$, $x(0) = h$, $\dot{x}(0) = 0$, put $x = x_0 + \varepsilon x_1 + O(\varepsilon^2)$. The unperturbed fall is $x_0 = h - gt^2/2$ and

$$x_1''(t) = F(h - gt^2/2, -gt), \qquad x_1(t) = \int_0^t (t-s)F(h - gs^2/2, -gs)\,\mathrm{d}s.$$

This is the first-order position correction for general smooth $F$. Let $T_0 = \sqrt{2h/g}$. The impact time is

$$T_\varepsilon = T_0 + \varepsilon\frac{x_1(T_0)}{gT_0} + O(\varepsilon^2),$$

by the implicit function theorem at the simple zero of $x_0$. For linear drag $F = -\dot{x}$, one obtains $x_1 = gt^3/6$ and the delay $T_\varepsilon - T_0 = \varepsilon h/(3g) + O(\varepsilon^2)$.

### 8.3/2: Completeness of the pendulum

*Source: printed page 109.*

Since $|\ddot{\theta}| = |\sin\theta| \le 1$, for any finite $|t - t_0| \le T$ we have

$$|\dot{\theta}(t)| \le |\dot{\theta}(t_0)| + T, \quad |\theta(t)| \le |\theta(t_0)| + |\dot{\theta}(t_0)|T + T^2/2.$$

The pair $(\theta, \dot{\theta})$ stays bounded on finite intervals, so the extension theorem gives existence for every real time. The same conclusion holds on the cylindrical phase space.

### 8.3/3: Completeness for a power-law acceleration

*Source: printed page 109.*

For positive integers $k$, all solutions of $\ddot{x} = x^k$ are complete precisely when $k = 1$: then $x = Ae^t + Be^{-t}$. For $k > 1$, set $p = 2/(k-1)$ and choose $A > 0$ with $A^{k-1} = p(p+1)$. The function

$$x(t) = A(T-t)^{-p}, \qquad t < T,$$

satisfies the equation and blows up at $T$, disproving completeness of all solutions. If the convention $0 \in \mathbb{N}$ is used, $k = 0$ must be added: $\ddot{x} = 1$ also has only complete solutions.

### 8.4/1: Eastward deflection in a mine shaft

*Source: printed page 110.*

Use unit vectors east, north, and up. At latitude $\lambda$, $\boldsymbol{\Omega} = \Omega(0, \cos\lambda, \sin\lambda)$. To first order in $\Omega$, the falling velocity is $(0, 0, -gt)$, so $2\boldsymbol{v} \times \boldsymbol{\Omega}$ has east component $2g\Omega t \cos\lambda$. Integrating twice with zero initial displacement and horizontal velocity gives

$$d_E(t) = \frac{g\Omega\cos\lambda}{3} t^3 + O(\Omega^2).$$

At $T = \sqrt{2h/g}$ this is $d_E = \frac{2}{3}\Omega h T \cos\lambda + O(\Omega^2)$. Taking $h = 10\,\mathrm{m}$, $\lambda = 60°$, $g = 9.81\,\mathrm{m/s^2}$ and $\Omega \simeq 7.3 \times 10^{-5}\,\mathrm{s^{-1}}$ gives $d_E \simeq 0.348\,\mathrm{mm}$, eastward. These are the idealized constants used for the exercise.

### 8.4/2: The lens cap thrown toward the Earth

*Source: printed page 111–112.*

Use the orbital radius and inverse orbital angular speed as units. The station has $r = 1$, $\varphi = t$, and the cap has initial radial velocity $-v$, with $v \ll 1$. Newton's equation in polar coordinates is

$$\ddot{r} - r\dot{\varphi}^2 = -r^{-2}, \qquad r\ddot{\varphi} + 2\dot{r}\dot{\varphi} = 0.$$

Write $r = 1 + \rho$, $\varphi = t + \psi$ and keep first-order terms:

$$\ddot{\rho} = 3\rho + 2\dot{\psi}, \qquad \ddot{\psi} + 2\dot{\rho} = 0.$$

The initial data are $\rho = \psi = \dot{\psi} = 0$, $\dot{\rho} = -v$. Thus $\dot{\psi} = -2\rho$, $\ddot{\rho} = -\rho$, and

$$\rho = -v\sin t, \qquad \psi = 2v(1 - \cos t).$$

Relative to the station, the radial and along-track displacements trace the ellipse

$$\frac{\rho^2}{v^2} + \frac{(\psi - 2v)^2}{4v^2} = 1$$

to first order. It initially goes Earthward, then ahead, then outward, and returns close to the station after one revolution. For the book's $v \simeq 1/800$ and orbital radius 6400 km, the semiaxes are about 8 and 16 km and the maximum lead about 32 km.

This relative ellipse is an approximation, not an exact closed orbit in the rotating frame. In the two-body model the cap's inertial orbit has energy $(-1 + v^2)/2$, semimajor axis $(1 - v^2)^{-1}$, and eccentricity $v$. Its period differs from $2\pi$ at order $v^2$, so exact reunion after one station revolution is not implied. The error in position on that fixed interval is $O(v^2)$ in orbital-radius units.

### 8.5/1: The cubic Clairaut equation

*Source: printed page 113.*

Put $p = \dot{x}$ in $x = tp - p^3$. For a twice differentiable solution,

$$0 = (t - 3p^2)\dot{p}.$$

On regular pieces $p = c$, giving the straight lines $x = ct - c^3$. The exceptional set is $t = 3p^2$, $x = 2p^3$, whose projection is the semicubical cusp $27x^2 = 4t^3$, $t \geq 0$. Its two envelope branches are

$$x = \pm \frac{2}{3\sqrt{3}} t^{3/2}, \qquad t > 0,$$

and each is a solution, with slope $p = \pm\sqrt{t/3}$. A line of slope $c$ is tangent to the envelope at $(3c^2, 2c^3)$.

At a given $(t, x)$, the possible slopes are the real roots of $p^3 - tp + x = 0$. There are three distinct roots when $4t^3 - 27x^2 > 0$, one when it is negative, and multiple roots on the cusp. Each simple root defines a smooth normal equation locally. One must specify a branch before invoking uniqueness. In the class of $C^1$ solutions, tangent line pieces can be joined to envelope pieces at the tangency points; these joins need not be $C^2$. At the cusp the envelope itself is $C^1$ from the right but not $C^2$.

### 8.5/2: Integral curves on a folded equation surface

*Source: printed page 115.*

The surface $p^2 = t$ has coordinates $(p, x)$. The contact equation $dx = p\,dt$ becomes $dx = 2p^2\,dp$. Consequently its lifted integral curves and their projections are

$$(t, x, p) = (p^2, C + 2p^3/3, p), \qquad (x - C)^2 = 4t^3/9.$$

For $t > 0$ the two projected branches are $x = C \pm 2t^{3/2}/3$. The lifted curve is smooth at $p = 0$, while the projection has a cusp there and is not one single-valued solution graph across both branches.

### 8.5/3: General Clairaut equations and duality

*Source: printed page 116.*

On $x = tp - f(p)$ use coordinates $(t, p)$. The contact equation reduces to $(t - f'(p))dp = 0$. Away from the exceptional set $t = f'(p)$, integral curves have $p = c$ and project to $x = ct - f(c)$. The discriminant/envelope is

$$t = f'(c), \qquad x = cf'(c) - f(c).$$

Where $f''(c) \neq 0$, invert $t = f'(c)$ and call the resulting graph $x = g(t)$. Differentiation gives $g'(t) = c$, proving tangency of the associated line. It also gives $tg'(t) - g(t) = f(c)$, so taking the Legendre transform again recovers $f$. This proves the stated local duality. At points with $f'' = 0$ the inversion must be considered separately; global single-valued duality needs additional assumptions such as strict convexity.

### 8.5/4: Legendre transform of a power

*Source: printed page 116.*

For $f(p) = |p|^\alpha/\alpha$, $\alpha > 1$, the relation $t = f'(p) = \operatorname{sgn}(p)|p|^{\alpha-1}$ has inverse $p = \operatorname{sgn}(t)|t|^{1/(\alpha-1)}$. Therefore

$$tp - f(p) = \left(1 - \frac{1}{\alpha}\right)|t|^{\alpha/(\alpha-1)} = \frac{|t|^\beta}{\beta}, \qquad \frac{1}{\alpha} + \frac{1}{\beta} = 1.$$

The formula includes zero by continuity. The strictly convex supremum definition gives the same transform even when the relevant higher derivatives at zero do not exist.

## 9 Autonomous phase curves (§9)

### 9.1/1: Is the variational equation autonomous?

*Source: printed page 117.*

Along a solution $x_0(t)$ of $\dot{x} = v(x)$, the variational equation is $\dot{z} = Dv(x_0(t))z$. At an equilibrium its coefficients are constant, hence it is autonomous. Generally they vary with time: for $\dot{x} = x^2$ and $x_0(t) = 1/(1-t)$, one gets $\dot{z} = 2z/(1-t)$. A nonstationary base solution can still give an autonomous variational equation, for example when $v$ is linear.

### 9.2/1: Unique phase curves do not imply autonomy

*Source: printed page 118.*

The nonautonomous equation $\dot{x} = 1 + t^2$ has solutions $x = t + t^3/3 + C$. Each maximal solution maps $\mathbb{R}$ onto $\mathbb{R}$, so there is only one phase curve as a subset of phase space: the entire line. Yet the velocity explicitly depends on time.

**Review note.** For an autonomous equation, the proof that intersecting maximal phase curves coincide also covers incomplete solutions. Shift the time parameter to match one common point. Uniqueness makes the shifted solutions equal on their common interval; if either interval extended beyond the other, it would extend the supposedly maximal solution. Thus their shifted intervals, and hence their images, agree.

### 9.3/1: Closed subgroups of the plane, space, and circle

*Source: printed page 120.*

Every closed additive subgroup of $\mathbb{R}^n$ has the form

$$G = V \oplus (\mathbb{Z}b_1 + \cdots + \mathbb{Z}b_k)\,,$$

where $V$ is a vector subspace and the $b_i$ are linearly independent vectors in $V^\perp$. Thus, up to an invertible linear change of coordinates, it is $\mathbb{R}^p \times \mathbb{Z}^k \times \{0\}^{n-p-k}$, with $p + k \le n$. For $n = 2$ the types are $\{0\}, \mathbb{Z}, \mathbb{Z}^2, \mathbb{R}, \mathbb{R} \times \mathbb{Z}, \mathbb{R}^2$; for $n = 3$ they are those six together with $\mathbb{Z}^3, \mathbb{R} \times \mathbb{Z}^2, \mathbb{R}^2 \times \mathbb{Z}, \mathbb{R}^3$. Lattices need not align with the original coordinate axes.

Here is a proof of the structural assertion. If a closed subgroup contains nonzero $g_j \to 0$ with $g_j/|g_j| \to e$, integer multiples of $g_j$ approximate every $se$, so it contains the line $\mathbb{R}e$. Let $V$ be a maximal vector subspace contained in $G$. Orthogonal projection onto $V^\perp$ identifies the remaining subgroup with $D = G \cap V^\perp$: subtracting a vector in $V$ stays in $G$. It is closed and must be discrete, since otherwise the preceding argument would enlarge $V$.

A discrete subgroup $D$ is a lattice in its linear span. To verify this by induction, choose a shortest nonzero vector $b_1$. The projection of $D$ to $b_1^\perp$ is discrete: otherwise choose projected vectors tending to zero, subtract integer multiples of $b_1$ to bound their components along $b_1$, and take a convergent subsequence. A discrete subgroup has only finitely many points in a compact set (differences of an accumulating sequence would tend to zero), giving a contradiction. Apply induction to the projected group and lift its basis. The kernel is $\mathbb{Z}b_1$ by the shortest-vector property, so these lifts together with $b_1$ generate $D$ freely.

Finally a closed subgroup $H \subset S^1$ has closed inverse image under $t \mapsto e^{it}$, containing $2\pi\mathbb{Z}$. The closed-subgroup classification of $\mathbb{R}$ gives either $\mathbb{R}$ or $(2\pi/m)\mathbb{Z}$ for a positive integer $m$. Hence $H = S^1$ or the group of all $m$th roots of unity, including the trivial group for $m = 1$.

### 9.3/2: Orbit closures of two incommensurate oscillators

*Source: printed page 121.*

For $\ddot{x}_1 = -x_1$, $\ddot{x}_2 = -2x_2$, the quantities $r_1^2 = x_1^2 + \dot{x}_1^2$ and $r_2^2 = x_2^2 + \dot{x}_2^2/2$ are constant. If both vanish, the orbit is a point. If just one vanishes, it is a periodic ellipse, diffeomorphic to a circle. If both are positive, the orbit lies on the product of the two ellipses, a two-torus, with angle velocities 1 and $\sqrt{2}$.

It is dense in that torus. To approximate any first angle, sample times $t = \alpha + 2\pi m$; the second angles then differ by $2\pi m\sqrt{2}$. Irrational rotation is dense: the pigeonhole principle supplies nonzero integer multiples arbitrarily close to zero modulo 1, and their successive multiples approximate any point of the circle. Thus the orbit closure is the whole two-torus, although the orbit itself is not closed.

## 10 Directional derivatives and first integrals (§10)

### 10.1/1: The Hamiltonian differentiated along its own field

*Source: printed page 122.*

For $X_H = \sum_i (H_{p_i}\partial_{q_i} - H_{q_i}\partial_{p_i})$, the chain rule gives

$$L_{X_H} H = \sum_i (H_{q_i} H_{p_i} - H_{p_i} H_{q_i}) = 0.$$

This is the differential form of conservation of the Hamiltonian.

### 10.3/1: Which directional-derivative identities hold?

*Source: printed page 123.*

The coordinate formula $L_v f = \sum_i v_i \partial_i f$ gives additivity in $f$ and the Leibniz rule by the corresponding rules for partial derivatives. It also gives $L_{u+v} = L_u + L_v$ and $L_{fu} = f L_u$ by distributing the coefficients. The fifth identity is false in general. For $u = \partial_x$, $v = x\partial_x$, and $f = x$,

$$L_u L_v f = 1, \qquad L_v L_u f = 0.$$

Commutation of coordinate partial derivatives does not cancel the derivatives of the variable coefficients of a field.

### 10.3/2*: Every smooth derivation comes from a vector field

*Source: printed page 124.*

Yes, for derivations of the $\mathbb{R}$-algebra $C^\infty(U)$, hence $\mathbb{R}$-linear derivations. Let $D$ be such a derivation and put $v_i = D(x_i)$; these are smooth functions. First $D1 = 2D1$, so $D$ annihilates constants. It is local at each point $p$: if $f = 0$ near $p$, choose a smooth bump $\chi$ equal to 1 near $p$ and supported where $f = 0$. From $D(\chi f) = 0$ evaluated at $p$, obtain $(Df)(p) = 0$.

On a small convex ball about $p$, Hadamard's formula gives

$$f(x) = f(p) + \sum_i (x_i - p_i) h_i(x), \qquad h_i(x) = \int_0^1 \partial_i f(p + s(x-p))\,\mathrm{d}s.$$

Use cutoffs to extend the $h_i$ to $U$ without changing their germs at $p$. Locality and the product rule then give

$$(Df)(p) = \sum_i (Dx_i)(p) h_i(p) = \sum_i v_i(p) \partial_i f(p).$$

Thus $D = L_v$. Uniqueness follows by applying $L_v$ to the coordinate functions. This proves that $v \mapsto L_v$ is an isomorphism of $C^\infty(U)$-modules.

### 10.4/1: The commutator is first order

*Source: printed page 124.*

Expanding twice gives

$$(L_a L_b - L_b L_a) f = \sum_j \left( \sum_i a_i \partial_i b_j - b_i \partial_i a_j \right) \partial_j f.$$

The second-derivative terms cancel because $\partial_i \partial_j f = \partial_j \partial_i f$. Hence this operator is $L_c$, where $c_j = \sum_i (a_i \partial_i b_j - b_i \partial_i a_j)$.

### 10.4/2: Linearity, skew symmetry, and Jacobi

*Source: printed page 124.*

Linearity and skew symmetry follow immediately from $[A, B] = AB - BA$ applied to $A = L_a$, $B = L_b$. In any associative operator algebra,

$$[[A, B], C] = ABC - BAC - CAB + CBA.$$

Adding its cyclic permutations cancels all six ordered triple products in pairs. Thus the Jacobi identity holds for the operators. Since $v \mapsto L_v$ is injective, all three identities hold for the corresponding vector fields.

### 10.4/3: The cross product is a Lie bracket

*Source: printed page 124.*

The cross product is bilinear and skew-symmetric. For Jacobi, the vector triple-product identity gives

$$(a \times b) \times c = b(a \cdot c) - a(b \cdot c).$$

Add the two cyclic analogues; symmetry of the dot product cancels every term. Therefore $\mathbb{R}^3$ with its cross product is a Lie algebra.

### 10.4/4: Matrices with the commutator bracket

*Source: printed page 125.*

Matrix multiplication is bilinear and associative. Therefore $[A, B] = AB - BA$ is bilinear, skew-symmetric, and satisfies Jacobi by the same six-term cancellation in 10.4/2. The space of all $n \times n$ matrices is closed under this operation and is a Lie algebra.

### 10.4/5: Symmetric and skew-symmetric matrices

*Source: printed page 125.*

For symmetric $A, B$, $[A, B]^T = -[A, B]$, so their commutator is generally not symmetric. For example,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad [A, B] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus symmetric matrices do not form a Lie algebra under this bracket for $n \geq 2$ (the $n = 1$ case is abelian). If $A^T = -A$ and $B^T = -B$, then again $[A, B]^T = -[A, B]$, now proving closure. Skew-symmetric matrices form the Lie algebra $\mathfrak{so}(n)$.

### 10.4/6: Coordinate components of the bracket

*Source: printed page 125.*

Apply the commutator to the coordinate function $x_j$. Since $L_b x_j = b_j$ and $L_a x_j = a_j$,

$$[a,b]_j = L_a b_j - L_b a_j = \sum_i \left( a_i \frac{\partial b_j}{\partial x_i} - b_i \frac{\partial a_j}{\partial x_i} \right).$$

This also checks the sign convention used in the other bracket exercises.

### 10.4/7: Commuting flows and vanishing brackets

*Source: printed page 125.*

Suppose the flows $g^t$ of $a$ and $h^s$ of $b$ commute. Differentiate $g^t(h^s(x)) = h^s(g^t(x))$ in $s$ at zero to obtain $Dg^t(x)b(x) = b(g^t(x))$. Differentiating in $t$ at zero yields $Da\,b = Db\,a$, or $[a,b] = 0$.

Conversely assume $Db\,a = Da\,b$. Along $g^t(x)$, the two vectors $Dg^t(x)b(x)$ and $b(g^t(x))$ solve the same linear equation $\dot{w} = Da(g^t(x))w$ with the same initial value. Hence they agree. It follows that $s \mapsto g^t(h^s(x))$ is a solution of $\dot{z} = b(z)$ starting at $g^t(x)$, so uniqueness gives $g^t h^s(x) = h^s g^t(x)$. This is a local statement where both compositions are defined; for complete fields it holds for all real $s, t$.

### 10.4/8: The sign of the rotation-field bracket

*Source: printed page 125.*

With the usual angular-velocity convention, put $r = x - o$ and $a_\omega(x) = \omega \times r$. Using the bracket defined in 10.4/6,

$$[a_\alpha, a_\beta](x) = \beta \times (\alpha \times r) - \alpha \times (\beta \times r) = -(\alpha \times \beta) \times r = -a_{\alpha \times \beta}(x).$$

**Review note.** The printed answer has the opposite sign. With $L_{[a,b]} = L_a L_b - L_b L_a$ and velocity $a_\omega = \omega \times r$, the minus sign above is required. A direct check uses $\alpha = e_1$, $\beta = e_2$: the bracket at $r = e_1$ equals $-e_2$. The printed plus sign corresponds instead to defining the rotation field as $r \times \omega$, or to reversing the bracket convention.

### 10.5/1: First integrals near a limit cycle

*Source: printed page 125.*

For an isolated planar periodic orbit, choose a small transverse segment and its increasing Poincaré return map. The fixed point representing the cycle is isolated, so on each side sufficiently close to it, the displacement has a fixed sign. Iterating the map in forward or backward time makes points on that side approach the fixed point. Flowing the segment gives a small annular neighborhood whose trajectories approach the cycle in at least one time direction.

A continuous first integral is constant on the cycle, say with value $c$. It is constant on every nearby trajectory; taking a sequence of points of that trajectory tending to the cycle proves its value is also $c$. Thus it is constant throughout that neighborhood. This includes a semistable planar cycle, using the appropriate time direction separately on each side.

**Review note.** The planar limit-cycle setting is essential. An arbitrary isolated periodic orbit in higher dimension may have transverse saddle directions and a nonconstant first integral: $\dot\theta = 1$, $\dot u = u$, $\dot v = -v$ on $S^1 \times \mathbb{R}^2$ has the isolated periodic orbit $u = v = 0$ and integral $uv$.

### 10.5/2: Global first integrals of a diagonal linear system

*Source: printed page 126.*

For $\dot x = x$, $\dot y = ky$, if $k > 0$, every trajectory tends to $(0, 0)$ as $t \to -\infty$. Continuity then forces any global first integral to equal its value at the origin. For $k = 0$, $y$ itself is a nonconstant integral.

For $k < 0$, put $a = -k > 0$. In the first quadrant $q = x^a y$ is invariant. Define on the whole plane

$$f(x,y) = \begin{cases} \exp(-1/(x^a y)), & x > 0,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

This is a nonconstant $C^\infty$ first integral. To check smoothness across the axes, each derivative inside the quadrant is a finite sum of powers of $x, y$ times $e^{-1/(x^a y)}$; on a bounded neighborhood, the exponential dominates every inverse power as either coordinate tends to zero. All derivatives therefore extend by zero, including at the origin. Thus precisely $k \le 0$ admit nonconstant global smooth first integrals.

### 10.5/3: The algebra of first integrals

*Source: printed page 126.*

If $L_v f = L_v g = 0$, then $L_v(f + g) = 0$ and $L_v(fg) = f L_v g + g L_v f = 0$. Constants are also first integrals, and scalar multiples remain first integrals. Therefore these functions form a unital $\mathbb{R}$-algebra (and, where defined, any smooth function of finitely many first integrals is again a first integral).

### 10.6/1: Why the shape of the domain matters

*Source: printed page 127.*

Take $U = \mathbb{R}^2 \setminus \{(0, y) : y \ge 0\}$ and the field $\partial_x$. This slit plane is connected. Define $f(x, y) = e^{-1/y}$ on $x > 0, y > 0$, and $f = 0$ elsewhere in $U$. The function is smooth because it is flat as $y \downarrow 0$, and $\partial_x f = 0$ everywhere in $U$. Yet $f(1, 1) = e^{-1}$ and $f(-1, 1) = 0$. It cannot be a function of $y$ alone: horizontal slices have two components above the slit.

### 10.7/1: Kepler's sectorial velocity

*Source: printed page 128.*

For planar motion let $r$ denote distance and $\varphi$ the polar angle. The determinant $\ell = x\dot y - y\dot x = r^2\dot\varphi$ satisfies

$$\dot\ell = x\ddot y - y\ddot x = x(-y/r^3) - y(-x/r^3) = 0.$$

Thus $r^2\dot\varphi$ is a first integral wherever the equation is defined. Geometric swept area per unit time is $\ell/2$.

### 10.7/2: Any central force preserves sectorial velocity

*Source: printed page 128.*

If $\ddot{\boldsymbol{r}} = a(r)\boldsymbol{r}$, then $\frac{d}{dt}\det(\boldsymbol{r}, \dot{\boldsymbol{r}}) = \det(\dot{\boldsymbol{r}}, \dot{\boldsymbol{r}}) + a(r)\det(\boldsymbol{r}, \boldsymbol{r}) = 0$. Hence the same integral $r^2\dot{\varphi}$ exists, independently of the radial force law.

### 10.7/3: Three-dimensional angular momentum

*Source: printed page 128.*

For a central force,

$$\frac{d}{dt}(\boldsymbol{r} \times \dot{\boldsymbol{r}}) = \dot{\boldsymbol{r}} \times \dot{\boldsymbol{r}} + \boldsymbol{r} \times a(r)\boldsymbol{r} = 0.$$

Every component is therefore a first integral. For unit mass this vector is the angular momentum; a nonzero value also confines the motion to its perpendicular plane.

### 10.7/4: A cyclic coordinate

*Source: printed page 128.*

Hamilton's equation is $\dot{p}_i = -\partial H/\partial q_i$. If $H$ is independent of $q_i$, this derivative vanishes identically, so $p_i$ is constant along every solution and is a first integral.

### 10.7/5: A complete equation has global time-dependent integrals

*Source: printed page 128.*

Fix the reference time 0 and write $G(s, t)$ for the evolution from $t$ to $s$. Completeness makes

$$a(t, x) = G(0, t)x$$

defined everywhere on the extended phase space. Its components are first integrals: by the evolution law, $a(t, G(t, 0)a_0) = a_0$. For fixed $t$ the map $x \mapsto a(t, x)$ is a diffeomorphism, with inverse $G(t, 0)$, so its $n$ components are functionally independent even with respect to $x$ alone. Finally any time-dependent first integral $F$ satisfies

$$F(t, x) = F(0, a(t, x)).$$

Thus every such integral is a function of these $n$ integrals. This uses the Euclidean phase coordinates in the problem; on a general manifold the initial-point map is globally manifold-valued and need not admit one global coordinate system.

# 11 First-order partial differential equations (§11)

### 11.1/1: Characteristics of a linear PDE

*Source: printed page 129.*

Write $u_x - yu_y = 0$. The characteristic field is $(1, -y)$, so its equations are $\dot{x} = 1$, $\dot{y} = -y$. Eliminating the parameter gives $y = Ce^{-x}$, including the characteristic $y = 0$.

### 11.1/2: The general solution along these characteristics

*Source: printed page 130.*

The coordinate $c = ye^x$ is constant on each characteristic, and $(x, y) \mapsto (x, c)$ is a global diffeomorphism. In these coordinates the equation is $u_x|_c = 0$, hence

$$u(x, y) = F(ye^x)$$

for an arbitrary differentiable $F$. On the whole plane this is the complete family; on a general restricted domain one must also account for disconnected characteristic segments.

### 11.1/3: Rotational characteristics

*Source: printed page 130.*

The equation $yu_x = xu_y$ is $L_{(y,-x)}u = 0$. Its nonzero characteristics are the circles centered at the origin. Therefore every solution on $\mathbb{R}^2$ is radial:

$$u(x, y) = F(x^2 + y^2).$$

For smooth solutions $F$ is smooth on $[0, \infty)$ in the one-sided sense and can be extended smoothly across zero. This follows by applying the even-function Taylor expansion to $u(r, 0)$ at $r = 0$; away from zero use $F(s) = u(\sqrt{s}, 0)$. Conversely any such $F$ gives a smooth solution. For only $C^1$ solutions, require the composite to be $C^1$ rather than unnecessarily requiring $F$ to be $C^1$ at zero.

### 11.1/4: One invariant need not separate all global characteristics

*Source: printed page 130.*

For $xu_x = yu_y$, the field is $(x, -y)$ and $xy$ is invariant, but a level $xy = c > 0$ has two different branches. The smooth function

$$u(x, y) = \begin{cases} e^{-1/(xy)}, & x > 0,\ y > 0, \\ 0, & \text{otherwise} \end{cases}$$

satisfies the PDE, with smoothness across the axes following from exponential flatness. Yet $u(1, 1) = e^{-1}$ and $u(-1, -1) = 0$, despite equal products. Hence not every global solution is a single function of $xy$.

### 11.2/1: Sinusoidal initial data on a transversal

*Source: printed page 131.*

Every characteristic $ye^x = c$ meets $x = 0$ at $(0, c)$. Transporting the initial value $\sin c$ gives $u(x, y) = \sin(ye^x)$. This is the unique global solution since every characteristic meets the initial line exactly once.

### 11.2/2: Noncharacteristic points of a vertical line

*Source: printed page 131.*

For $yu_x = xu_y$, the characteristic vector at $(1, y)$ is $(y, -1)$. It is transverse to the vertical line $x = 1$ exactly when its horizontal component $y$ is nonzero. Thus the only characteristic point of this line is $(1, 0)$.

### 11.2/3: A globally nonunique Cauchy problem

*Source: printed page 131.*

The solution must be $u = F(x^2 + y^2)$, and its restriction to $x = 1$ gives $F(1 + y^2) = y^2$. Thus $F(s) = s - 1$ for $s \geq 1$, with no values specified for $0 \leq s < 1$ except smooth matching. One solution is $u = x^2 + y^2 - 1$. Infinitely many others are

$$u = x^2 + y^2 - 1 + B(x^2 + y^2),$$

where $B$ is any smooth nonzero bump supported in $(0, 1)$. Existence holds, but global uniqueness fails because the initial line misses all circles of radius less than one.

### 11.3/1: An inhomogeneous transport equation

*Source: printed page 132.*

For $u_x - yu_y = y$ and $u(0, y) = \sin y$, use the characteristic $y = ce^{-x}$. Along it, $du/dx = ce^{-x}$, giving $u = \sin c + c(1 - e^{-x})$. Substituting $c = ye^x$ yields

$$\boxed{u(x, y) = \sin(ye^x) + y(e^x - 1)}.$$

Direct differentiation verifies both the PDE and the data.

### 11.4/1: Burgers evolution of arctangent data

*Source: printed page 133.*

For the printed data $u(0, x) = \arctan x$ in $u_t + uu_x = 0$, label a particle by its initial coordinate $a$. Characteristics give

$$x = a + t \arctan a, \qquad u = \arctan a.$$

Equivalently the graph at time $t$ is $x = \tan u + tu$, $-\pi/2 < u < \pi/2$. Its projection derivative is

$$\frac{\partial x}{\partial a} = 1 + \frac{t}{1 + a^2}, \qquad u_x = \frac{1}{1 + a^2 + t}.$$

For every $t > -1$ the map $a \mapsto x$ is an increasing diffeomorphism of $\mathbb{R}$. Thus the classical solution exists for all forward time, with progressively flatter increasing graphs. At $t = -1$ its slope blows up at $a = 0$; for $t < -1$ the characteristic surface folds.

**Review note.** The decreasing graph and positive breaking time $\pi/2$ printed with this problem are inconsistent with the displayed initial condition $\arctan x$. The formula above solves the stated data. A decreasing profile instead breaks forward in time at $t_* = -1/\min u_0'$ when that minimum is negative; even the unscaled $\operatorname{arccot} x$ would give $t_* = 1$, not $\pi/2$.

### 11.4/2: A velocity field in an external force

*Source: printed page 133.*

If a particle follows $\dot{x} = u(t, x)$ and has acceleration $F(x)$, the chain rule gives $\ddot{x} = u_t + u_x \dot{x} = u_t + u u_x$. Therefore the Eulerian velocity equation is $u_t + u u_x = F(x)$ (unit mass, or $F$ interpreted as force per unit mass).

### 11.4/3: Harmonic focusing

*Source: printed page 133.*

For $F(x) = -x$ with zero initial velocity, a particle starting at $a$ has $x = a \cos t$, $u = -a \sin t$. While $\cos t \neq 0$, eliminate $a$ to obtain $u(t, x) = -x \tan t$. The maximal time strip containing $t = 0$ is $|t| < \pi/2$. At its endpoints all particle positions focus at zero and no single smooth velocity field represents the family.

### 11.4/4: The first breakdown time for sinusoidal forcing

*Source: printed page 133.*

Let $X(t, a)$ solve $X'' = \sin X$, $X(0, a) = a$, $X_t(0, a) = 0$. Then $u(t, X(t, a)) = X_t(t, a)$ as long as $J = X_a$ stays positive. The variational equation is

$$J'' = (\cos X) J, \quad J(0) = 1, \ J'(0) = 0.$$

Before its first zero and for $0 \leq t < \pi/2$, variation of constants yields

$$J(t) = \cos t + \int_0^t \sin(t - s)(1 + \cos X(s, a)) J(s) \, \mathrm{d}s \geq \cos t > 0.$$

A first-zero argument therefore shows no zero occurs before $\pi/2$. Also $|X(t, a) - a| \leq t^2/2$, so the increasing map $a \mapsto X(t, a)$ is onto, giving a global smooth velocity field on each such strip.

For $a = \pi$, the particle is stationary, and $J = \cos t$, $J_t = -\sin t$. Hence $u_x(t, \pi) = J_t / J = -\tan t$ diverges as $t \uparrow \pi/2$. The largest width is consequently $C = \pi/2$.

### 11.5/1: Characteristics in the extended space

*Source: printed page 134.*

The characteristic vector for $u_t + uu_x = 0$ in $(t, x, u)$-space is $(1, u, 0)$. Integrating gives $(t, x, u) = (s, x_0 + u_0 s, u_0)$ after choosing the initial time as zero. These are straight lines; their projections to $(t, x)$ are the particle trajectories.

### 11.5/2: A local normal form for a quasilinear equation

*Source: printed page 135.*

For $\sum a_i(x, u)u_{x_i} = b(x, u)$, the characteristic vector is $A = (a, b)$. If $a \neq 0$, relabel coordinates so $a_1 \neq 0$. Use the hypersurface $x_1 = x_1^0$ as a transversal and label points on it by $(x_2^0, \ldots, x_n^0, u^0)$. Following the local characteristic flow defines smooth coordinates $(s, x_2^0, \ldots, x_n^0, u^0)$ by the inverse function theorem. In these coordinates $A = \partial_s$ and integral hypersurfaces that are graphs take the form $u^0 = U(s, x_2^0, \ldots, x_n^0)$ with $U_s = 0$.

For a solution graph through the reference point, its intersection with the transversal is a graph in $(x_2^0, \ldots, x_n^0, u^0)$; flowing it gives the required representation locally. The proviso about graphs is necessary for a change of coordinates mixing dependent and independent variables.

### 11.6/1: Local existence and uniqueness for noncharacteristic data

*Source: printed page 136.*

Parametrize the initial hypersurface by $x = \gamma(z)$, $z \in \mathbb{R}^{n-1}$, and prescribe $u = \phi(z)$. Solve the characteristic system $X_s = a(X, U)$, $U_s = b(X, U)$ with $(X, U)(0, z) = (\gamma(z), \phi(z))$. The derivative of $(s, z) \mapsto X(s, z)$ at the initial point has columns $a$ and the tangent vectors $\partial_{z_j}\gamma$. Noncharacteristicity says these columns form a basis. Thus this map is locally invertible, and $u(x) = U(s(x), z(x))$ is smooth. Its graph is tangent to $(a, b)$, so it solves the PDE and the data. Any other solution with the same data contains the same characteristic segments by ODE uniqueness, and hence gives the same graph locally.

### 11.7/1: The characteristic system preserves the equation surface

*Source: printed page 136.*

For $F(x, p, y) = 0$, the characteristic system is

$$\dot{x} = F_p, \quad \dot{p} = -F_x - pF_y, \quad \dot{y} = p \cdot F_p.$$

The chain rule gives $\dot{F} = F_x \cdot F_p + F_p \cdot (-F_x - pF_y) + F_y(p \cdot F_p) = 0$. Thus a characteristic starting on $E = \{F = 0\}$ remains on $E$ throughout its interval of existence.

### 11.7/2: Solution jets and characteristics

*Source: printed page 137.*

Let $u$ be a $C^2$ solution and consider its first-jet graph $j^1u = (x, Du(x), u(x))$. Differentiating $F(x, Du, u) = 0$ gives

$$F_x + D^2u\, F_p + (Du)F_y = 0.$$

The tangent vector to $j^1u$ above $\dot{x} = F_p$ is therefore $(F_p, D^2u\, F_p, Du \cdot F_p) = (F_p, -F_x - pF_y, p \cdot F_p)$. The characteristic field is tangent to the jet graph, so uniqueness makes its short trajectories stay in that graph. Conversely, if the jet graph is made of characteristics of the equation, these characteristics lie in $E$ by definition, so $F(x, Du, u) = 0$ pointwise.

**Review note.** Tangency to the displayed vector field alone, without membership in $E$, is not enough: it would only constrain which level of $F$ the jet graph occupies. The phrase "characteristics of the equation" includes the condition $F = 0$.

### 11.7/3: Projection to quasilinear characteristics

*Source: printed page 137.*

Write a quasilinear equation as $F(x, p, y) = a(x, y) \cdot p - b(x, y) = 0$. Then $\dot{x} = F_p = a(x, y)$ and, on $E$, $\dot{y} = p \cdot F_p = a \cdot p = b$. The projection to $(x, y)$ is exactly the characteristic system $(\dot{x}, \dot{y}) = (a, b)$ of the quasilinear equation.

### 11.7/4*: Coordinate invariance of characteristics

*Source: printed page 137.*

Let $\alpha = dy - \sum_i p_i\, dx_i$ be the contact form. Its restriction to each contact plane $C = \ker \alpha$ has the nondegenerate skew form $d\alpha = \sum_i dx_i \wedge dp_i$. On $E = \{F = 0\}$, set $K = TE \cap C$. Since $F_p \neq 0$, $K$ is a hyperplane in the symplectic vector space $C$. The kernel of $d\alpha|_K$ is a one-dimensional line and is intrinsic to $E$ and the contact planes.

The displayed characteristic vector $X_F$ lies in $K$: $\alpha(X_F) = 0$ and $dF(X_F) = 0$. For any $Z \in K$,

$$d\alpha(X_F, Z) = F_p \cdot Z_p + (F_x + pF_y) \cdot Z_x = dF(Z) = 0,$$

where $Z_y = p \cdot Z_x$ was used. Thus $X_F$ spans this intrinsic line.

A change $(x, y) \mapsto (X, Y)$ lifts to jet coordinates wherever a transformed graph remains a graph, defining $P$ by $dY - P \cdot dX = \lambda(dy - p \cdot dx)$ with $\lambda \neq 0$. It carries contact planes to contact planes, and on them the exterior derivatives differ only by the nonzero factor $\lambda$. It therefore carries the kernel line above to the corresponding kernel line on the transformed equation surface. Its unparametrized characteristic curves are preserved. Finally replacing $F$ by $hF$, $h \neq 0$, changes $X_F$ on $E$ to $hX_F$, so it changes speed but not characteristics. This proves both requested invariances, and indeed the contact-transformation version as well.

### 11.7/5: Distance to a smooth plane curve

*Source: printed page 137.*

Near a point of a smooth embedded curve use tubular coordinates $x = \gamma(s) + rn(s)$, with $s$ arclength and $n$ the unit normal. In a sufficiently small tube the nearest point is $\gamma(s)$ and the distance is $u = |r|$. Since $n \cdot \partial_s x = 0$ and $n \cdot \partial_r x = 1$, one has $\nabla r = n$. Off the curve, $\nabla u = \operatorname{sgn}(r) n$, so $u_{x_1}^2 + u_{x_2}^2 = 1$. The unsigned distance is not differentiable on the curve itself, which is why it is excluded.

### 11.7/6: Distance to a smooth submanifold

*Source: printed page 137–138.*

In a tubular neighborhood where the nearest-point projection $\pi(x)$ to the embedded submanifold is unique and smooth, write $\nu = x - \pi(x)$, $u = |\nu| > 0$. The vector $\nu$ is perpendicular to the tangent space containing $D\pi(x)h$. Therefore

$$Du(x)h = \frac{\nu \cdot (h - D\pi(x)h)}{|\nu|} = \frac{\nu \cdot h}{|\nu|}.$$

It follows that $\nabla u = \nu/|\nu|$ and $|\nabla u|^2 = 1$. This is a local tubular-neighborhood assertion, excluding the submanifold and any cut locus; a uniform tube need not exist for a noncompact submanifold.

### 11.7/7: Every smooth eikonal solution is locally a distance plus a constant

*Source: printed page 138.*

Suppose $|\nabla u| = 1$. Differentiation gives $D^2 u \, \nabla u = 0$. Along an integral curve of $\nabla u$, the gradient is consequently constant, so the curve is a straight unit-speed line, and $u$ increases along it at rate one. A nearby regular level hypersurface $S = \{u = c\}$ is orthogonal to these lines. The tubular-neighborhood theorem then identifies their parameter with signed distance from $S$.

To obtain the literal unsigned distance on a whole neighborhood of the chosen point $x_0$, choose $c < u(x_0)$ sufficiently close and restrict the neighborhood to the side $u > c$. There $u(x) = c + \operatorname{dist}(x, S)$. Choosing $S$ through $x_0$ instead would give signed distance and would not equal unsigned distance on both sides.

### 11.7/8: Hamilton–Jacobi characteristics

*Source: printed page 138.*

If $F = H(x, p)$ is independent of $y$, the characteristic equations reduce to $\dot{x} = H_p$, $\dot{p} = -H_x$, $\dot{y} = p \cdot H_p$. Their $(x, p)$ projections are precisely Hamiltonian trajectories, and 11.7/1 ensures they remain on $H = 0$. Conversely any such Hamiltonian trajectory lifts by integrating $\dot{y} = p \cdot \dot{x}$ with an arbitrary initial value of $y$.

# 12 Conservative systems with one degree of freedom (§12)

Throughout this section write $y = \dot{x}$ and $H(x, y) = y^2/2 + U(x)$. A level is drawn by taking both graphs $y = \pm\sqrt{2(E - U(x))}$ over $\{U \le E\}$. Its upper half moves right and its lower half moves left. A noncritical turning point joins these halves smoothly; an equilibrium is a separate orbit even when it belongs to a separatrix level. The drawings following the solutions implement this rule.

### 12.3/1: Pendulum, center, and saddle energy levels

*Source: printed page 142.*

For the pendulum, $U = -\cos x$ and $y^2 = 2(E + \cos x)$. There are no levels for $E < -1$. At $E = -1$ there are the centers $(2k\pi, 0)$. For $-1 < E < 1$, each potential well contains a closed oval. At $E = 1$ the level has the saddles $((2k + 1)\pi, 0)$ and the connecting separatrix arcs $y = \pm 2|\cos(x/2)|$. For $E > 1$ the upper and lower graphs extend across the whole real axis without turning. On the cylinder $x$ mod $2\pi$ these are periodic rotations; on the plane their lifts are not closed.

For $\ddot{x} = -x$, the levels are circles $x^2 + y^2 = 2E$ (or the point zero when $E = 0$). For $\ddot{x} = x$, they are hyperbolas $y^2 - x^2 = 2E$. At $E = 0$ the two lines $y = \pm x$ consist of the equilibrium and four separate separatrix rays. Figure 1 sketches all three cases.

### 12.3/2: Kepler levels and the four potential profiles

*Source: printed page 142.*

For the radial Kepler potential $U = -1/x + C/x^2$ take the physical domain $x > 0$ and $C > 0$. Its unique minimum is at $x = 2C$ with value $-1/(4C)$; $U \to +\infty$ as $x \downarrow 0$ and $U \to 0^-$ as $x \to \infty$. Thus:

- Below $-1/(4C)$ the level is empty; at that value it is the equilibrium.
- For $-1/(4C) < E < 0$, it is a closed oval between two positive turning points, the roots of $Ex^2 + x - C = 0$.
- For $E = 0$, it has one turning point $x = C$, and two branches tending to $y = 0$ at infinity, joined at that turning point.
- For $E > 0$, it has one positive turning point and branches tending to $y = \pm\sqrt{2E}$ at infinity.

The exact equation is $y^2 = 2E + 2/x - 2C/x^2$. If $C = 0$, $U = -1/x$ is monotone: levels with $E < 0$ turn once and run into the collision boundary $x = 0$; nonnegative levels have no finite turning point. For $C < 0$, the potential is likewise strictly increasing from $-\infty$ to 0 and has the same qualitative alternatives, with a stronger collision singularity.

For the four schematic profiles of Figure 96, the corresponding sketches are in Figure 3. In profile I, a center and a saddle bound one well; the saddle level has a homoclinic loop on the well side and escape branches on the other side. Profile II also has one saddle and one center, but the finite-height tails change which branches escape as the energy passes a tail height. Profile III has, in order, a center, a lower saddle, a second center, and a higher saddle. Crossing the lower saddle energy joins the two wells; crossing the higher one opens escape to the right. Profile IV has two centers separated

by one saddle; its unequal tail heights control which outer branches escape. For each profile the full answer at any specified energy is obtained by the two square-root graphs above every allowed interval. The source specifies shapes rather than formulas, so the plotted numerical heights and horizontal scales are schematic, not additional data attributed to the book.

### 12.4/1: Tangents to saddle separatrices

*Source: printed page 144.*

At a nondegenerate maximum $\xi$, expand

$$U(x) = U(\xi) + \tfrac{1}{2}U''(\xi)(x-\xi)^2 + o((x-\xi)^2).$$

On the separatrix level $E = U(\xi)$ this gives $y^2 = -U''(\xi)(x-\xi)^2 + o((x-\xi)^2)$. Therefore the two tangent lines are $y = \pm\sqrt{-U''(\xi)}(x-\xi)$. They are also the stable and unstable eigendirections of the linearized system.

### 12.5/1: Completeness for positive many-body potential energy

*Source: printed page 145.*

For $m_i\ddot{x}_i = -\partial_i U$, $m_i > 0$, on all of $\mathbb{R}^N$, energy conservation gives

$$E = \sum_i \frac{m_i\dot{x}_i^2}{2} + U(x), \qquad |\dot{x}_i| \le \sqrt{2E/m_i}$$

when $U > 0$. Hence on every finite time interval each coordinate and velocity stays bounded. The phase point lies in a compact subset of the smooth equation's full phase space, so the extension theorem excludes a finite maximal endpoint. Every solution is complete. A lower bound $U \ge B$ suffices by replacing $E$ with $E - B$; no upper bound on $U$ or its derivatives is required.

### 12.9/1: Time graphs near the pendulum separatrix

*Source: printed page 149.*

Let $E = 1 - \delta$, $0 < \delta \ll 1$. The turning angle $A$ obeys $\cos A = -E$, so $A = \pi - \sqrt{2\delta} + O(\delta^{3/2})$. Starting at $x = 0$ with positive velocity, the pendulum crosses the bottom rapidly, then spends a long time near $A$, returns rapidly through zero, and waits near $-A$. The angle graph has long rounded plateaus near $\pm A$, while $y(t)$ has alternating narrow positive and negative pulses, vanishing at the turning points. The maximum speed is $\sqrt{2(E+1)} \to 2$.

The limiting positive separatrix passage is exactly

$$x_s(t) = 4\arctan(e^t) - \pi, \qquad y_s(t) = 2\operatorname{sech} t.$$

Its negative counterpart describes the return passage. Finite-energy oscillations concatenate well-separated approximations to these passages; Figure 9 shows numerical quadrature of a near-critical periodic motion.

### 12.9/2: The logarithmic divergence of the pendulum period

*Source: printed page 149.*

Write $k = \sin(A/2) = \sqrt{(E+1)/2}$. Substituting $\sin(x/2) = k \sin\phi$ into the period quadrature gives

$$T(E) = 4 \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Set $\eta = \sqrt{1-k^2} = \sqrt{(1-E)/2}$. Only the endpoint $\phi = \pi/2$ is singular. With $s = \pi/2 - \phi$, its denominator is $\sqrt{\eta^2 + (1-\eta^2)\sin^2 s}$. On a fixed small interval, replacing $\sin s$ by $s$ changes the integral by a bounded amount uniformly in $\eta$; this follows from $\sin s = s + O(s^3)$. The model integral is $\operatorname{arsinh}(c/\eta) = \log(1/\eta) + O(1)$. The remainder away from the endpoint stays bounded. Consequently

$$\boxed{T(E) = 2 \log \frac{1}{1-E} + O(1)}, \qquad E \uparrow 1.$$

Thus the divergence is logarithmic, with positive sign. More precise evaluation gives $T(E) = 2\log(32/(1-E)) + o(1)$, but the coefficient and bounded remainder already answer the problem. The period is not proportional to a positive multiple of $\log(1-E)$, which tends to minus infinity.

### 12.9/3: Six oscillatory potentials

*Source: printed page 149.*

For each sign take $y^2 = 2(E - U(x))$; Figures 5–6 show both signs. Minima of $U$ are centers, nondegenerate maxima are saddles, and the corresponding critical energy levels are their separatrices.

1. $U = x \sin x$ is even, with a minimum at zero ($U''(0) = 2$). The remaining critical points solve $\sin x + x \cos x = 0$, or $\tan x = -x$. The alternating wells and barriers grow in height in absolute value. For any fixed finite energy there are distant forbidden barriers, so all allowed components are bounded, although there are infinitely many distant wells. The negative potential reverses centers and saddles and the barrier ordering.

2. $U = \sin x/x$, defined as 1 at zero, is even and tends to zero. Zero is a maximum, since $U = 1 - x^2/6 + O(x^4)$. The other extrema solve $x\cos x - \sin x = 0$, or $\tan x = x$, with decaying heights. For $E > 0$, all sufficiently distant points are allowed; for $E < 0$, only finitely many wells are accessible. At $E = 0$ there are infinitely many alternating allowed and forbidden intervals. Negating $U$ interchanges maxima and minima and changes which intervals are allowed. These statements concern accessibility; the square-root rule determines the exact branches.

3. $U = \sin(x^2)$ is even and lies between $-1$ and 1. Zero is a minimum. The other critical points satisfy $x^2 = \pi/2 + k\pi$, $k \geq 0$, and alternate maxima of height 1 and minima of height $-1$. For $-1 < E < 1$, allowed intervals give closed ovals; the central component is present only for $E \geq 0$, with a singleton at $E = 0$. At $E = 1$ all maxima lie on the separatrix level. For $E > 1$ there are two graphs over the entire line. The wells become narrower at large $|x|$. For the negative potential, zero becomes a maximum of height zero, and the roles of the other extrema reverse.

The exact energy equations, together with the rule at the start of this section, specify every level, including empty levels and equilibrium singletons not individually visible at the scale of a sketch.

### 12.9/4: Six oscillatory force laws

*Source: printed page 149.*

Integrate $U' = -F$, taking an irrelevant additive constant to be zero. For the plus force in each pair,

| $F(x)$ | $U(x)$ |
|---|---|
| $x \sin x$ | $x \cos x - \sin x$ |
| $\sin x / x$ | $-\displaystyle\int_0^x \frac{\sin s}{s}\,\mathrm{d}s$ |
| $\sin(x^2)$ | $-\displaystyle\int_0^x \sin(s^2)\,\mathrm{d}s$ |

Use the continuous value $F(0) = 1$ in the middle row. Negating the force negates the potential. The energy levels again follow from $y^2 = 2(E - U(x))$; Figures 7–8 show representative levels for all six cases.

For a simple zero $\xi$ of $F$, the equilibrium is a center if $F'(\xi) < 0$ and a saddle if $F'(\xi) > 0$. This gives a precise classification for the sketches:

- For $F = x \sin x$, the nonzero zeros are $\xi = k\pi$, and $F'(\xi) = k\pi(-1)^k$. At zero, $F = x^2 + O(x^4)$ and $U = -x^3/3 + O(x^5)$, so this is a degenerate equilibrium rather than a center or ordinary saddle.
- For $F = \sin x / x$, the zeros are $k\pi$, $k \neq 0$, and $F'(k\pi) = (-1)^k/(k\pi)$. Zero is not an equilibrium. The potential is bounded and approaches finite, generally different, limits at the two ends of the line; energies above its global maximum give two through-going branches.
- For $F = \sin(x^2)$, the nonzero zeros are $\xi = \pm\sqrt{k\pi}$, $k \geq 1$, with $F'(\xi) = 2\xi(-1)^k$. At zero, again $F = x^2 + O(x^6)$ and $U = -x^3/3 + O(x^7)$, giving a degenerate equilibrium. Its potential also approaches finite limits at infinity.

For each negative force, reverse the signs in this test. In particular the two signs exchange centers and saddles at every simple zero. At the degenerate origin the zero-energy level has a cusp with two branches on one side and the origin as a separate orbit; negating the force reflects which side is accessible.

### 12.10/1: Energy drift and cycles under a small perturbation

*Source: printed page 149–151.*

Consider $\dot{x} = y + \varepsilon f_1(x, y)$, $\dot{y} = -x + \varepsilon f_2(x, y)$. On a fixed annulus $0 < a \leq \sqrt{x^2 + y^2} \leq b < R$, small $\varepsilon$ preserves rotation and permits a smooth first-return map. With $H = (x^2 + y^2)/2$,

$$\dot{H} = \varepsilon(x f_1 + y f_2).$$

A return takes time $2\pi + O(\varepsilon)$ and its trajectory differs by $O(\varepsilon)$ from $(A\cos t, -A\sin t)$. Therefore

$$\Delta H = \varepsilon \mathcal{F}(A) + O(\varepsilon^2), \quad \mathcal{F}(A) = \int_0^{2\pi} (x f_1 + y f_2)(A\cos t, -A\sin t)\,\mathrm{d}t = \oint_{\text{counterclockwise}} (f_1\,dy - f_2\,dx).$$

The last orientation is important: the unperturbed motion itself is clockwise. For $\varepsilon > 0$, positive $\mathcal{F}$ gives outward drift and negative $\mathcal{F}$ inward drift, wherever the leading term is bounded away from zero. A simple positive root $A_0$ yields a unique nearby periodic orbit by applying the implicit function theorem to the return displacement divided by $\varepsilon$. Its radius on the transversal is $A_0 + O(\varepsilon)$. Since

$$P_\varepsilon(A) = A + \varepsilon \frac{\mathcal{F}(A)}{A} + O(\varepsilon^2),$$

this cycle is attracting if $\varepsilon \mathcal{F}'(A_0) < 0$ and repelling if it is positive. A multiple root needs higher-order analysis, and behavior arbitrarily near the origin needs a separate equilibrium analysis.

For the Van der Pol example, $f_1 = 0$, $f_2 = y(1 - x^2)$, so

$$\mathcal{F}(A) = A^2 \int_0^{2\pi} \sin^2 t \, \mathrm{d}t - A^4 \int_0^{2\pi} \sin^2 t \cos^2 t \, \mathrm{d}t = \pi(A^2 - A^4/4).$$

Its simple positive root is $A_0 = 2$, and $\mathcal{F}'(2) = -4\pi$. Thus small positive $\varepsilon$ produces an attracting limit cycle near the circle of radius two, provided the domain contains that circle and a neighborhood. For negative $\varepsilon$ the stability reverses.

## 12.10/2*: Sustained rotations of a pendulum with weak friction and torque

*Source: printed page 151.*

Write the equation on the cylinder as

$$\dot{x} = y, \qquad \dot{y} = M - \sin x - \varepsilon y, \qquad \varepsilon > 0.$$

Changing $(x, y, M)$ to $(-x, -y, -M)$ reverses the torque, so it suffices to study $M \geq 0$. We give the small-friction regimes, an exact shooting characterization of the transition, and the leading transition formula. A rotation is a periodic orbit on the cylinder, not a periodic angle on its real-line cover.

**Resting states and the absence of periodic librations.** For $0 \leq M < 1$, equilibria occur at $x_c = \arcsin M$ and $x_s = \pi - \arcsin M$ modulo $2\pi$, with $y = 0$. The eigenvalues obey $\lambda^2 + \varepsilon\lambda + \cos x_* = 0$. Thus $x_c$ is an attracting equilibrium (a focus for $\varepsilon^2 < 4 \cos x_c$) and $x_s$ is a saddle. At $M = 1$ they merge in a saddle-node; for $M > 1$ there is no equilibrium.

On the real-line cover the tilted energy

$$W = \tfrac{1}{2} y^2 - \cos x - Mx$$

satisfies $\dot{W} = -\varepsilon y^2$. Thus there is no nonconstant periodic motion with zero winding number: integrating over a period would give $\int y^2 dt = 0$. Any self-sustained periodic motion must rotate. In particular at $M = 0$ no rotation can persist either, since ordinary energy strictly decreases off rest.

**The weak-torque scale and the rotation balance.** Set $M = \varepsilon\mu$. For the untilted pendulum energy $H = y^2/2 - \cos x$, one has $\dot{H} = \varepsilon(\mu y - y^2)$. On an unperturbed positive rotation of energy $E > 1$, $y = \sqrt{2(E + \cos x)}$. During one revolution,

$$\Delta H = \varepsilon\{2\pi\mu - I(E)\} + O(\varepsilon^2), \qquad I(E) = \int_{-\pi}^{\pi} \sqrt{2(E + \cos x)} \, \mathrm{d}x.$$

This expansion is uniform on compact energy intervals strictly above 1. Here

$$I(1) = \int_{-\pi}^{\pi} 2\cos(x/2)\,\mathrm{d}x = 8, \quad I'(E) = \int_{-\pi}^{\pi} \frac{dx}{\sqrt{2(E+\cos x)}} > 0,$$

and $I(E) \to \infty$. Hence for each fixed $\mu > 4/\pi$ there is exactly one $E_\mu > 1$ solving $I(E_\mu) = 2\pi\mu$. The return-map implicit function theorem gives a nearby attracting rotation for sufficiently small $\varepsilon$, with energy $E_\mu + O(\varepsilon)$ on a fixed transversal. Its leading period is $I'(E_\mu)$. The restoring sign follows because $2\pi\mu - I(E)$ decreases strictly with $E$.

**Separatrix transition and retrapping torque.** The rotation is born through a saddle connection near the pendulum separatrix, where the preceding regular-return expansion is not uniform. Parametrize the positive unperturbed separatrix by $x_s(t) = 4\arctan e^t - \pi$, $y_s(t) = 2\operatorname{sech} t$. The first-order difference in energy between the unstable and stable saddle branches is

$$\varepsilon \int_{-\infty}^{\infty} (\mu y_s - y_s^2)\,\mathrm{d}t = \varepsilon(2\pi\mu - 8).$$

Indeed $\int y_s dt = 2\pi$ and $\int y_s^2 dt = 8$. Smooth dependence of the saddle manifolds on parameters, measured on a transversal away from the saddle, makes this the first variation of their splitting. Its derivative in $\mu$ is $2\pi \neq 0$, so the implicit function theorem gives a transition torque

$$\boxed{M_r(\varepsilon) = \frac{4}{\pi}\varepsilon + o(\varepsilon)}.$$

At this torque, the saddle's unstable branch joins its next translate on the cover (a homoclinic loop on the cylinder). The rotation period tends to infinity there. The formula is asymptotic, not an exact threshold for finite damping.

For an exact implicit specification at a chosen small $\varepsilon$, integrate the positive branch of

$$y\frac{dy}{dx} = M - \sin x - \varepsilon y$$

from the outgoing saddle at $x_s$ and compare it on a transverse line with the incoming branch of the saddle at $x_s + 2\pi$. The value $M = M_r$ makes the branches agree, with $y \to 0$ at both saddles. The local slope at the outgoing endpoint is the positive root of $\lambda^2 + \varepsilon\lambda + \cos x_s = 0$, so the singular endpoint can be initialized by this slope or by the smooth time-parametrized system.

**Why the transition gives the phase portrait.** Increasing $M$ at fixed $\varepsilon$ increases the right-hand side of the $y(x)$ equation on $y > 0$. The outgoing saddle branch moves toward passage over the next barrier rather than return into the well; at equality it is the saddle connection. Above the connection a rotating return orbit exists, whereas below it the outgoing branch is captured by the next well. The same comparison on the positive transverse section gives an interval as the domain of rotating returns. Every fixed point of its return map has multiplier

$$P' = \exp\left(\int_0^T \operatorname{div}(y, M - \sin x - \varepsilon y)\,\mathrm{d}t\right) = e^{-\varepsilon T} < 1.$$

Consequently every rotation is attracting, and there can be at most one: two attracting simple fixed points on an interval would require an intervening fixed point where the return displacement

crosses in the opposite direction. This also gives the stability without averaging. A negative rotating periodic orbit is impossible for $M > 0$, since the exact energy balance over a period would be $-2\pi M = \varepsilon \int y^2 dt > 0$. Contractible cycles have already been excluded.

For sufficiently small positive friction the resulting regimes are:

- $0 \le M < M_r$: no sustained rotation; apart from saddle trajectories, motions settle into a resting well.
- $M = M_r$: the saddle connection is the threshold orbit and is not a finite-period oscillation.
- $M_r < M < 1$: an attracting resting state and an attracting rotating cycle coexist. The saddle's stable manifold separates their basins. This is hysteresis: starting from rest and reducing the torque from a rotating state need not produce the same outcome.
- $M > 1$: there is no resting state and a unique attracting positive rotation. At $M = 1$ the resting saddle-node remains as a nonhyperbolic exception while the rotating attractor already exists for small friction.

For $M > 1$, existence can also be seen directly from the trapping horizontal strip $(M-1)/\varepsilon \le y \le (M+1)/\varepsilon$: the vector field points inward and $x$ advances, so a transverse return map has a fixed point. For a fixed $0 < M < 1$ and $\varepsilon \to 0$, the rotating state is likewise well above the threshold and coexists with rest. Its large speed is $y = M/\varepsilon + O(\varepsilon/M)$ and its period is asymptotic to $2\pi\varepsilon/M$. The formulas for $M = \varepsilon\mu$ describe the different, near-threshold scale. Negative torque gives the reflected portrait. No claim about large-friction bifurcations is needed for this small-friction exercise.

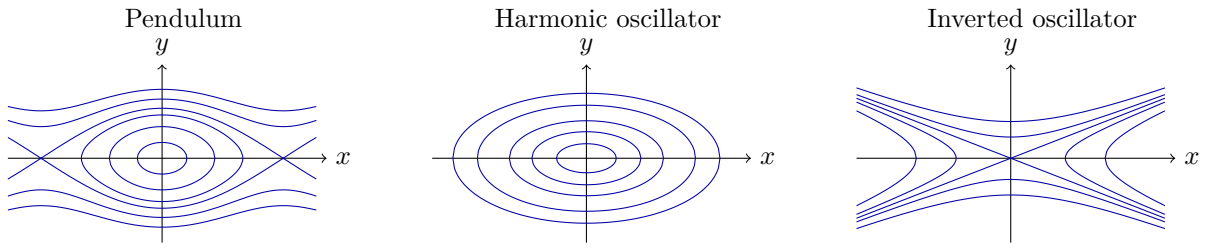



Figure 1: Energy levels for 12.3/1 (axes independently scaled). Upper branches move right and lower branches move left; the middle portrait is a center and the right portrait a saddle.

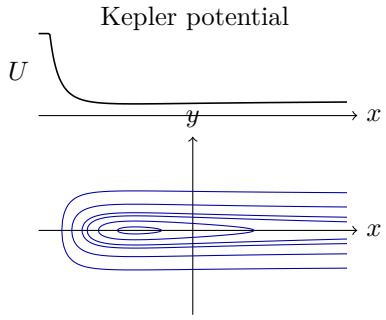



Figure 2: Radial Kepler levels with C=1. The black upper curve is the potential; blue curves below are phase levels. The horizontal plotting window is x from 0.18 to 6. The singular wall at zero lies outside the window.




Figure 3: Schematic reconstructions of Figure 96, profiles I and II: potential above, energy levels below. Tail shapes are shown only over a finite window.




Figure 4: Schematic reconstructions of Figure 96, profiles III and IV. Each local minimum produces a center; each local maximum produces a saddle separatrix.

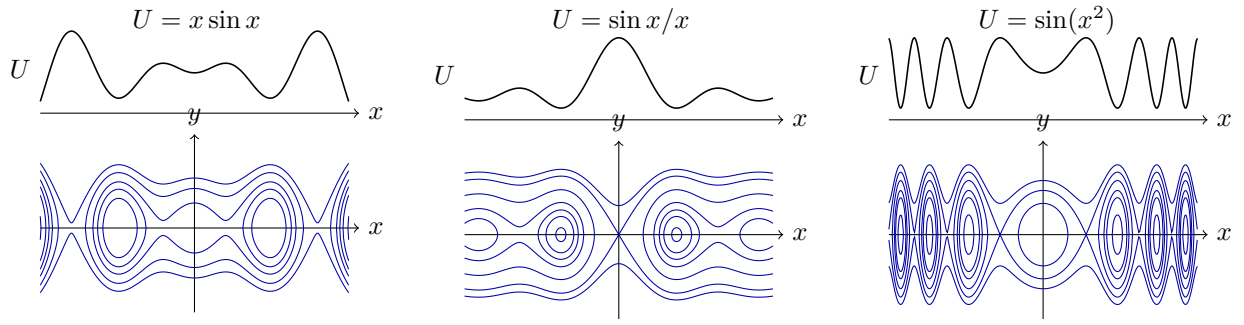



Figure 5: Representative levels for 12.9/3, plus potentials. The upper black curves show the potentials; the lower blue curves show energy levels. Windows are respectively $|x| \leq 10$, 12, and 4.5.

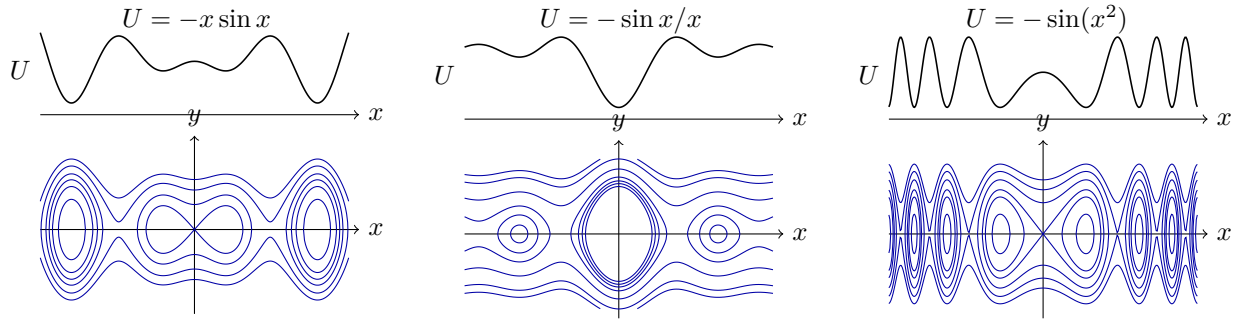



Figure 6: Representative levels for 12.9/3, minus potentials. The upper black curves show the potentials; the lower blue curves show energy levels. Windows are respectively $|x| \leq 10$, 12, and 4.5.

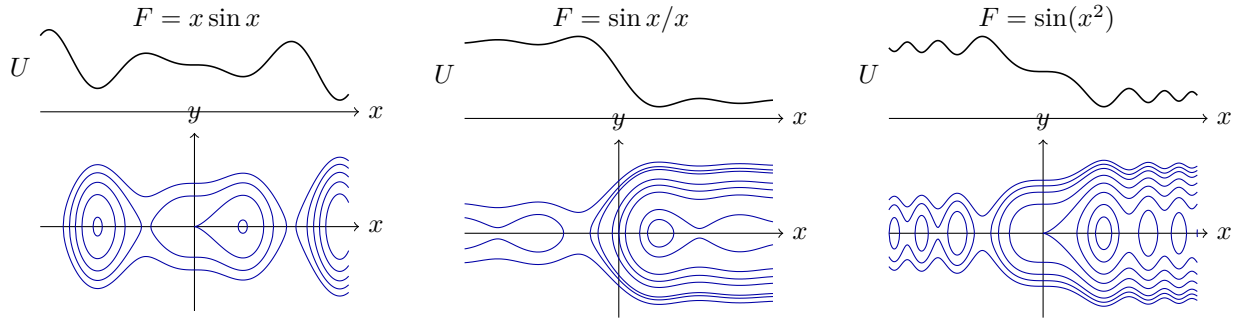



Figure 7: Representative levels for 12.9/4, plus force laws. The black curves are the integrated potentials, not the forces. Windows are respectively $|x| \leq 10$, 12, and 4.5.

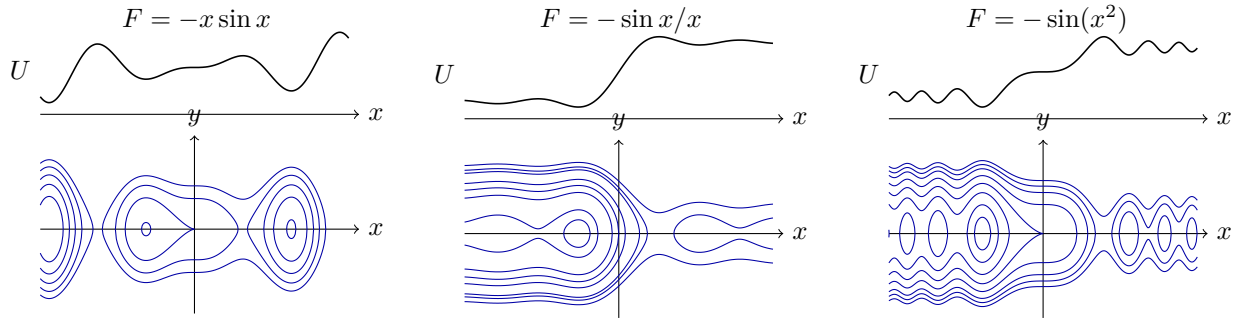



Figure 8: Representative levels for 12.9/4, minus force laws. The black curves are the integrated potentials, not the forces. Windows are respectively $|x| \leq 10$, 12, and 4.5.

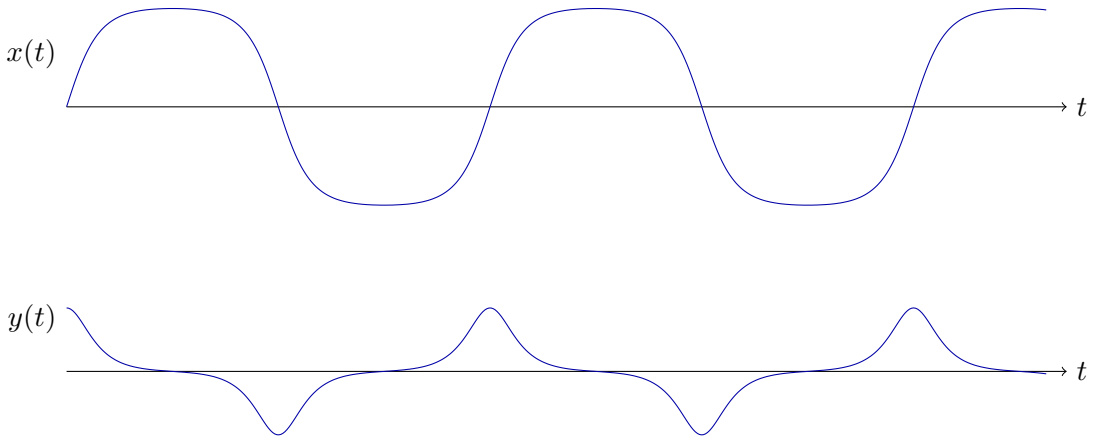



Figure 9: Near-separatrix pendulum motion at $E = 0.999$ (12.9/1), over $0 \le t \le 48$. Long angle plateaus alternate with rapid transitions and velocity pulses. The drawing is obtained by numerical integration; the proof and limiting formula are given in the solution.

## Chapter 2: review guide and coverage

| Book section | Subsections containing numbered problems | Number |
|---|---|---|
| §7 | 1, 3, 4, 5, 6, 7 | 22 |
| §8 | 1, 2, 3, 4, 5 | 17 |
| §9 | 1, 2, 3 | 4 |
| §10 | 1, 3, 4, 5, 6, 7 | 20 |
| §11 | 1, 2, 3, 4, 5, 6, 7 | 23 |
| §12 | 3, 4, 5, 9, 10 | 10 |
| Chapter 2 | Explicitly numbered problems | 96 |
| Chapters 1–2 | Combined total | 204 |

All subparts of the numbered exercises are included. The starred problems are 7.1/4, 10.3/2, 11.7/4, and 12.10/2. Figures are original drawings; numerical sampling is used only to illustrate formulas established in the text. The source is self-contained: the drawings are included as TikZ paths and require no separate image files.

**Suggested review order.** Start with 7.1/4–7 (global and time-preserving straightening), 7.3/5 (the valid long-time scale), 8.5/1–3 (implicit equations and envelopes), 9.3/1 (closed subgroups), and 10.3/2 (derivations). Then check 10.4/8 (the bracket sign), 10.5/1 (the planar hypothesis), 11.4/1 (the mismatch between the printed data and illustration), and 11.7/4 (contact invariance). Finish with 12.9/2 (the logarithmic period) and 12.10/2 (the asymptotic, rather than exact, retrapping threshold).

**Source.** Vladimir I. Arnold, *Ordinary Differential Equations*, third edition, translated by Roger Cooke, Springer-Verlag, 1992, Chapter 2, pp. 89–151. Exercise labels and equations were checked against the library scan. These solutions are an independent worked companion and are intended for mathematical review.

# Chapter 3 — Linear Systems

**Guide.** The problems below follow the numbering in §§13–29. We write $I$ for the identity, $A^T$ for the transpose, and $J_m(\lambda) = \lambda I + N_m$ for the Jordan block with ones just above the diagonal. Equivalence of flows preserves time. Statements and figures are summarized or redrawn rather than reproduced.

## 13 Linear problems (Section 13)

### 13.1/1: Coordinate independence of linearization

*Source: printed page 153.*

Let $v(0) = 0$, and change coordinates by $y = h(x)$. The new field is $w(h(x)) = Dh(x)v(x)$. Differentiating at zero gives

$$Dw(h(0))Dh(0) = Dh(0)Dv(0),$$

because the term containing $D^2h$ is multiplied by $v(0) = 0$. Thus the two matrices are conjugate by $Dh(0)$. They represent the same linear map on the tangent space at the equilibrium.

### 13.1/2: Linearizing the pendulum

*Source: printed page 153.*

The equilibria of $\dot{x} = y$, $\dot{y} = -\sin x$ are $(k\pi, 0)$. Put $x = k\pi + \xi$, $y = \eta$. The linearization is

$$\dot{\xi} = \eta, \qquad \dot{\eta} = -(-1)^k \xi.$$

For even $k$ this is a harmonic oscillator with eigenvalues $\pm i$; for odd $k$ it is a saddle with eigenvalues $\pm 1$.

### 13.2/1: The generator of a linear group

*Source: printed page 154.*

Let $A = (d/dt)g^t|_{t=0}$. The group law gives

$$\frac{g^{t+h} - g^t}{h} = \frac{g^h - I}{h} g^t \longrightarrow A g^t.$$

Hence $x(t) = g^t x_0$ satisfies $\dot{x} = Ax$, $x(0) = x_0$. Interchanging the factors also gives $Ag^t = g^t A$.

### 13.2/2: The velocity field of a rotation

*Source: printed page 154.*

For an oriented angular-velocity vector $\omega$ and a point $o$ on the axis, the field is $v(x) = \omega \times (x - o)$. The axial component is fixed; on the perpendicular plane the velocity is tangent to the circle, with speed $|\omega|$ times its radius. With origin $o$, this is a skew-symmetric linear operator.

## 14 The exponential (Section 14)

### 14.1/1: Finiteness of the operator norm

*Source: printed page 155.*

For a linear operator on a finite-dimensional Euclidean space, $x \mapsto \|Ax\|$ is continuous on the compact unit sphere. Its maximum exists and is finite and nonnegative. This maximum is $\|A\|$. Moreover, $\|A\| = 0$ precisely when $A = 0$.

### 14.1/2: Norm inequalities

*Source: printed page 155.*

Taking suprema over $\|x\| = 1$ proves $\|\lambda A\| = |\lambda|\|A\|$ and $\|A + B\| \le \|A\| + \|B\|$. The bound $\|Ay\| \le \|A\|\|y\|$, also valid at $y = 0$, gives $\|ABx\| \le \|A\|\|B\|\|x\|$, hence $\|AB\| \le \|A\|\|B\|$.

### 14.1/3: Bounds in terms of matrix entries

*Source: printed page 156.*

In an orthonormal basis, $\|Ae_j\|^2 = \sum_i |a_{ij}|^2 \le \|A\|^2$. Conversely, Cauchy–Schwarz in each row gives

$$\|Ax\|^2 = \sum_i \left| \sum_j a_{ij} x_j \right|^2 \le \sum_{i,j} |a_{ij}|^2 \|x\|^2.$$

Thus $\max_j \sum_i |a_{ij}|^2 \le \|A\|^2 \le \sum_{i,j} |a_{ij}|^2$.

### 14.2/1: Dimension of the operator space

*Source: printed page 156.*

The $n^2$ elementary matrices $E_{ij}$, each with one nonzero entry equal to one, form a basis: $A = \sum a_{ij} E_{ij}$ uniquely. Therefore $\dim \mathrm{End}(\mathbb{R}^n) = n^2$.

### 14.3/1: Entrywise convergence and completeness

*Source: printed page 157.*

The preceding bounds show that norm convergence is equivalent to convergence of every entry in an orthonormal basis. In any other fixed basis the matrix is $S^{-1}AS$; multiplication by fixed matrices is continuous, so the equivalence persists. A norm-Cauchy sequence has Cauchy entries, which converge in $\mathbb{R}$ (or $\mathbb{C}$). Their matrix limit is also its norm limit. Thus the operator space is complete.

### 14.4/1: Uniform convergence and differentiation of series

*Source: printed page 158.*

Suppose $\|f_j(t)\| \le M_j$ and $\sum M_j < \infty$. For all $t$,

$$\left\| \sum_{j=p}^{q} f_j(t) \right\| \le \sum_{j=p}^{q} M_j.$$

Completeness gives a pointwise limit, and the same estimate makes convergence uniform.

For the differentiation assertion, work on a compact interval, assume each $f_j$ differentiable, $\sum f_j(t_0)$ convergent, and $\sum f_j'$ uniformly convergent to $g$. Write $S_N = \sum_{j \le N} f_j$. The mean-value inequality bounds

$$\|(S_M - S_N)(t) - (S_M - S_N)(t_0)\| \le |t - t_0| \sup_s \|S_M'(s) - S_N'(s)\|.$$

Consequently $S_N$ converges uniformly to a function $S$. The same inequality, on the interval between $t$ and $t + h$, bounds the difference quotient of $S - S_N$ by $\sup_s \|g(s) - S_N'(s)\|$. First choose $N$ to make this small, then let $h \to 0$ in the finite sum. This proves $S'(t) = g(t)$. This argument does not require continuity of the individual derivatives.

### 14.5/1: Four matrix exponentials

*Source: printed page 158–159.*

Direct substitution in $e^{tA} = \sum_{j \ge 0} t^j A^j / j!$ gives, in the order listed,

$$\begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix}, \quad \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}, \quad \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The first follows entrywise; the second has $A^2 = 0$; the third has $A^2 = -I$; the fourth has $A^3 = 0$ and $A^2$ has only its upper-right entry nonzero.

### 14.6/1: The polynomial space

*Source: printed page 159.*

The monomials $1, x, \ldots, x^{n-1}$ span the polynomials of degree less than $n$, and uniqueness of polynomial coefficients makes them independent. The dimension is $n$.

### 14.6/2: Differentiation on polynomials

*Source: printed page 159.*

Differentiation is linear. Its kernel is the constants, and its image is all polynomials of degree less than $n-1$, since each has a polynomial antiderivative of degree less than $n$. For $n = 1$ the image is zero.

### 14.6/3: Translation on polynomials

*Source: printed page 159.*

$H^t p(x) = p(x+t)$ is linear and preserves the degree bound. Since $H^{-t}H^t = I$, its kernel is zero and its image is the entire polynomial space. Taylor's finite formula also proves $H^t = e^{tD}$, where $D = d/dx$.

### 14.7/1: A nondiagonalizable operator

*Source: printed page 160.*

Take $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. It is nonzero but $N^2 = 0$, so every eigenvalue is zero. If it were diagonalizable, its diagonal form would be the zero matrix, a contradiction.

### 14.7/2: Real diagonalization

*Source: printed page 160.*

A diagonal matrix in a real basis has real diagonal entries, and these entries are precisely its eigenvalues, counted with multiplicity. Thus diagonalizability over $\mathbb{R}$ requires a real spectrum.

### 14.7/3: Distinct real eigenvalues

*Source: printed page 160.*

Choose an eigenvector $v_j$ for each of the $n$ distinct real eigenvalues $\lambda_j$. In a relation $\sum c_j v_j = 0$, apply $\prod_{k \neq j}(A - \lambda_k I)$ to obtain $c_j \prod_{k \neq j}(\lambda_j - \lambda_k) v_j = 0$. Thus every $c_j = 0$. These eigenvectors form a real diagonalizing basis.

### 14.8/1: Strictly triangular matrices are nilpotent

*Source: printed page 161.*

A nonzero contribution to $(A^k)_{ij}$ for a strictly upper-triangular matrix requires a chain $i < i_1 < \cdots < i_{k-1} < j$. No chain of length $n$ exists among $n$ indices, so $A^n = 0$. In particular, the displayed $2 \times 2$ block squares to zero.

### 14.8/2: Nilpotence of differentiation

*Source: printed page 161.*

On polynomials of degree less than $n$, $D^n = 0$. Since $D^{n-1}x^{n-1} = (n-1)! \neq 0$, the nilpotency index is $n$ (including $n = 1$).

### 14.8/3: Exponentiating a nilpotent Jordan block

*Source: printed page 161.*

For $N_n$, $(N_n^k)_{ij} = 1$ if $j - i = k$ and is zero otherwise. Hence

$$(e^{tN_n})_{ij} = \begin{cases} t^{j-i}/(j-i)!, & j \geq i, \\ 0, & j < i. \end{cases}$$

The basis $1, x, x^2/2!, \ldots, x^{n-1}/(n-1)!$ identifies $N_n$ with differentiation and the exponential with translation.

### 14.9/1: A space of quasi-polynomials

*Source: printed page 161.*

Multiplication by the nowhere-zero function $e^{\lambda x}$ is a linear isomorphism from the polynomials of degree less than $n$ onto the specified space. A basis is $e^{\lambda x}, xe^{\lambda x}, \ldots, x^{n-1}e^{\lambda x}$, and its dimension is $n$.

### 14.9/2: Uniqueness of coefficients

*Source: printed page 161–162.*

For the fixed exponent $\lambda$, an identity $e^{\lambda x} \sum c_j x^j = 0$ implies $\sum c_j x^j = 0$ after division by the exponential. All coefficients vanish. Thus the polynomial factor, and its coefficients, are unique.

### 14.9/3: The exponential of a Jordan block

*Source: printed page 162.*

In the basis $e^{\lambda x}x^j/j!$, differentiation has matrix $J_n(\lambda)$. Equivalently, $\lambda I$ commutes with $N_n$, giving

$$e^{tJ_n(\lambda)} = e^{\lambda t} \sum_{j=0}^{n-1} \frac{t^j N_n^j}{j!}.$$

In particular, $\exp\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right) = e\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$.

# 15 Properties of the exponential (Section 15)

### 15.1/1: When the exponential addition rule fails

*Source: printed page 164.*

Let $A = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$ and $B = A^T$. Then

$$e^A e^B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad e^{A+B} = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}.$$

They differ. If $AB = BA$, however, the binomial theorem and absolute convergence of the exponential series prove $e^{A+B} = e^A e^B$.

### 15.1/2: Invertibility

*Source: printed page 164.*

$A$ commutes with $-A$, so $e^A e^{-A} = e^0 = I$. Therefore $e^A$ is invertible and its determinant is nonzero.

### 15.1/3: Skew-symmetric generators

*Source: printed page 164.*

Transposing the power series gives $(e^A)^T = e^{A^T}$. If $A^T = -A$, then $(e^A)^T e^A = e^{-A} e^A = I$, so $e^A$ is orthogonal.

### 15.3/1: Uniqueness of the generator

*Source: printed page 165.*

If $g^t = e^{tA}$, differentiation at $t = 0$ gives $A = (g^t)'|_0$. Thus no other operator can generate the same differentiable one-parameter group.

### 15.5/1: Complex multiplication as a real matrix

*Source: printed page 166.*

Multiplication by $u + iv$ sends $x + iy$ to $(ux - vy) + i(vx + uy)$. Its matrix in the basis $(1, i)$ is $\left(\begin{smallmatrix}u&-v\\v&u\end{smallmatrix}\right)$.

### 15.5/2: Modulus and argument near one

*Source: printed page 166.*

For $z = a + ib$ and large $n$, the real part of $1 + z/n$ is positive. Taylor expansion yields

$$|1 + z/n| = \sqrt{1 + 2a/n + |z|^2/n^2} = 1 + a/n + O(n^{-2}),$$

$$\arg(1 + z/n) = \arctan \frac{b/n}{1 + a/n} = b/n + O(n^{-2}).$$

These imply the requested $o(n^{-1})$ estimates.

### 15.5/3: Five exponential values

*Source: printed page 167.*

Euler's formula gives $e^0 = 1$, $e^1 = e$, $e^i = \cos 1 + i \sin 1$, $e^{\pi i} = -1$, and $e^{2\pi i} = 1$.

### 15.5/4: The complex addition law

*Source: printed page 167.*

The absolutely convergent product series can be grouped by total degree:

$$e^z e^w = \sum_{n \geq 0} \sum_{j=0}^{n} \frac{z^j w^{n-j}}{j!(n-j)!} = \sum_{n \geq 0} \frac{(z+w)^n}{n!} = e^{z+w}.$$

### 15.5/5: Series for sine and cosine

*Source: printed page 167.*

Separating the even and odd powers in $e^{it}$ and comparing real and imaginary parts gives

$$\cos t = \sum_{j \geq 0} \frac{(-1)^j t^{2j}}{(2j)!}, \qquad \sin t = \sum_{j \geq 0} \frac{(-1)^j t^{2j+1}}{(2j+1)!}.$$

Absolute convergence justifies the separation for every real $t$.

### 15.6/1: Uniform convergence of Euler polygons

*Source: printed page 168.*

Fix $T > 0$, put $h = T/N$, and let $x_{j+1} = (I + hA)x_j$. For the exact solution $u(t) = e^{tA}x_0$, Taylor's formula on $[0, T]$ gives $u((j+1)h) = (I + hA)u(jh) + r_j$, with $\|r_j\| \leq Ch^2$. Thus the errors satisfy

$$\|e_{j+1}\| \leq (1 + h\|A\|)\|e_j\| + Ch^2, \quad e_0 = 0,$$

and iteration gives $\max_{j \leq N} \|e_j\| \leq CTe^{T\|A\|}h$. Between vertices, the polygon and the exact curve differ by the vertex error plus $O(h)$, uniformly. Hence the polygons converge uniformly on $[0, T]$. Reverse time to obtain the same conclusion on any compact interval.

# 16 The determinant of the exponential (Section 16)

### 16.1/1: Determinant and oriented volume

*Source: printed page 170.*

If a parallelepiped has edge matrix $B$, its oriented volume is $\det B$. Its image under $A$ has edge matrix $AB$, hence volume $\det(AB) = \det A \det B$. The ratio is $\det A$, independently of the initial basis. For singular $A$ the image has zero volume.

### 16.2/1: Trace and determinant from the spectrum

*Source: printed page 170.*

Over $\mathbb{C}$, factor $\det(tI - A) = \prod_{j=1}^{n}(t - \lambda_j)$. Comparing the coefficient of $t^{n-1}$ and the constant term gives $\operatorname{tr} A = \sum_j \lambda_j$ and $\det A = \prod_j \lambda_j$, with algebraic multiplicities. Diagonalizability is unnecessary.

### 16.4/1: Contraction of volume

*Source: printed page 173.*

If every eigenvalue has negative real part, the real number $\operatorname{tr} A = \sum_j \operatorname{Re} \lambda_j$ is negative. The flow multiplies volumes by $\det e^{tA} = e^{t \operatorname{tr} A} < 1$ for $t > 0$. This assertion concerns volume, not monotonic decrease of every Euclidean distance.

### 16.4/2: Eigenvalues of an exponential

*Source: printed page 173.*

Triangularize $A$ over $\mathbb{C}$. Each power, and hence its exponential, is upper triangular; the diagonal entries of $e^A$ are $e^{\lambda_j}$. Consequently its eigenvalues are these numbers, including multiplicities, and $\det e^A = \prod_j e^{\lambda_j} = e^{\operatorname{tr} A}$.

# 17 Distinct real eigenvalues (Section 17)

### 17.2/1: Three overdamped pendulum motions

*Source: printed page 175–176.*

For $\ddot{x} + (10/3)\dot{x} + x = 0$,

$$x(t) = Ce^{-t/3} + De^{-3t}, \quad C = (9x_0 + 3y_0)/8, \quad D = -(x_0 + 3y_0)/8.$$

In Fig. 116 all three marked starting points have $x_0 < 0$, $y_0 > 0$. Curve II is the fast eigenline $y = -3x$, so $C = 0$ and $x$ increases exponentially to zero. Curve I lies to its left, so $C < 0$: it also approaches zero from below, but with the slow $e^{-t/3}$ tail. Its velocity cannot first cross from positive to negative while $x < 0$, since at $y = 0$ one has $\dot{y} = -x > 0$. Curve III has $C > 0$ and $D < 0$: it crosses $x = 0$, reaches one positive maximum, and then decreases to zero. A zero or a critical point

is determined by a linear equation in $e^{-8t/3}$, so there can be at most one of each. Figure 10 sketches representatives.

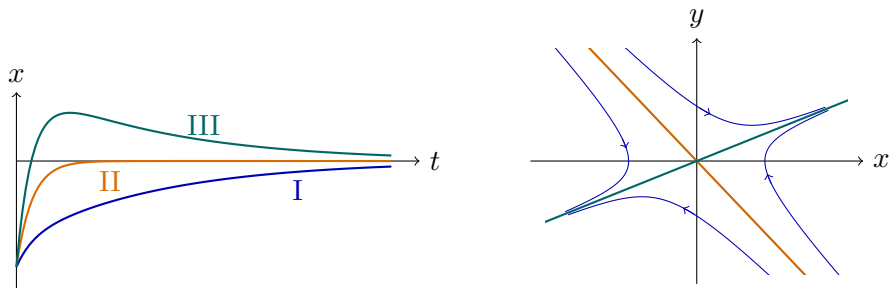
### 17.2/2: The inverted pendulum with friction

*Source: printed page 175.*

The equation $\ddot{x} = x - k\dot{x}$ has roots

$$\lambda_{\pm} = \frac{-k \pm \sqrt{k^2+4}}{2}, \qquad \lambda_- < 0 < \lambda_+.$$

Thus $x = C_+ e^{\lambda_+ t} + C_- e^{\lambda_- t}$, $y = \dot{x}$. The lines $y = \lambda_- x$ and $y = \lambda_+ x$ are respectively stable and unstable. All other orbits have both components and pass between these separatrices. The origin is a saddle for every real $k$; friction alone cannot stabilize the upper equilibrium.




Overdamped motion  Inverted pendulum ($k = 1$)

Figure 10: Left: the three motions in 17.2/1, with equal negative initial displacement. Right: saddle separatrices and representative trajectories in 17.2/2. Orange is the stable direction and teal is the unstable direction.

### 17.3/1: Fibonacci numbers

*Source: printed page 176.*

Let $\phi = (1+\sqrt{5})/2$ and $\psi = (1-\sqrt{5})/2$. Both satisfy $r^2 = r + 1$, so $(\phi^n - \psi^n)/\sqrt{5}$ satisfies the recurrence and the initial values $a_0 = 0$, $a_1 = 1$. Uniqueness of a recursively determined sequence proves

$$a_n = \frac{\phi^n - \psi^n}{\sqrt{5}} \sim \frac{\phi^n}{\sqrt{5}}, \qquad \lim_{n\to\infty} \frac{\log a_n}{n} = \log \phi.$$

# 18 Complexification and realification (Section 18)

### 18.1/1: The real matrix of a complex operator

*Source: printed page 177.*

Write $A = \alpha + i\beta$ in the specified complex bases. For $z = x + iy$,

$$Az = (\alpha x - \beta y) + i(\beta x + \alpha y).$$

In the real bases ordered as $(e_1, \ldots, e_m, ie_1, \ldots, ie_m)$ and similarly in the target, the matrix is $A_{\mathbb{R}} = \left(\begin{smallmatrix} \alpha & -\beta \\ \beta & \alpha \end{smallmatrix}\right)$.

### 18.1/2: Realification respects sums and products

*Source: printed page 177.*

Realification changes the allowed scalars but not the underlying map. Thus $(A+B)_{\mathbb{R}}z = A_{\mathbb{R}}z + B_{\mathbb{R}}z$ and $(AB)_{\mathbb{R}}z = A_{\mathbb{R}}(B_{\mathbb{R}}z)$ for every $z$. These prove both identities, with compatible domains for the product.

### 18.2/1: The complexified matrix

*Source: printed page 178.*

If $Ae_j = \sum_i a_{ij} f_i$, complexification keeps this identity and extends it complex-linearly. Hence the matrix of $A_{\mathbb{C}}$ is the same real array $(a_{ij})$, now acting on complex coordinates.

### 18.2/2: Complexification respects sums and products

*Source: printed page 178.*

Evaluate the operators on $x + iy$: $(A + B)_{\mathbb{C}}(x + iy) = (A + B)x + i(A + B)y$, and $(AB)_{\mathbb{C}}(x + iy) = ABx + iABy$. These equal $(A_{\mathbb{C}} + B_{\mathbb{C}})(x + iy)$ and $A_{\mathbb{C}}B_{\mathbb{C}}(x + iy)$, respectively.

### 18.3/1: The matrix of multiplication by i

*Source: printed page 178.*

Multiplication by $i$ sends $(x, y)$ to $(-y, x)$. Its real matrix is $\left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$, whose square is $-I$.

### 18.3/2: Linearity of the conjugate operator

*Source: printed page 179.*

By definition $\overline{A}z = \overline{A\bar{z}}$. For $c, d \in \mathbb{C}$,

$$\overline{A}(cz + dw) = \overline{A(\bar{c}\bar{z} + \bar{d}\bar{w})} = c\overline{A}z + d\overline{A}w.$$

Thus $\overline{A}$ is complex-linear, even though conjugation itself is only real-linear.

### 18.3/3: Conjugating matrix entries

*Source: printed page 179.*

For a real basis vector $e_j = \overline{e}_j$, conjugating $Ae_j = \sum_i a_{ij} e_i$ gives $\overline{A}e_j = \sum_i \overline{a_{ij}} e_i$. This is entrywise conjugation of the matrix.

### 18.3/4: Algebraic identities for conjugation

*Source: printed page 179.*

Entrywise conjugation commutes with addition and with matrix multiplication, and conjugates scalar factors. Therefore

$$\overline{A + B} = \overline{A} + \overline{B}, \qquad \overline{AB} = \overline{A}\,\overline{B}, \qquad \overline{\lambda A} = \bar{\lambda}\,\overline{A}.$$

### 18.3/5: Recognizing a complexification

*Source: printed page 179.*

A complexified real operator has a real matrix, hence equals its conjugate. Conversely, if $A = \overline{A}$, its entries in a real basis are real. Its restriction $B$ to the real subspace maps that subspace into itself, and complex linearity gives $A(x + iy) = Bx + iBy = B_{\mathbb{C}}(x + iy)$.

### 18.4/1: Exponential and change of scalars

*Source: printed page 179.*

Realification, complexification, and conjugation preserve powers and the identity. They are continuous real-linear operations on their finite-dimensional matrix spaces. Applying them term by term to the absolutely convergent exponential series gives

$$(e^A)_{\mathbb{R}} = e^{A_{\mathbb{R}}}, \qquad \overline{e^A} = e^{\overline{A}}, \qquad (e^B)_{\mathbb{C}} = e^{B_{\mathbb{C}}}.$$

### 18.4/2: Determinants

*Source: printed page 179.*

After complexifying the real block matrix of $A = \alpha + i\beta$, the coordinate change $(x, y) \mapsto (x+iy, x-iy)$ turns it into $\operatorname{diag}(A, \overline{A})$. Hence

$$\det A_{\mathbb{R}} = \det A \det \overline{A} = |\det A|^2.$$

The determinant polynomial has integer coefficients, so $\det \overline{A} = \overline{\det A}$. Complexification leaves a real matrix unchanged, giving $\det B_{\mathbb{C}} = \det B$.

### 18.4/3: Traces

*Source: printed page 179.*

The real block matrix has trace $2 \operatorname{tr} \alpha = \operatorname{tr} A + \overline{\operatorname{tr} A}$. Summing diagonal entries also gives $\operatorname{tr} \overline{A} = \overline{\operatorname{tr} A}$ and $\operatorname{tr} B_{\mathbb{C}} = \operatorname{tr} B$.

### 18.4/4: The complex determinant identity

*Source: printed page 179–180.*

Triangularize $A$ over $\mathbb{C}$. The exponential has diagonal entries $e^{\lambda_j}$, so $\det e^A = \prod_j e^{\lambda_j} = e^{\sum_j \lambda_j} = e^{\operatorname{tr} A}$. This proves the full complex identity, including its argument, not merely its absolute value.

### 18.5/1: Sum and product rules

*Source: printed page 180.*

The difference quotient of a sum is the sum of the difference quotients. For a product, write

$$\frac{f(t+h)g(t+h) - f(t)g(t)}{h} = \frac{f(t+h) - f(t)}{h} g(t+h) + f(t) \frac{g(t+h) - g(t)}{h}.$$

Differentiability implies continuity, so the limit is $f'g + fg'$. The argument applies equally to complex-valued functions of real time.

### 18.5/2: Vector-valued differentiation

*Source: printed page 180.*

Realification, complexification, and conjugation commute with the difference quotient because its denominator $h$ is real; their continuity lets them pass through its limit. Addition is componentwise. Applying the preceding product argument to each coordinate gives $(f\varphi)' = f'\varphi + f\varphi'$. These prove all five identities.

# 19 Complex linear equations (Section 19)

There are no explicitly numbered problems in this section. The main formula needed below is $z(t) = e^{tA}z_0$. For a simple spectrum, $z(t) = \sum_j c_j e^{\lambda_j t} v_j$. For $\lambda = \alpha + i\omega$, one has $r(t) = r_0 e^{\alpha t}$ and $\theta(t) = \theta_0 + \omega t$: rays when $\omega = 0$, circles when $\alpha = 0 \neq \omega$, and logarithmic spirals when $\alpha\omega \neq 0$.

# 20 Complexifying real equations (Section 20)

### 20.5/1: The nonlinear pendulum with friction

*Source: printed page 192.*

Write $\dot{x} = y$, $\dot{y} = -\sin x - ky$, with $k > 0$. Its energy satisfies

$$E = \tfrac{1}{2}y^2 + 1 - \cos x, \qquad \dot{E} = -ky^2 \leq 0.$$

The equilibria $(2m\pi, 0)$ are sinks: foci for $0 < k < 2$ and nodes for $k \geq 2$ (a repeated eigenvalue at $k = 2$). The points $((2m+1)\pi, 0)$ are saddles. Above the horizontal axis trajectories move right; below it they move left. Energy decreases except at instantaneous turning points, excluding nonconstant periodic orbits and saddle connections between equal-energy saddles. Each saddle's unstable branches descend into the adjacent potential wells. Its stable branches form basin boundaries. Initial rotation loses energy and ultimately gives way to oscillation, except on the stable manifolds approaching an upper equilibrium. On the cylinder, compactness of energy sublevels and $\dot{E} = -ky^2$ imply that every forward limit lies in $y = 0$ and then in $\sin x = 0$. The resulting phase portrait is shown in Fig. 11. For $k = 0$ the energy levels are the familiar conservative curves; negative $k$ reverses the dissipative behavior under time reversal and $y \mapsto -y$.

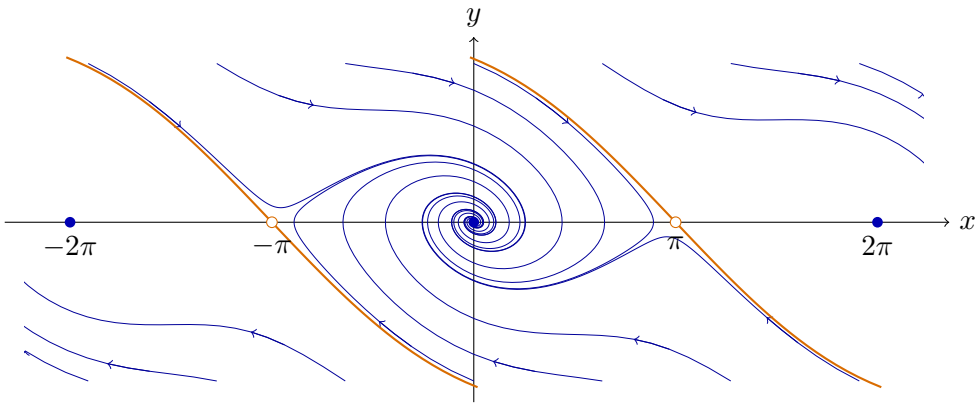



Figure 11: Damped nonlinear pendulum with $k = 1/2$. Blue trajectories lose energy and settle into wells. Orange curves are stable manifolds of the upper saddles. The portrait repeats under $x \mapsto x + 2\pi$.

### 20.6/1: Three scalar equations with simple roots

*Source: printed page 194.*

The three real general solutions, with arbitrary real constants, are

$$\begin{aligned} x^{(4)} + 4x = 0: &\quad x = e^t(C_1 \cos t + C_2 \sin t) + e^{-t}(C_3 \cos t + C_4 \sin t), \\ x^{(4)} = x: &\quad x = C_1 e^t + C_2 e^{-t} + C_3 \cos t + C_4 \sin t, \\ \dddot{x} + x = 0: &\quad x = C_1 e^{-t} + e^{t/2}\left(C_2 \cos \frac{\sqrt{3}t}{2} + C_3 \sin \frac{\sqrt{3}t}{2}\right). \end{aligned}$$

The characteristic roots are respectively $\{1+i, 1-i, -1+i, -1-i\}$, $\{1, -1, i, -i\}$, and $\{-1, (1+i\sqrt{3})/2, (1-i\sqrt{3})/2\}$. Distinctness and the initial-value dimension show that these lists exhaust the solutions.

**Review note.** Embedded question on p. 190: a repeated real eigenvalue with one eigenvector gives $\dot{u} = \lambda u + v$, $\dot{v} = \lambda v$, hence $(u, v) = e^{\lambda t}(u_0 + tv_0, v_0)$. This produces the sheared, degenerate node. A scalar matrix $\lambda I$ instead gives straight radial trajectories.

## 21 Equivalence and classification (Section 21)

### 21.1/1: Four degenerate three-dimensional portraits

*Source: printed page 196.*

Choose coordinates adapted to the invariant line and plane in Fig. 143. These give the following complete parametrizations and sketches in Fig. 12.

6. For a negative real root $a$ and a pair $\pm i\omega$, use $z = z_0 e^{at}$ and $(u, v) = r(\cos(\omega t + \theta), \sin(\omega t + \theta))$. For $r > 0$ trajectories wind on circular cylinders toward the periodic orbit in $z = 0$; on the axis they approach the origin.

7. For $a < 0$ and $a \pm i\omega$, use $z = z_0 e^{at}$ and $(u, v) = r_0 e^{at}(\cos(\omega t + \theta), \sin(\omega t + \theta))$. The ratio $r/|z|$ is constant when $z \neq 0$, so curves spiral on cones toward the vertex. The invariant plane contains spirals and the axis contains rays.

8. For 0 and $a \pm i\omega$ with $a > 0$, $z = z_0$ while $r = r_0 e^{at}$. Each plane $z = z_0$ contains outward spirals around its equilibrium $(0, 0, z_0)$; the entire axis is fixed.

9. For $a < 0$, 0, $b > 0$, the solutions are $(u, z, v) = (u_0 e^{at}, z_0, v_0 e^{bt})$. Every plane $z = z_0$ contains a saddle centered on the fixed axis. Its stable and unstable lines are $v = 0$ and $u = 0$.

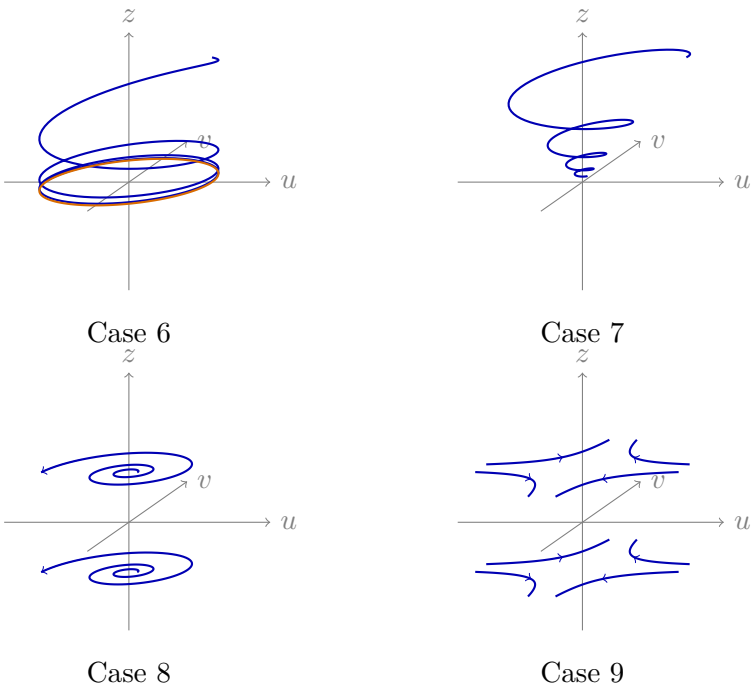



Figure 12: Projected representatives of the four cases in Fig. 143, expressed in invariant coordinates. Case 6 approaches a periodic circle; case 7 approaches a vertex on a cone; case 8 spirals away within a fixed plane; case 9 is a family of planar saddles.

### 21.2/1: Implications between equivalences

*Source: printed page 197.*

A linear isomorphism is a diffeomorphism, and a diffeomorphism is a homeomorphism. The same conjugacy equation $hf^t = g^t h$ is retained, proving both implications.

### 21.2/2: Orbit preservation is insufficient

*Source: printed page 197–198.*

For $f^t(x) = e^t x$ and $g^t(x) = e^{2t}x$, the orbits are the same two open rays and the origin. The identity therefore preserves every orbit. It does not conjugate the flows, since $e^t x \neq e^{2t}x$ for $x \neq 0$, $t \neq 0$. Indeed, no nonzero linear map $h(x) = cx$ can do so.

### 21.2/3: Equivalence relations

*Source: printed page 198.*

The identity conjugates each flow to itself. If $hf^t = g^t h$, then $h^{-1}g^t = f^t h^{-1}$. If also $kg^t = \ell^t k$, then $(kh)f^t = \ell^t(kh)$. Each of the three classes of admissible maps is closed under inverse and composition. Thus reflexivity, symmetry, and transitivity hold.

### 21.3/1: Equal eigenvalues do not suffice

*Source: printed page 198.*

The two generators are $I$ and $I + N_2$. Every conjugate of $I$ is $I$, whereas $N_2 \neq 0$. Hence they are not linearly equivalent, despite having the same eigenvalues. Equivalently their eigenspaces have dimensions two and one.

## 22 Topological classification (Section 22)

### 22.5/1: Openness of a fixed signature

*Source: printed page 204.*

Let a nondegenerate symmetric form have positive and negative subspaces $E_+$ and $E_-$. On their unit spheres it is bounded below by $c > 0$ and above by $-c < 0$, after reducing $c$ if necessary. A perturbation of operator norm less than $c$ remains positive on $E_+$ and negative on $E_-$. The maximal dimensions of positive and negative subspaces are therefore at least the original ones; since these dimensions already sum to $n$, they must be exactly the original ones, with no null direction. This proves openness.

### 22.5/2: Differentiation on quadratic forms

*Source: printed page 205.*

Write $q(x) = x^T P x$, $P = P^T$. Then

$$L_A q(x) = Dq(x)Ax = x^T(A^T P + PA)x.$$

The matrix on the right is symmetric and depends linearly on $P$. The space of symmetric matrices has $n(n+1)/2$ independent entries, proving the assertion about the induced linear operator.

### 22.5/3: The spectrum on quadratic forms

*Source: printed page 206.*

Triangularize the dual operator over $\mathbb{C}$, with diagonal entries $\lambda_1, \ldots, \lambda_n$. On the products of its basis linear forms the product rule gives a triangular matrix for $L_A$, with diagonal entries

$$\boxed{\lambda_i + \lambda_j, \qquad 1 \leq i \leq j \leq n.}$$

This argument includes nondiagonalizable $A$. If the dual basis is an eigenbasis, its products are actual eigenvectors. The restriction $i \leq j$ avoids double-counting; there are $n(n+1)/2$ entries.

### 22.5/4: Solving the quadratic-form equation

*Source: printed page 206.*

If no sum $\lambda_i + \lambda_j$ is zero (including $i = j$), then zero is absent from the spectrum of $L_A$, so it is invertible. Thus $L_A q = f$ has exactly one quadratic solution for each quadratic $f$. If all eigenvalues have strictly positive real parts, or all have strictly negative real parts, every sum has that same strict sign in its real part and is nonzero.

### 22.5/5: Positivity of the inverse

*Source: printed page 206.*

Assume $\operatorname{Re}\lambda_j > 0$ and $L_A q = f$ is positive-definite. Since $e^{tA}x \to 0$ as $t \to -\infty$, integration along the orbit gives

$$q(x) = \int_{-\infty}^{0} f(e^{tA}x)\,\mathrm{d}t.$$

Convergence follows from exponential decay with at most polynomial factors in the Jordan form. For $x \neq 0$ the integrand is positive at every finite time, proving $q(x) > 0$. In matrix notation the unique solution of $A^T P + PA = Q > 0$ is $P = \int_0^\infty e^{-tA^T} Q e^{-tA}\,\mathrm{d}t > 0$.

### 22.8/1: The incoming subspace

*Source: printed page 208.*

For the standard saddle, $g^t(x_-, x_+) = (e^{-t}x_-, e^t x_+)$. This tends to zero as $t \to +\infty$ precisely when $x_+ = 0$, that is, when $x$ belongs to the incoming subspace. The analogous backward-time condition characterizes the outgoing subspace.

### 22.8/2: Four inequivalent three-dimensional saddles

*Source: printed page 209.*

A conjugacy preserves the sets of points tending to the equilibrium forward and backward in time, and maps their individual orbits bijectively. A zero-dimensional incoming set consists only of the equilibrium; a one-dimensional one consists of the equilibrium and two nonconstant orbits; an incoming set of dimension at least two contains infinitely many nonconstant orbits. For $(m_-, m_+) = (3,0), (2,1), (1,2), (0,3)$, the ordered pairs of these orbit counts are respectively $(\infty, 0), (\infty, 2), (2, \infty), (0, \infty)$, counting only nonconstant orbits. All differ. No theorem on invariance of dimension is needed.

### 22.8/3: Discrete hyperbolic operators

*Source: printed page 209.*

Here equivalence means $hA = Bh$, not equivalence of their continuous-time generators. First suppose $A$ invertible and has no eigenvalues of modulus one. Decompose $\mathbb{R}^n = E_s \oplus E_u$ according to $|\lambda| < 1$

and $|\lambda| > 1$. A complete list of invariants is

$$\dim E_s, \quad \dim E_u, \quad \operatorname{sgn}\det(A|_{E_s}), \quad \operatorname{sgn}\det(A|_{E_u}),$$

with sign $+1$ in dimension zero. Thus dimensions alone, sufficient for flows, are insufficient for maps: $x \mapsto x/2$ and $x \mapsto -x/2$ are not conjugate.

*Necessity.* The stable and unstable sets are characterized by $A^k x \to 0$ and $A^{-k} x \to 0$. A conjugacy maps each onto its counterpart. Their dimensions and the orientation signs of the restricted homeomorphisms are invariant under conjugation.

*Sufficiency for contractions.* Write a real invertible contraction as $A = Re^L$, where $R$ is an involution commuting with $L$ and all eigenvalues of $L$ have negative real part. To construct this, take $R = -I$ on the real generalized eigenspaces of negative real eigenvalues and $R = I$ on the other real blocks. Then $RA$ has a real logarithm: use the real logarithm series on positive Jordan blocks, and conjugate branches on each nonreal pair. Its eigenvalues have real parts $\log|\lambda| < 0$.

Choose a positive-definite Lyapunov form for $L$ and average it with its pullback by $R$. The resulting form is $R$-invariant and strictly decreases under $e^{tL}$. Every nonzero orbit meets its unit ellipsoid $S$ once. The map

$$h(e^{tL}u) = e^{-t}u \quad (u \in S), \qquad h(0) = 0$$

is a homeomorphism by the exponential Lyapunov bounds, and it commutes with $R$. It conjugates $A$ to $e^{-1}R$. Identify the ellipsoid with a round sphere, so $R$ becomes an orthogonal involution.

If two such involutions $R, S$ have the same determinant, there is a path $K : [0,1] \to SO(n)$ from $I$ to $SR$. Extend it to all real $s$ by $K(s-1) = SK(s)R$. The endpoint condition makes this continuous. Then

$$H(ru) = rK(\log r)u \quad (r > 0,\ \|u\| = 1), \qquad H(0) = 0$$

is a homeomorphism, with inverse using $K(\log r)^{-1}$, and $H(e^{-1}Rx) = e^{-1}SH(x)$. This proves classification of contractions by dimension and determinant sign. Apply it to $A^{-1}|_{E_u}$ for expansions, and take products.

*If singular operators are included in the question*, add the sizes of all nilpotent Jordan blocks at zero. Indeed, the invariant core $\bigcap_{k\geq 0} \operatorname{im} A^k$ carries the invertible part; its stable and unstable invariants are as above. The dimensions $d_k = \dim\ker A^k$ are preserved by a conjugacy, and $d_k - d_{k-1}$ counts the zero-eigenvalue blocks of size at least $k$. They determine the nilpotent part up to linear conjugacy. Conversely, combine that linear conjugacy with the homeomorphism of the invertible parts. This also handles the literal wording, which does not exclude zero eigenvalues.

# 23 Stability of equilibria (Section 23)

### 23.1/1: Five stability tests

*Source: printed page 211.*

For $\dot{x} = 0$, every equilibrium is stable because all solutions are constant. For $\dot{x} = x$, the origin is unstable because $x_0 e^t$ eventually leaves any fixed neighborhood. For $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1$, the conserved radius gives stability. For $\dot{x}_1 = x_1$, $\dot{x}_2 = -x_2$, any nonzero first component grows, so the origin is unstable. Finally, for the nonlinear pendulum the lower equilibria $(2k\pi, 0)$ are stable because the energy has a strict local minimum there; the upper equilibria $((2k+1)\pi, 0)$ are unstable saddles. For example, a displacement from an upper equilibrium with zero velocity rolls away, reaching a fixed smaller angle with nonzero speed.

### 23.1/2: Coordinate independence of stability

*Source: printed page 211.*

Let $h$ be a coordinate homeomorphism taking the equilibrium to the new origin. Given a target neighborhood, continuity of $h$ supplies a smaller original neighborhood mapped inside it. Original stability supplies an initial neighborhood whose trajectories remain there; continuity of $h^{-1}$ supplies an initial neighborhood in the new coordinates. This is exactly stability in the new coordinates. Reversing $h$ proves the converse. The argument is local, so all neighborhoods may be chosen within the coordinate charts.

### 23.1/3: Arbitrarily large amplification can coexist with stability

*Source: printed page 211.*

No. For $\ddot{x} = -x^3$, the conserved energy $E = y^2/2 + x^4/4$ has a strict minimum, giving Lyapunov stability. Start at $(x, y) = (0, a)$ with $a > 0$. At the first positive turning point, energy conservation gives $x = 2^{1/4}\sqrt{a}$, $y = 0$. The norm ratio is $2^{1/4}/\sqrt{a}$, arbitrarily large as $a \to 0$, although the maximum displacement itself tends to zero. Choose $a < \varepsilon$ and $a < \sqrt{2}/N^2$ to obtain the requested inequalities.

### 23.2/1: The corresponding asymptotic-stability questions

*Source: printed page 211.*

None of the equilibria in 23.1/1 is asymptotically stable: the unstable ones already fail stability, while the stable ones have neighboring constant or nonconstant periodic solutions. Coordinate independence follows from 23.1/2 and preservation of limits by a homeomorphism.

The amplification condition still does not rule out asymptotic stability. Consider $\dot{x} = y$, $\dot{y} = -x^3 - y^3$. Here $E = y^2/2 + x^4/4$ satisfies $\dot{E} = -y^4$. Its sublevels are compact, and the only trajectory contained in $\{y = 0\}$ is the origin; hence the origin is asymptotically stable. To justify the last step directly, any limit point generates a complete trajectory in the limit set on which $E$ is constant, forcing $y = 0$ and then $x = 0$.

Nevertheless, start at $(0, a)$ and put $x = \sqrt{a}X$, $y = aY$, $s = \sqrt{a}\,t$. Then $X' = Y$, $Y' = -X^3 - a^{3/2}Y^3$, with $(X, Y)(0) = (0, 1)$. As $a \to 0$, continuous dependence on a fixed $s$-interval gives the undamped solution, which reaches $X = 1/2$. Thus for small $a$ the original solution reaches at least $x = \sqrt{a}/3$, and amplification is at least $1/(3\sqrt{a})$. This is unbounded.

### 23.2/2: Attraction without stability

*Source: printed page 211.*

No. Here is a smooth planar counterexample with every solution tending to zero. On the punctured plane use polar coordinates and first consider the auxiliary time $s$:

$$\frac{dr}{ds} = r\sin\theta - r^3, \qquad \frac{d\theta}{ds} = 1 - \cos\theta.$$

Multiply both right-hand sides by $\eta(r) = e^{-1/r^2}$ to obtain the actual $t$-time field, and set it equal to zero at the origin. In Cartesian coordinates all possible factors $r^{-j}$ are dominated by $\eta$ and its derivatives, so this extension is $C^\infty$. Multiplication by a positive factor away from zero preserves oriented trajectories.

On the ray $\theta = 0$, $dr/ds = -r^3$, so $r \to 0$. For $0 < \theta_0 < 2\pi$, $\theta \to 2\pi$, and solving the linear equation for $u = r^{-2}$ yields

$$\frac{(1-\cos\theta)^2}{r(\theta)^2} = \frac{(1-\cos\theta_0)^2}{r_0^2} + 2(\theta - \theta_0 - \sin\theta + \sin\theta_0).$$

The numerator on the left tends to zero while the right tends to a positive number, so $r \to 0$. Also $dr/ds \le r - r^3$, so trajectories stay bounded. The actual forward time is infinite as a nonconstant trajectory approaches the smooth equilibrium; equivalently $dt/ds = 1/\eta(r) \ge 1$.

Take $r_0 = \theta_0 = a \downarrow 0$. At $\theta = \pi$, the formula gives $r(\pi)^2 \to 2/\pi$. Thus arbitrarily small initial points make excursions of a fixed positive size before returning to zero. The equilibrium attracts every point but is not Lyapunov stable.

### 23.3/1: An unstable equilibrium with no positive linear eigenvalue

*Source: printed page 212.*

The scalar equation $\dot{x} = x^3$ has derivative zero at the origin, so its only linearized eigenvalue is zero. But $x(t) = x_0/\sqrt{1 - 2x_0^2 t}$ for $x_0 > 0$ grows and exits any prescribed small neighborhood. Thus the equilibrium is unstable.

### 23.3/2: A power-law restoring force

*Source: printed page 212.*

For the positive integer $n$, the energy is $E = y^2/2 + x^{n+1}/(n+1)$. If $n$ is odd, it has a strict minimum at zero and its nearby nonzero levels are closed curves: the origin is stable but not asymptotically stable. If $n$ is even, start with $x_0 < 0$, $y_0 = 0$. Then $\dot{y} = -x^n < 0$, so $x$ decreases and continues accelerating left. While $x \le x_0$, $\dot{y} \le -|x_0|^n$, forcing exit from a fixed neighborhood in finite time. Hence the origin is unstable. (For $n = 0$ it is not an equilibrium.)

### 23.4/1: Closing the continuation gap

*Source: printed page 213.*

The decay estimate was derived only while the solution remains in the neighborhood where the Lyapunov inequality holds. Choose a compact sublevel $\{V \le c\}$ contained in that neighborhood. Starting with $V < c$, the inequality $\dot{V} \le -\gamma V$ prevents exit through its boundary. A solution confined to this compact subset of the domain extends for all forward time. Now integration is legitimate for every $t \ge 0$, and $V(t) \le V(0)e^{-\gamma t}$ gives both stability and convergence.

### 23.4/2: Local conjugacy to the stable linearization

*Source: printed page 214.*

Choose a common positive quadratic form $V$ with

$$-bV \le L_v V \le -aV < 0, \qquad -bV \le L_{Ax} V \le -aV < 0$$

in a sufficiently small punctured sublevel, for some $b \ge a > 0$. This is possible because $v(x) = Ax + O(\|x\|^2)$ and $DV(x) = O(\|x\|)$. Every nonzero interior point lies uniquely on a forward trajectory from $S = \{V = c\}$: backward integration must reach $S$ in finite time by the same exponential bounds. If $f^t$ is the nonlinear flow and $g^t = e^{tA}$, define

$$h(f^t u) = g^t u \quad (u \in S,\ t \ge 0), \qquad h(0) = 0.$$

Transversality to $S$ makes the coordinates $(t, u)$ continuous with continuous inverse. Uniform exponential bounds show that $t \to \infty$ precisely as the point tends to zero, for both flows; hence $h$ and its inverse are continuous there too. Finally $hf^s = g^s h$ whenever the relevant orbit segment stays in these neighborhoods. This is the required local, time-preserving conjugacy.

## 24 Purely imaginary eigenvalues (Section 24)

### 24.1/1: Topological classification of planar centers

*Source: printed page 215.*

Every nonconstant orbit of a real planar system with eigenvalues $\pm i\omega$ has least period $2\pi/|\omega|$. A time-preserving conjugacy preserves the set of return times and thus this least period. Therefore two such systems can be conjugate only when their unordered eigenvalue pairs agree. Conversely, equal eigenvalue pairs give real linear conjugacy to the same rotation generator. Reversing the sign of $\omega$ is implemented by a reflection.

### 24.3/1: Repairing the irrational-rotation argument

*Source: printed page 219.*

The small rotation furnished by the pigeonhole principle must be nonzero modulo $2\pi$. This is exactly where irrationality is needed. For any $\varepsilon > 0$, find $s > 0$ and an integer $m$ with $0 < |s\alpha - 2\pi m| < \varepsilon$. Let the signed difference be $\beta$. The points $0, \beta, 2\beta, \ldots, \lfloor 2\pi/|\beta| \rfloor \beta$ modulo $2\pi$ form an $\varepsilon$-net, whether $\beta$ is positive or negative. They are all nonnegative multiples of $s\alpha$ modulo $2\pi$, hence belong to the forward orbit. Translating the net by the initial angle proves density.

### 24.4/1: Leading digits of powers of two

*Source: printed page 219.*

$\log_{10} 2$ is irrational: a rational equality $\log_{10} 2 = p/q$ would imply $2^q = 10^p$, impossible by prime factorization. Thus the fractional parts of $n \log_{10} 2$ are dense modulo one. A positive decimal block representing an integer $d$ with $k$ digits is the leading block of $2^n$ exactly when

$$\{n \log_{10} 2\} \in [\log_{10} d - (k-1),\ \log_{10}(d+1) - (k-1)).$$

This interval has positive length, so it is visited infinitely often. In particular, the digit 7 occurs. A block with an initial zero cannot be a leading block in the usual decimal notation.

### 24.4/2: A supremum attained only as a limit

*Source: printed page 219.*

The expression is at most two. The orbit $(t, \sqrt{2}t)$ modulo $2\pi$ is dense on the two-torus, so it comes arbitrarily close to $(0, \pi/2)$, where the continuous function $\cos\theta + \sin\varphi$ equals two. Therefore the supremum for $t > 0$ is two. Equality at a finite $t$ would require $t = 2\pi k$ and $\sqrt{2}t = \pi/2 + 2\pi m$, making $\sqrt{2} = (4m+1)/(4k)$; this is impossible.

### 24.5/1*: Density on a multidimensional torus

*Source: printed page 220.*

We prove the stronger assertion that the forward trajectory $\theta(t) = \theta_0 + \omega t$ is uniformly distributed on the torus. For any $k \in \mathbb{Z}^m \setminus \{0\}$, rational independence gives $k \cdot \omega \neq 0$, and

$$\frac{1}{T} \int_0^T e^{ik\cdot\theta(t)}\, \mathrm{d}t = e^{ik\cdot\theta_0} \frac{e^{i(k\cdot\omega)T} - 1}{iT(k\cdot\omega)} \longrightarrow 0.$$

The mean of the constant character is one. By linearity, time averages of trigonometric polynomials equal their normalized space integrals in the limit. Such polynomials approximate continuous periodic functions uniformly (for example by the multidimensional Fejer kernels), so the statement extends to all continuous functions. In any nonempty open set choose a nonnegative continuous function supported there with positive integral. Its time average is positive, forcing the trajectory to enter that set. Hence the trajectory is dense. Only nonzero-radius oscillator factors are included in the torus when some initial amplitudes vanish.

### 24.6/1*: Uniform distribution of an irrational rotation

*Source: printed page 220.*

For any nonzero integer $k$, the geometric-sum identity gives

$$\frac{1}{N}\sum_{j=0}^{N-1} e^{ik(\varphi+j\alpha)} = e^{ik\varphi}\frac{1-e^{ikN\alpha}}{N(1-e^{ik\alpha})} \longrightarrow 0,$$

since $\alpha/(2\pi)$ irrational makes the denominator nonzero. Uniform approximation by trigonometric polynomials gives convergence of averages for continuous functions. Bound an arc's indicator above and below by continuous functions differing only in arbitrarily short neighborhoods of its endpoints. Their integrals approach the arc's length divided by $2\pi$. Squeezing proves the stated visit-frequency formula.

### 24.6/2: The frequency ratio for leading 7 and 8

*Source: printed page 221.*

The corresponding intervals for the fractional logarithm have lengths $\log_{10}(8/7)$ and $\log_{10}(9/8)$. Hence

$$\lim_{N\to\infty}\frac{N_7(N)}{N_8(N)} = \frac{\log(8/7)}{\log(9/8)} > 1,$$

because $8/7 > 9/8 > 1$. Uniform distribution, rather than mere density, is what justifies this frequency comparison.

## 25 Multiple eigenvalues (Section 25)

### 25.3/1: Jordan form of a fourth-order equation

*Source: printed page 224.*

For $x^{(4)} - 2\ddot{x} + x = 0$, the state $(x, \dot{x}, \ddot{x}, \dddot{x})$ has companion matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 2 & 0 \end{pmatrix}.$$

Its characteristic polynomial is $(\lambda-1)^2(\lambda+1)^2$. For an eigenvalue $\lambda$, the first three rows force an eigenvector to be a multiple of $(1, \lambda, \lambda^2, \lambda^3)$, so each eigenspace is one-dimensional. The Jordan form is therefore $J_2(1) \oplus J_2(-1)$. More explicitly, $v(\lambda) = (1, \lambda, \lambda^2, \lambda^3)^T$ and $v'(\lambda)$ form a Jordan chain at each root, since $(A - \lambda I)v'(\lambda) = v(\lambda)$ there.

### 25.4/1: Every allowed quasi-polynomial is a solution

*Source: printed page 225.*

Yes. Factor the characteristic polynomial as $p(z) = \prod_j (z - \lambda_j)^{\nu_j}$. For $0 \le r < \nu_j$,

$$(D - \lambda_j)^{\nu_j}(t^r e^{\lambda_j t}) = e^{\lambda_j t} D^{\nu_j} t^r = 0.$$

All factors of $p(D)$ commute, so $p(D)$ annihilates each such function and every linear combination. This also explains why the bound on the polynomial degree is essential.

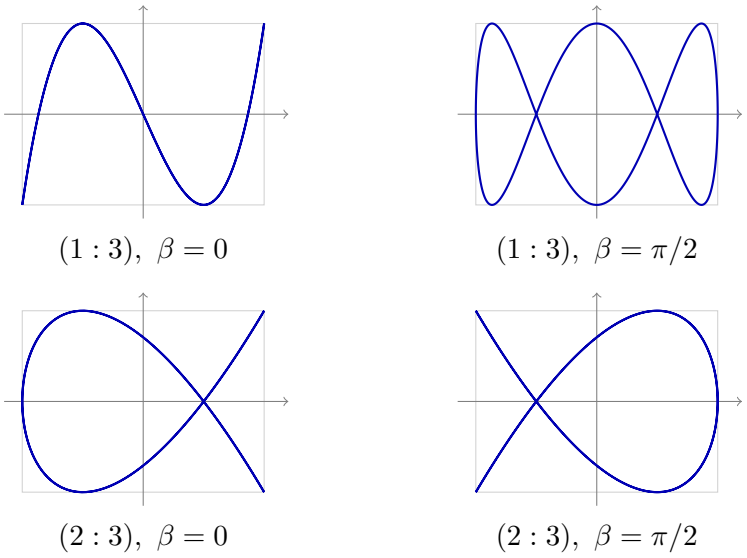
### 25.6/1: Two Lissajous families

*Source: printed page 229.*

After independent scalings of the axes, the two families are

$$(x_1, x_2) = (\cos(t + \alpha), \cos(3t + \beta)), \qquad (x_1, x_2) = (\cos(2t + \alpha), \cos(3t + \beta)).$$

The relative phase changes the curve; frequencies alone do not determine a unique drawing. Figure 13 gives representatives at two relative phases. With both phases zero, the first is the cubic graph $x_2 = 4x_1^3 - 3x_1$. For the second, $x_2^2 = (x_1 + 1)(2x_1 - 1)^2/2$, with $-1 \le x_1 \le 1$. Both parametrizations are $2\pi$-periodic; for nonzero amplitudes this is the least period of the parametrized motion.



$(1:3),\ \beta = 0$ $(1:3),\ \beta = \pi/2$

$(2:3),\ \beta = 0$ $(2:3),\ \beta = \pi/2$

Figure 13: Lissajous curves $(\cos nt, \cos(mt + \beta))$ for two frequency ratios and two relative phases. Frequencies alone do not fix the portrait.

### 25.6/2: Chebyshev polynomials

*Source: printed page 229.*

Take $x = \cos t$, $y = \cos nt$. Then $y = T_n(x)$, where the cosine addition identity gives

$$T_0 = 1, \quad T_1 = x, \quad T_{n+1} = 2xT_n - T_{n-1}.$$

Induction shows that $T_n$ is a polynomial of degree $n$, with leading coefficient $2^{n-1}$ for $n \geq 1$. As $\cos t$ covers $[-1, 1]$, the Lissajous curve traces its graph on that interval, generally back and forth.

### 25.6/3: An indefinite quadratic potential

*Source: printed page 229.*

For $U = x_1^2 - x_2^2$ the equations are $\ddot{x}_1 = -2x_1$ and $\ddot{x}_2 = 2x_2$. Thus

$$x_1 = A\cos(\sqrt{2}t) + B\sin(\sqrt{2}t), \qquad x_2 = Ce^{\sqrt{2}t} + De^{-\sqrt{2}t}.$$

The configuration curve stays in the strip $|x_1| \leq \sqrt{A^2 + B^2}$ and generally oscillates across it while escaping exponentially in $x_2$. If $C = 0$ it approaches the line $x_2 = 0$ forward in time while continuing to oscillate; if $C = D = 0$ it repeatedly traces a segment. These formulas include all exceptional cases and specify the curve without suggesting it must be a conic.

### 25.6/4: Stability for a quadratic mechanical system

*Source: printed page 229.*

The equilibrium $(x, \dot{x}) = (0, 0)$ is Lyapunov stable exactly when $U$ is positive-definite. In principal axes, positive eigenvalues of its Hessian give independent bounded oscillators. A negative eigenvalue gives an exponentially growing mode. A zero eigenvalue gives a free mode $x_j = c_j + d_j t$, which is unstable under arbitrarily small nonzero velocity. No such system in positive dimension is asymptotically stable: in the only stable case nonzero energy is conserved, so nearby nonzero solutions cannot converge to zero.

## 26 Quasi-polynomials (Section 26)

### 26.1/1: Independence of two sine functions

*Source: printed page 230.*

They are independent exactly when $\alpha\beta \neq 0$ and $\alpha^2 \neq \beta^2$ (for real or complex parameters). A relation $c\sin\alpha t + d\sin\beta t = 0$ gives, on differentiating once and three times at zero, $c\alpha + d\beta = 0$ and $c\alpha^3 + d\beta^3 = 0$. The determinant is $\alpha\beta(\beta^2 - \alpha^2)$, so both coefficients vanish in the stated case. Otherwise one function is zero or the two differ only by sign.

### 26.1/2: Independence of distinct exponentials

*Source: printed page 230.*

For each $j$, apply $\prod_{k\neq j}(D-\lambda_k)$ to a relation $\sum c_k e^{\lambda_k t}=0$. All terms except the $j$th disappear, leaving $c_j\prod_{k\neq j}(\lambda_j-\lambda_k)e^{\lambda_j t}=0$. Thus $c_j=0$ for every $j$.

### 26.1/3: Uniqueness of a sum of quasi-polynomials

*Source: printed page 230.*

Suppose $\sum_j e^{\lambda_j t}p_j(t)=0$, with $\deg p_j<\nu_j$ and distinct exponents. Fix $j$ and apply $Q_j(D)=\prod_{k\neq j}(D-\lambda_k)^{\nu_k}$. After division by $e^{\lambda_j t}$, this gives $Q_j(D+\lambda_j)p_j=0$. On the finite-dimensional polynomial space, each factor $D+(\lambda_j-\lambda_k)$ is invertible: it is a nonzero scalar plus a nilpotent operator, whose inverse is a finite geometric series. Hence $p_j=0$. This proves uniqueness of every coefficient.

### 26.3/1*: Rotation-invariant function spaces on the circle

*Source: printed page 232.*

Over $\mathbb{C}$, all such finite-dimensional spaces are

$$V=\operatorname{span}_{\mathbb{C}}\{e^{ik\theta}:k\in F\},\qquad F\subset\mathbb{Z}\text{ finite.}$$

To prove necessity, average a Hermitian inner product on $V$ over rotations. The rotations then act as commuting unitary operators and admit a common orthonormal eigenbasis. A one-dimensional character of the circle has the form $e^{iks}$ with integer $k$: lift it to the real parameter, where the continuous character is $e^{ics}$, and use $2\pi$-periodicity. If $f(\theta+s)=e^{iks}f(\theta)$, set $\theta=0$ to obtain $f(s)=f(0)e^{iks}$, with $f(0)\neq 0$ for a nonzero function. Conversely, each listed span is rotation-invariant. For real-valued functions, the possibilities are sums of the constant space and any finite selection of the two-dimensional spaces $\operatorname{span}_{\mathbb{R}}\{\cos k\theta,\sin k\theta\}$, $k\geq 1$.

### 26.5/1: Trial forms for thirteen forced equations

*Source: printed page 235.*

Let $P_d, Q_d$ denote independent polynomials of degree at most $d$ with coefficients to be determined. The rule is: for forcing $e^{\lambda t}P_d(t)$, multiply the trial polynomial by $t^m$, where $m$ is the multiplicity of $\lambda$ as a characteristic root. This follows by shifting $D$ to $D+\lambda$ and factoring out its zero of order $m$. In each pair below the first sign is plus.

| Number | Suitable particular-solution form |
|---|---|
| 1, 2 | $P_2(t)$ |
| 3, 4 | $Ce^{2t}$ |
| 5 | $e^{-t}P_1(t)$ |
| 6 | $te^{-t}P_1(t)$ |
| 7 | $t\big(P_3(t)\cos t + Q_3(t)\sin t\big)$ |
| 8 | $P_3(t)\cos t + Q_3(t)\sin t$ |
| 9, 10 | $e^{t}\big(P_1(t)\cos t + Q_1(t)\sin t\big)$ |
| 11 | $e^{(1+i)t}P_2(t) + te^{(1-i)t}Q_2(t)$ |
| 12 | $te^{(1+i)t}P_2(t) + e^{(1-i)t}Q_2(t)$ |
| 13 | $te^{t}\big(P_2(t)\cos t + Q_2(t)\sin t\big)$ |

For 11 and 12 the coefficients may be complex. Indeed, $(1+i)^2 = 2i$ and $(1-i)^2 = -2i$, so only one of the two forcing exponents resonates for each complex-coefficient operator. For 13, both $1 \pm i$ are simple roots of $\lambda^4 + 4$, so both terms acquire a factor $t$. The question asks for trial forms; substitution uniquely determines their coefficients once the homogeneous overlap has been removed.

### 26.6/1: Forced systems preserve quasi-polynomial form

*Source: printed page 239.*

For $\dot{x} = Ax + f(t)$, variation of constants gives

$$x(t) = e^{tA}\left(c + \int_0^t e^{-sA} f(s)\,\mathrm{d}s\right).$$

Every entry of $e^{\pm sA}$ is a sum of quasi-polynomials by Jordan form. Products preserve that class. Its antiderivatives also remain in the class: integrate polynomials directly, and use integration by parts finitely many times for $s^m e^{\lambda s}$ when $\lambda \neq 0$. Thus the formula proves the assertion for every component of every solution.

### 26.6/2: Forced recurrences

*Source: printed page 239.*

Let $E$ be the forward shift. A forcing term $\lambda^n p(n)$ with $\lambda \neq 0$ and $\deg p = d$ is annihilated by $(E-\lambda)^{d+1}$, since applying $E - \lambda$ takes a finite difference of the polynomial. A finite product of these annihilators kills a sum of forcing terms. Apply that product to the recurrence, written with a polynomial in $E$, to obtain a homogeneous constant-coefficient recurrence. Its solutions are finite sums of $\mu^n q(n)$, with the polynomial degrees determined by characteristic multiplicities. If the recurrence has zero characteristic roots and arbitrary one-sided initial data, finite initial transients must also be included; the quasi-polynomial description is exact after those nilpotent transients have vanished.

For the displayed sequence, index its first term as $x_1 = 0$. Then $x_n = 2x_{n-1} + n$ for $n \geq 2$, and a particular solution is $-n - 2$. Thus

$$\boxed{x_n = 3\,2^{n-1} - n - 2 \quad (n \geq 1),}$$

which gives $0, 2, 7, 18, 41, 88, \ldots$. With zero-based indexing of the same list the formula is $x_n = 3\,2^n - n - 3$, and its forcing is $n + 1$.

### 26.7/1: The small-amplitude pendulum period

*Source: printed page 240.*

With $x(0) = A$, $\dot{x}(0) = 0$, oddness in $A$ gives $x = A\cos t + A^3 u(t) + O(A^5)$ on fixed bounded time intervals. The cubic equation is

$$u'' + u = \tfrac{1}{6}\cos^3 t = \tfrac{1}{8}\cos t + \tfrac{1}{24}\cos 3t, \quad u(0) = u'(0) = 0,$$

so $u = (\cos t - \cos 3t)/192 + t\sin t/16$. The next maximum is near $T = 2\pi$. Dividing the condition $\dot{x}(T) = 0$ by $A$ and writing $T = 2\pi + \delta$ yields $-\delta + A^2(2\pi/16) + O(A^4) = 0$. Therefore

$$T(A) = 2\pi\left(1 + \frac{A^2}{16} + O(A^4)\right).$$

Smoothness of the return time follows from the implicit function theorem for the divided equation; its evenness in $A$ justifies the fourth-order remainder. The secular term in the finite-time expansion does not assert unbounded pendulum motion.

### 26.7/2: Quadratic and cubic corrections to the period

*Source: printed page 241.*

Assume $\omega > 0$ and $x(0) = A$, $\dot{x}(0) = 0$ for $x'' + \omega^2 x + ax^2 + bx^3 = 0$. Introduce $\tau = \Omega t$, $x = Au_1 + A^2 u_2 + \cdots$, $\Omega = \omega(1 + \sigma A^2 + O(A^3))$. The order-$A^2$ forcing has no fundamental harmonic, so there is no order-$A$ frequency correction. With primes denoting $\tau$ derivatives,

$$u_1 = \cos\tau, \qquad u_2'' + u_2 = -\frac{a}{\omega^2}\cos^2\tau, \qquad u_2 = \frac{a}{3\omega^2}\cos\tau - \frac{a}{2\omega^2} + \frac{a}{6\omega^2}\cos 2\tau.$$

At order $A^3$, the coefficient of $\cos\tau$ in the forcing for $u_3'' + u_3$ is

$$2\sigma + \frac{5a^2}{6\omega^4} - \frac{3b}{4\omega^2}.$$

It must vanish for a periodic correction. Hence $\sigma = 3b/(8\omega^2) - 5a^2/(12\omega^4)$, and

$$\boxed{T(A) = \frac{2\pi}{\omega}\left[1 + \left(\frac{5a^2}{12\omega^4} - \frac{3b}{8\omega^2}\right)A^2 + O(A^3)\right].}$$

For an asymmetric potential, the positive turning point is the amplitude convention here; the negative turning point need not be $-A$. The energy derivation below independently justifies the expansion.

### 26.7/3: Recovering the period from the energy integral

*Source: printed page 241.*

For the pendulum, put $k = \sin(A/2)$ in the standard energy integral to obtain

$$T = 4\int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - k^2 \sin^2 u}} = 2\pi\left(1 + \frac{k^2}{4} + O(k^4)\right) = 2\pi\left(1 + \frac{A^2}{16} + O(A^4)\right).$$

For $U(x) = \omega^2 x^2/2 + ax^3/3 + bx^4/4$, use the smooth local coordinate $z$ defined by $U(x) = \omega^2 z^2/2$, $z'(0) = 1$. Inverting its Taylor series gives

$$x = z - \frac{a}{3\omega^2} z^2 + \left(\frac{5a^2}{18\omega^4} - \frac{b}{4\omega^2}\right) z^3 + O(z^4).$$

If $\rho = \sqrt{2E}/\omega$, the turning points become $z = \pm\rho$ and

$$T = \frac{2}{\omega}\int_{-\rho}^{\rho} \frac{x'(z)}{\sqrt{\rho^2 - z^2}}\,\mathrm{d}z = \frac{2\pi}{\omega}\left[1 + \left(\frac{5a^2}{12\omega^4} - \frac{3b}{8\omega^2}\right)\rho^2 + O(\rho^4)\right].$$

Odd terms integrate to zero and the mean of $z^2$ in this weighted integral is $\rho^2/2$. Since $\rho^2 = 2U(A)/\omega^2 = A^2 + O(A^3)$, this reproduces 26.7/2 without expanding an integral at moving singular endpoints.

## 27 Nonautonomous linear equations (Section 27)

### 27.3/1: A fundamental oscillator matrix

*Source: printed page 244.*

The columns of

$$X(t) = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$$

satisfy $\dot{X} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) X$. Since $X(0) = I$, they have independent initial values and form a fundamental system; equivalently $\det X(t) = 1$ for every $t$.

### 27.4/1: Independent vector functions with zero determinant

*Source: printed page 246.*

Yes: $\varphi_1(t) = (1, 0)^T$ and $\varphi_2(t) = (t, 0)^T$ are linearly independent as functions, since $c_1 + c_2 t \equiv 0$ forces both constants to vanish. But at each time their values are collinear, so their vector Wronskian (the determinant of their value columns) is identically zero. They cannot be a fundamental system of one common homogeneous linear ODE.

### 27.4/2: Wronskian and transition determinant

*Source: printed page 246.*

Let $X(t)$ have the fundamental solutions as columns and let $\Phi(t, t_0)$ be the transition operator. Each column satisfies $\varphi_j(t) = \Phi(t, t_0)\varphi_j(t_0)$, so $X(t) = \Phi(t, t_0)X(t_0)$. Taking determinants gives $W(t) = \det \Phi(t, t_0)W(t_0)$.

### 27.5/1: A scalar fundamental system

*Source: printed page 247.*

For constant coefficients, each characteristic root $\lambda$ of multiplicity $m$ supplies $e^{\lambda t}, te^{\lambda t}, \dots, t^{m-1}e^{\lambda t}$. The annihilator calculation and independence proof in 25.4/1 and 26.1/3 show these form a basis. For real coefficients replace each conjugate pair by $t^j e^{\alpha t}\cos\omega t$ and $t^j e^{\alpha t}\sin\omega t$. In particular, $x'' + ax = 0$ has bases

$$\begin{cases} (\cos\sqrt{a}\,t,\ \sin\sqrt{a}\,t), & a > 0, \\ (e^{\sqrt{-a}\,t},\ e^{-\sqrt{-a}\,t}), & a < 0, \\ (1, t), & a = 0. \end{cases}$$

Their Wronskians are respectively $\sqrt{a}$, $-2\sqrt{-a}$, and 1, all nonzero in their cases.

### 27.5/2: Vanishing at a single time

*Source: printed page 247.*

No. Take the analytic functions $f(t) = 1$ and $g(t) = t^2$. Their Wronskian is $fg' - f'g = 2t$, which vanishes at zero but not identically.

**Review note.** The printed answer's qualification "unless the functions are analytic" is false for the question as written: the counterexample is polynomial. Analyticity makes an identically zero Wronskian imply dependence, but does not make a single zero imply identical vanishing. Solutions of the same regular homogeneous linear equation do have the latter property.

### 27.5/3: An identically zero Wronskian without dependence

*Source: printed page 248.*

For smooth functions the answer is still no. Define $f(t) = e^{-1/t^2}$ for $t > 0$ and zero for $t \le 0$; define $g(t) = f(-t)$. Both are smooth, their supports are disjoint except for the point where both are flat, and $fg' - f'g = 0$ everywhere. But evaluating a constant linear relation at positive and negative times shows that they are independent. For analytic functions on a connected interval the answer is yes: where $f \neq 0$, $(g/f)' = W/f^2 = 0$, and analytic continuation extends the resulting identity; the case $f \equiv 0$ is immediate.

### 27.6/1: The volume after unit time

*Source: printed page 249.*

The system matrix is $\begin{pmatrix} 2 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix}$, whose trace is 2. Liouville's formula gives the volume multiplier $e^{2t}$. The initial unit cube therefore has volume $e^2$ after unit time, without any need to compute its individual transformed edges.

### 27.6/2: Damping and parametric instability

*Source: printed page 251.*

For $x'' + \alpha(t)x' + \omega(t)^2 x = 0$, the state matrix has trace $-\alpha(t)$, so

$$\det \Phi(t, 0) = \exp\left( - \int_0^t \alpha(s)\, \mathrm{d}s \right).$$

If $\alpha(t) < 0$, this determinant is at least one for $t \geq 0$. Asymptotic stability would force both columns of $\Phi$ to tend to zero and hence its determinant to tend to zero, a contradiction.

Positive damping does not guarantee stability when the stiffness varies with time. To give an explicit construction, first let $y'' + q(t)y = 0$, with $q = 1$ for a quarter-period $\pi/2$ and $q = 4$ for a quarter-period $\pi/4$, repeated with $T = 3\pi/4$. The two transition matrices are

$$M_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 1/2 \\ -2 & 0 \end{pmatrix}, \quad M_2 M_1 = \operatorname{diag}(-1/2, -2).$$

Put $x = e^{-\alpha t/2} y$, $\alpha = 1/10$. Then $x'' + \alpha x' + (q(t) + \alpha^2/4)x = 0$, with positive damping and positive stiffness. Its growing multiplier has modulus $2e^{-\alpha T/2} > 1$. For a smooth example, smooth $q$ in arbitrarily short neighborhoods of the two jumps, preserving positivity and periodicity. The transition matrix depends continuously on the coefficient in the integral norm, by its integral equation and Gronwall's inequality. The strict inequality for the multiplier therefore persists. This supplies smooth coefficients satisfying the book's hypotheses.

### 27.6/3*: Liouville's theorem for a nonlinear flow

*Source: printed page 251.*

Let $F_t$ be the flow of a $C^1$ vector field, defined on the region under consideration. Its derivative $Y(t, a) = D_a F_t(a)$ satisfies $\dot{Y} = Dv(F_t(a))Y$, $Y(0, a) = I$. Jacobi's determinant formula gives

$$\partial_t J(t, a) = (\operatorname{div} v)(F_t(a)) J(t, a), \qquad J = \det Y > 0.$$

For a bounded measurable region $D$ whose closure stays within the flow domain on the time interval, change of variables and differentiation under the integral give

$$\frac{d}{dt} \operatorname{vol} F_t(D) = \int_D (\operatorname{div} v)(F_t(a)) J(t, a)\, \mathrm{d}a = \int_{F_t(D)} \operatorname{div} v(x)\, \mathrm{d}x.$$

The same result holds under the usual integrability conditions for unbounded regions. In particular, zero divergence preserves volume. For Hamilton's equations the mixed derivatives cancel in the divergence, yielding the Hamiltonian corollary.

### 27.7/1: The Dirichlet spectrum with zero potential

*Source: printed page 255.*

For $x'' + \lambda x = 0$, $x(0) = x(l) = 0$, integration by parts gives $\lambda \int_0^l x^2 = \int_0^l (x')^2 > 0$ for a nonzero solution. Thus $\lambda > 0$, and the left boundary condition leaves $x = C\sin(\sqrt{\lambda}t)$. The right one gives

$$\lambda_k = (k\pi/l)^2, \qquad x_k(t) = C\sin(k\pi t/l), \quad k = 1, 2, \ldots, \ C \neq 0.$$

Each eigenspace is one-dimensional.

### 27.7/2: The high-frequency eigenvalue asymptotic

*Source: printed page 256.*

Let $q_{\min} \le q(t) \le q_{\max}$ on $[0, l]$. The eigenvalues in increasing order of $-D^2 - q(t)$ with Dirichlet endpoints satisfy

$$(k\pi/l)^2 - q_{\max} \le \lambda_k \le (k\pi/l)^2 - q_{\min}.$$

One way to see this using the section's comparison theorem is to shoot with $x(0) = 0$, $x'(0) = 1$: increasing $q + \lambda$ makes the $k$th zero occur no later. The two constant potentials thus bracket the parameter producing its $k$th zero at $l$. The parameters are simple and ordered: at a zero $l$, differentiating in $\lambda$ gives $x'(l)\partial_\lambda x(l) = \int_0^l x(t)^2\,\mathrm{d}t > 0$, so the angle crosses the endpoint transversely and in the positive direction. Dividing the bounds by $k^2$ proves $\lambda_k/k^2 \to (\pi/l)^2$.

### 27.7/3: The self-adjoint variable-coefficient equation

*Source: printed page 256.*

For $(px')' + qx = 0$, with $p > 0$, use $y = px'$. Then

$$x' = y/p, \qquad y' = -qx, \qquad \varphi' = \frac{qx^2 + y^2/p}{x^2 + y^2},$$

where $\varphi$ is measured from the positive $y$-axis toward the positive $x$-axis. At a zero of $x$, the crossing direction is determined by the nonzero $y$ exactly as before. For two solutions of the same equation, $x_1y_2 - y_1x_2$ is constant. Independent solutions therefore have alternating simple zeros, with exactly one zero of one between consecutive zeros of the other. For the same $p$, increasing $q$ increases the angular velocity at a fixed angle, so Sturm comparison follows. With unequal $p$, the sufficient comparison conditions are $Q \ge q$ and $0 < P \le p$. Non-strict inequalities allow coincident endpoint zeros; strict comparison somewhere between the endpoints gives an interior zero, unless the exceptional equality case applies.

The change of independent variable $s = \int_0^t p(u)^{-1}\,\mathrm{d}u$ turns the equation into $x_{ss} + p(t(s))q(t(s))x = 0$, transferring the zero-spacing comparisons to $s$. For example, if $pq \ge \omega^2 > 0$, consecutive zeros have $s$-spacing at most $\pi/\omega$; if $pq \le \Omega^2$ with $\Omega > 0$, it is at least $\pi/\Omega$. If $q \le 0$, multiplication by $x$ and integration between two zeros would give $\int p(x')^2 = \int qx^2 \le 0$, excluding two zeros of a nonzero solution.

For the spectral problem $(px')' + (q + \lambda)x = 0$ with Dirichlet conditions, set

$$s = \int_0^t p(u)^{-1/2}\,\mathrm{d}u, \quad L = s(l), \quad x(t) = p(t)^{-1/4}z(s).$$

A direct differentiation gives

$$z_{ss} + (\lambda + \widetilde{q}(s))z = 0, \qquad \widetilde{q} = q - \frac{p''}{4} + \frac{(p')^2}{16p},$$

with the right-hand coefficients evaluated at $t = t(s)$. On the compact interval this potential is bounded. Thus the eigenvalues are simple, form an infinite increasing sequence, and satisfy

$$\frac{\lambda_k}{k^2} \longrightarrow \left( \frac{\pi}{\int_0^l p(t)^{-1/2}\,\mathrm{d}t} \right)^2.$$

Its $k$th eigenfunction has $k-1$ interior zeros, inherited from the transformed Sturm problem.

# 28 Periodic coefficients (Section 28)

### 28.2/1: Rotations and hyperbolic rotations

*Source: printed page 258.*

An ordinary rotation preserves the Euclidean norm under all iterates, proving stability. A nontrivial hyperbolic rotation has real reciprocal eigenvalues $e^a, e^{-a}$ with $a \neq 0$; a nonzero component in the expanding direction grows geometrically. Hence its fixed point is unstable. The identity ($a = 0$) is the excluded trivial case.

### 28.3/1: The unperturbed monodromy

*Source: printed page 260–261.*

For $x'' + \omega^2 x = 0$, the solutions with initial states $(1, 0)$ and $(0, 1)$ give

$$M(2\pi) = \begin{pmatrix} \cos 2\pi\omega & \sin(2\pi\omega)/\omega \\ -\omega \sin 2\pi\omega & \cos 2\pi\omega \end{pmatrix}.$$

Its trace is $2\cos 2\pi\omega$, and determinant one. Thus $|\operatorname{tr} M| < 2$ for $\omega \notin \frac{1}{2}\mathbb{Z}$, proving strong stability within the periodic Hamiltonian class. At a positive half-integer $M = \pm I$, which is stable but not strongly stable against arbitrary such perturbations. At $\omega = 0$, take the continuous limit $M = \left(\begin{smallmatrix} 1 & 2\pi \\ 0 & 1 \end{smallmatrix}\right)$, which is unstable.

### 28.4/1: Stability regions for a square-wave coefficient

*Source: printed page 262.*

There is a parameter mismatch between the printed equation and its supplied solution. We first solve the stated equation, in which $f(t) = \omega + \varepsilon$ on the first half-period and $f(t) = \omega - \varepsilon$ on the second. Define the entire real functions

$$C(q) = \sum_{j \geq 0} \frac{(-q)^j \pi^{2j}}{(2j)!}, \qquad S(q) = \sum_{j \geq 0} \frac{(-q)^j \pi^{2j+1}}{(2j+1)!}.$$

For $q > 0$ these are $\cos(\pi\sqrt{q})$ and $\sin(\pi\sqrt{q})/\sqrt{q}$; for negative $q$ they are the corresponding hyperbolic functions. The half-period transition is

$$F(q) = \begin{pmatrix} C(q) & S(q) \\ -qS(q) & C(q) \end{pmatrix}, \qquad \det F(q) = 1.$$

Consequently the exact answer, valid even when a half-period coefficient is zero or negative, is

$$M = F(\omega - \varepsilon)F(\omega + \varepsilon), \qquad D = \operatorname{tr} M = 2C(\omega - \varepsilon)C(\omega + \varepsilon) - 2\omega S(\omega - \varepsilon)S(\omega + \varepsilon).$$

The open stable region is $|D| < 2$, and the open unstable region is $|D| > 2$. On $D = \pm 2$, stability holds exactly when $M = \pm I$; otherwise the repeated multiplier has a nontrivial Jordan block and its powers grow linearly. These formulas fully specify the boundaries as well as the interior regions.

For their local shape when $\omega > |\varepsilon|$, put $a = \sqrt{\omega + \varepsilon}$, $b = \sqrt{\omega - \varepsilon}$, $s = (a+b)/2$, $d = (a-b)/2$. Then $\omega = s^2 + d^2$, $\varepsilon = 2sd$, and

$$D = (2 + \Delta)\cos 2\pi s - \Delta \cos 2\pi d, \qquad \Delta = \frac{2d^2}{s^2 - d^2}.$$

At $\varepsilon = 0$ the possible tongues start at $\omega = \kappa^2$, $\kappa = k/2$. Near a positive half-integer $\kappa = k + 1/2$, expansion of $D = -2$ yields

$$s = \kappa \pm \frac{|d|}{\pi\kappa} + O(d^2), \qquad \omega = \kappa^2 \pm \frac{|\varepsilon|}{\pi\kappa} + O(\varepsilon^2).$$

The region between these branches is unstable. Near a positive integer $k$, expansion of $D = 2$ yields $s = k \pm d^2/k + O(d^4)$, hence the two branches

$$\omega = k^2 - \frac{\varepsilon^2}{4k^2} + O(\varepsilon^4), \qquad \omega = k^2 + \frac{3\varepsilon^2}{4k^2} + O(\varepsilon^4).$$

The region between the two integer branches is unstable: near $s = k$, $D - 2 = 4\pi^2(d^4/k^2 - (s-k)^2) +$ higher-order terms. Its adjacent regions are stable. The exact formula above also covers the origin, where these positive-frequency expansions are not uniform.

**Review note.** The book's calculation instead sets the two frequencies to $\omega \pm \varepsilon$, which corresponds to $f = (\omega \pm \varepsilon)^2$. Under that intended frequency convention, use $s = \omega, d = \varepsilon$ above: the integer branches are $\omega = k \pm \varepsilon^2/k + O(\varepsilon^4)$, and the half-integer branches are $\omega = \kappa \pm |\varepsilon|/(\pi\kappa) + O(\varepsilon^2)$. Figure 14 uses this frequency convention. The exact trace formula avoids all these convention ambiguities.

### 28.4/2: Stabilizing the upper pendulum equilibrium

*Source: printed page 263–264.*

Yes, in the small-oscillation model studied in this chapter. Use the book's explicit piecewise-constant vertical acceleration of the support, $\pm c$ for successive intervals of length $\tau$, with displacement amplitude $a$ and $c = 8a/\tau^2$. Let $\omega^2 = g/l$, $\alpha^2 = c/l$, and suppose $\alpha^2 > \omega^2$. The linearized equations alternate between $x'' = k^2 x$ and $x'' = -\Omega^2 x$, where $k^2 = \alpha^2 + \omega^2$ and $\Omega^2 = \alpha^2 - \omega^2$. Their monodromy has determinant one and trace

$$D = 2\cosh(k\tau)\cos(\Omega\tau) + \left(\frac{k}{\Omega} - \frac{\Omega}{k}\right)\sinh(k\tau)\sin(\Omega\tau).$$

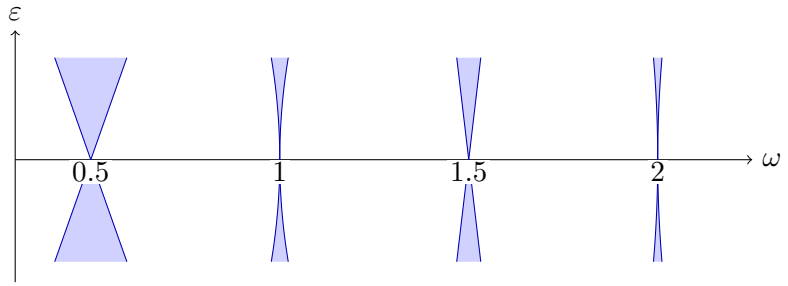



Figure 14: Leading-order shapes of the instability tongues (shaded) for the frequency convention $f = (\omega \pm \varepsilon)^2$. Half-integer tongues open linearly and integer tongues quadratically. Use the exact trace for quantitative boundaries.

Thus $|D| < 2$ is an exact sufficient and necessary condition for strict elliptic stability. Put $\delta^2 = a/l$ and $\mu^2 = g/c$. Then $k\tau = 2\sqrt{2}\delta\sqrt{1+\mu^2}$ and $\Omega\tau = 2\sqrt{2}\delta\sqrt{1-\mu^2}$. Expanding gives

$$D = 2 + 32\delta^2\mu^2 - \frac{64}{3}\delta^4 + O(\delta^6 + \delta^4\mu^2 + \delta^2\mu^4).$$

For sufficiently small fixed $a/l > 0$ and sufficiently large $c$ the negative fourth-order term prevails, while $D$ remains far above $-2$. This proves the existence of a stable high-frequency regime. To leading order its condition is $g/c < 2a/(3l)$, or

$$N = \frac{1}{2\tau} > \sqrt{\frac{3}{64}}\,\frac{\sqrt{gl}}{a}.$$

For $l = 20\,\mathrm{cm}$, $a = 1\,\mathrm{cm}$, $g = 980\,\mathrm{cm/s^2}$, the leading-order threshold is 30.3 cycles per second. The printed value near 43 results from a factor-of-two discrepancy in the fourth-order expansion; its recommendation of 50 cycles per second is nevertheless within the stable regime for these parameters. Strict stability persists after sufficiently small smoothing of the acceleration jumps. This Floquet argument proves stability of the linearized periodic equation; nonlinear stability of the full driven pendulum is a further question and does not follow just from unit-modulus multipliers.

# 29 Variation of constants (Section 29)

### 29.3/1: The forced oscillator for arbitrary forcing

*Source: printed page 266.*

For continuous $f$ and data $x(0) = x_0$, $\dot{x}(0) = v_0$, use the homogeneous basis $\cos t, \sin t$. Varying its coefficients gives $c_1' = -f(t)\sin t$, $c_2' = f(t)\cos t$. Hence

$$\boxed{x(t) = x_0\cos t + v_0\sin t + \int_0^t \sin(t-s)f(s)\,\mathrm{d}s.}$$

Differentiating once makes the endpoint term zero, and differentiating twice produces $f(t)$; the remaining integral is $-x(t) + x_0\cos t + v_0\sin t$. Thus $x'' + x = f$, and the initial values are immediate. For data at $t_0$, replace $t$ in the homogeneous factors by $t - t_0$ and replace the lower integration limit by $t_0$. The formula works on the entire interval where $f$ is continuous.

## Chapter 3 review and coverage

All 119 explicitly numbered problems in Chapter 3 are covered, including the starred problems. Multi-part exercises retain one heading per textbook problem; in particular 26.5/1 includes all thirteen equations. Section 19 has no numbered problems. The five figures above are original drawings; the pendulum portrait uses numerical integration, while the other plotted curves use explicit formulas or the indicated leading-order expansions.

| Section | Problems | Section | Problems |
|---|---|---|---|
| 13 | 4 | 22 | 8 |
| 14 | 19 | 23 | 9 |
| 15 | 10 | 24 | 7 |
| 16 | 4 | 25 | 6 |
| 17 | 3 | 26 | 10 |
| 18 | 15 | 27 | 12 |
| 19 | 0 | 28 | 4 |
| 20 | 2 | 29 | 1 |
| 21 | 5 | Total | 119 |

**Points to retain.** The exponential is always invertible, but its addition law requires commuting generators. Conjugacies of flows preserve time and periods, whereas orbit equivalences need not. Asymptotic stability includes Lyapunov stability; pointwise attraction alone is weaker. Jordan blocks affect stability at the spectral boundary. For periodic equations, use the monodromy, not the instantaneous eigenvalues. Variation of constants works without a small-forcing assumption.

**Corrections and qualifications.** Problem 27.5/2 has a false analyticity qualification in the printed answer. Problem 28.4/1 uses different parameters in its statement and supplied calculation; both conventions are treated explicitly here. The fourth-order expansion in 28.4/2 changes the leading numerical threshold, as checked against the exact trace. Discrete recurrences may have finite nilpotent transients when zero characteristic roots are allowed. Assertions about strict interior zeros in Sturm comparison require the appropriate independence or strictness conditions.

**Cumulative coverage.** Chapters 1–3 now contain 323 numbered problem solutions: 108 in Chapter 1, 96 in Chapter 2, and 119 in Chapter 3. This is a worked-solutions draft for mathematical review.

**Source.** V. I. Arnold, *Ordinary Differential Equations*, third edition, translated by Roger Cooke, Springer, 1992, Chapter 3, printed pp. 152–266 (library PDF pages 156–270). Problem identifiers preserve the section, subsection, and problem numbers of that edition.